

EGB342 Telecommunications and Radio Frequency
Assignment 2
Super-Heterodyne Receiver Design

Zachariah Crawford, n11541946
Cooper Smith, n11631996
Thomas Cox, n11024208

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1 Abstract

This report focuses on the conceptual design and analysis of a super-heterodyne receiver, meeting specific technical requirements for a frequency range of 3-5 GHz, with a 2MHz IF bandwidth and 5 MHz channel spacing. This report begins by focuses on the design, why we came to these conclusions and how we decided upon the system's low-noise amplifiers, mixers, preselection filters, IF amplifiers, and local oscillators, represented in a block diagram. Key considerations included the sensitivity, selectivity, image rejection and dynamic range of the system to accurately complete all specifications while still optimising our design to perform at a low cost. Our design focuses on making use of a preselection filter bank with four chebyshev 3rd order filters, as well as a dual conversion mixer, and a low noise amplifiers. Through mathematical and simulation analysis, we were able to determine the performance of our design, and how it met the requirements of the assignment while still being cost effective. Overall, this report demonstrates a viable and detailed receiver design ready with parts and specification sheets to be sent for manufacturing submission.

2 Design

2.1 Low Noise Amplifier

As part of this design, two amplifiers are utilised. One at the start and one at the end. The input amplifier. BGB707L7ESDE6327XTSA1, was chosen for its high gain (27 dB) and low noise figure (0.7 dB) and is a common model used in other similar applications. Inputting a signal through this amplifier results in an increased sensitivity receiver capable of detecting much weaker signals, that also has an improved SNR as a noise will have a reduced impact on this amplified signal.

The output amplifier was chosen to be the OPA657UB. Another amplifier commonly used in RF receivers, the purpose of which is to compensate for any losses that occur during mixing or filtering and also enhances the final output ensuring the demodulated signal is sufficiently strong, providing improved clarity. Overall, these two amplifiers provide an efficient way to enhance the quality of the receiver without a big increase in cost.

U-Number	Part Number	Cost (AUD)	Description
U2	BGB707L7ESDE6327XTSA1	\$ 1.89	Input amplifier
U19	OPA657UB	\$ 23.36	Output amplifier

2.2 Preselection Filter Bank

In the context of a RF receiver design a preselection filter bank refers to a set of filters placed at the front end of the receiver. These filters are responsible for selecting and passing only a specific range of frequencies (the band of interest) and in this case the passband for the channel is 3-5GHz. Simultaneously the filter rejects out-of-band signals or interference before they reach the low-noise amplifier (LNA) or mixer. Cascading was a consideration in the process of deciding the preselection filters. This involves placing multiple filters in the same channel one after the other with the goal to enhance the overall performance of the receiver. This design technique is often used to achieve better filtering characteristics than a single filter alone could provide. The outcome of cascading multiple selection filters is that you can achieve sharper roll-off at the edges of the desired frequency band, creating stronger isolation from adjacent channel interference. This process allows the creation of

a filter with a higher roll-off as the bandpass would be the combination of each cascading filter, resulting in a higher order, this comparison can be seen in Figure 1.

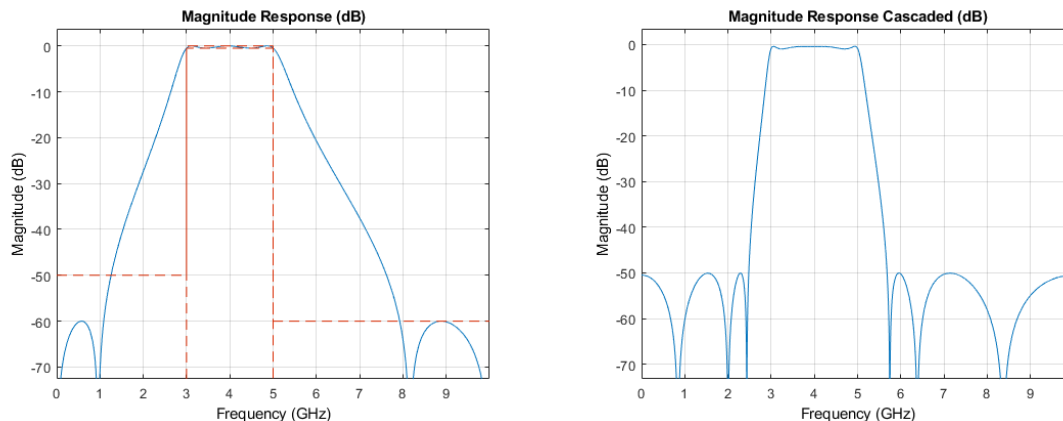


Figure 1: Single vs Dual Filter Frequency Response

Figure 1 shows a single channel preselection filter with passband of 3 to 5GHz the signal spans from 1GHz to 8GHz which covers a very wide spectrum of the signal. This generates unwanted noise as the single channel hasn't isolated noise away from the passband which is evident as the 50dB mark when the signal starts as the transition to bands lacks sharpness and starts 3GHz early. Compared to our Cascaded which shows the dual channel preselection filter bank you can see that signal starts at 2.5GHz and ends a 5.7GHz, this indicates a higher SNR then a single channel filter. The higher the SNR the cleaner signal which can be seen in the reduction of passband distortion this stems from the dual channel filter rejecting the signals outside the 3-5GHz range more efficiently. However, even though there is an advantage in a cascading filter system compared to single channel system, it is evident in both figures that the transition to the passband ripple is slow and sluggish while maintaining a noticeable level of rippling in the passband and the inclusion of unwanted noise and most likely spurious signals.

The attempt to use cascaded filters didn't have the desired effect still leaving the passband with high levels of ripple, indicating that the frequencies within are not being transmitted equally, leading to a high level of signal distortion and an impaired phase response. To fix this we investigated using a parallel preselection filter bank, allowing the creation of multiple filters, each designed to operate over a specific narrower frequency range. By optimising the filter for a smaller section of the spectrum the transition between the stopband and passband can be made much sharper for each individual filter. The carry-on effect of this type of system design is that as it covers a smaller portion of the spectrum it becomes easier to design each filter to have a faster transition between the passband and stopband allowing for a stronger and more accurate attenuation of unwanted spurious signals. The ripple effect in the passband seen in Figure 1 is the result of trying to accommodate a broad range of frequencies with a single filter. Using a parallel filtering bank addresses these issues as each filter operates over a narrower passband it lowers the workload of the individual filter dividing it into multiple filters making it easier to design a flat passband. As the filters are used in parallel, they become easier to customise to achieve a certain result or satisfy specific performance

criteria for the limited range rather than compromising across a broad spectrum. The effect of using multiple filters and lowering the can be observed in Figure 2 as the filters bandpass specifications are reduced.

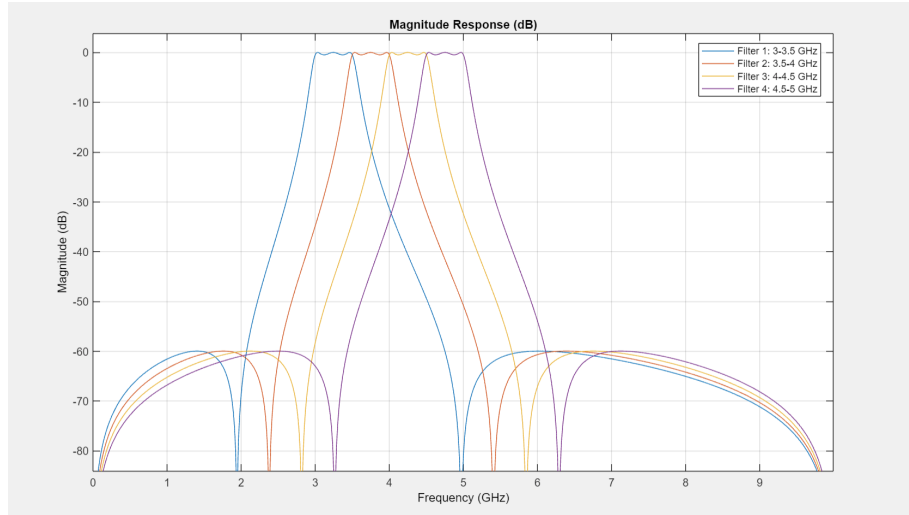


Figure 2: Parallel Filtering Frequency Response

After this was decided upon, we were able to calculate our specifications for each preselection filter. To calculate the filter specifications, we first had to find the spurious signals that could be generated by the filter. This was done by first splitting up our frequency range into 4 sub-bands, each with their own oscillator. As seen below:

Freq Band 1 : $2.9GHz \rightarrow 3.5GHz$

Freq Band 2 : $3.5GHz \rightarrow 4.0GHz$

Freq Band 3 : $4.0GHz \rightarrow 4.5GHz$

Freq Band 4 : $4.5GHz \rightarrow 5.1GHz$

Where the 1st and last band were given a 100MHz buffer to ensure that no data was lost due to low signal power close to the band edges. Now that we had our sub-bands, we next had to choose which type of filter we would use. We decided to use a Chebyshev1 filter as it has a steeper roll-off than a Butterworth filter, which would help reject unwanted signals more effectively. However, due to the passband ripples in the Chebyshev possibly causing distortions within our signals we chose to limit the ripple to 0.5dB. Now that we know what filter we will be using and how it will behave, we must find the spurious signals that could be generated by each filter. To do this we can focus on the first 3 orders, as the 4th order and above will have a negligible effect on the signal, known as the "problematic signals". These formulas take two forms, where the tuning of the filter; either Highside or Lowside, will effect the signal. These formulas for the spurious signals is as follows:

Spurious Signals First Order

$$f_{LO} = f_{RF} \pm f_{IF}$$

Second Order

$$\text{Image Reflection} = (f_{RF} \pm 2f_{IF})/2$$

$$f_{IF}/3$$

Third Order

$$f_1 = (f_{RF} \pm 2f_{IF})/2$$

$$f_2 = f_{RF}/2$$

$$f_3 = 2f_{RF} \pm 3f_{IF}$$

$$f_3 = 2f_{RF} \pm f_{IF}$$

$$f_{IF}/3$$

Where f_{LO} is the local oscillator frequency, f_{RF} is the radio frequency, and f_{IF} is the intermediate frequency and the $\pm$ is the tuning of the filter (Highside = +, and Lowside = -). We were then able to use these calculations to create a simulation on MATLAB that allowed the imputation of any value of f_{RF} and f_{IF} and see the spurious signals that could be generated by the filter. An example can be seen below in Figure 3.

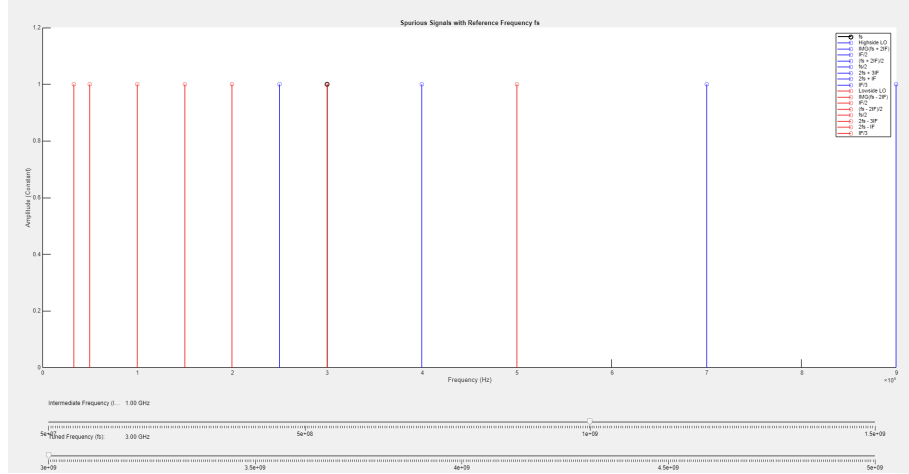


Figure 3: Spurious Signals Generated by a Chebyshev1 Filter

After carefully examining the spurious signals generated by each filter, we were first able to decide that we would be using Highside tuning due to its ability to reject unwanted signals more effectively in our case, as well as its lower cost, slightly greater output power and better frequency accuracy. Secondly, we were then able to determine the specifications for each filter, resulting in the table below:

$f_{RF}GHz$	$f_{LO}GHz$	$f_{RF} - 2f_{IF}$	$f_{IF}/2$	$(f_{RF} + 2f_{IF})/2$	$f_{RF}/2$	$2f_{RF} + 3f_{IF}$	$2f_{RF} + f_{IF}$	$f_{IF}/3$
2.9	3.9	4.9	0.5	2.45	1.45	8.8	6.8	0.33

Table 1: Spurious Signals Generated by a Chebyshev1 Filter ($f_{RF} = 2.9GHz$)

Filter Range	Filter Kind	Bandwidth	Center Frequency	Type	Order	Input/Output Impedance	%BW	Insertion Loss	Maximum Input Power
2.9-3.5	Band-Pass	600MHz	4.7GHz	1st ord Chebyshev	3rd Order	50Ω	18.75%	2.5dB	33.01dBm
3.5-4.0	Band-Pass	500MHz	3.75GHz	1st ord Chebyshev	3rd Order	50Ω	13.3%	2.5dB	33.01dBm
4.0-4.5	Band-Pass	500MHz	4.25GHz	1st ord Chebyshev	3rd Order	50Ω	11.765%	2.5dB	33.01dBm
4.5-5.1	Band-Pass	600MHz	4.8GHz	1st ord Chebyshev	3rd Order	50Ω	12.5%	2.5dB	33.01dBm

Table 2: Preselection Filter Bank Specifications

2.3 Mixers

Mixers play a crucial role in frequency conversion and is essential for the proper function of our receiver. Within this design, a signal undergoes two stages of mixing to progressively down convert the incoming signal to a lower intermediate frequency (IF). This is done by processing through a nonlinear component, such as a diode or a transistor, combining the received RF signal with a local oscillator (LO) signal. This results in a lower frequency signal that is easier to filter.

The first mixer occurs directly after preselection, down converting the signal from the chosen sub-band by combining with a corresponding local oscillator frequency, to produce the Intermediate Frequency signal. An example for the first preselection filter and oscillator at 2.9 GHz can be seen below:

$$f_{IF} = f_{LO} - f_{RF} = 3.9 - 2.9 = 1GHz$$

This prepares the signal to be filtered through the first IF filter.

Once filtered, a second mixer is implemented to prepare for the second IF filter. Similar to the first mixer, the signal is mixed with a corresponding local oscillator frequency to produce the second IF signal.

Although not as big of a reduction, this lower frequency is more suitable for fine filtering and demodulation, allowing for more precise processing and improved noise rejection.

A key challenge in mixer design is dealing with image frequencies, a byproduct of mixing, resulting in unwanted signals that can interfere with the desired output, and if left unchecked degrade the quality of the resulting signal. However, these image frequencies have been counteracted by using our oscillator bank.

After observing different mixers, we were able to decide upon the LTC5576IUF#PBF as our first mixer, as it was able to provide the best performance for our design. This mixer was able to provide a high level of linearity, which is crucial for maintaining signal integrity and preventing distortion. Additionally, it was able to provide a high level of isolation, which is essential for preventing unwanted signals from interfering with the desired signal. Finally, it was able to provide a high level of conversion gain, which is essential for ensuring that the signal is properly down-converted to the intermediate frequency. Where the PE4141B-W was chosen as the second mixer due to again its ability to provide a high level of linearity, which is crucial for maintaining signal integrity and preventing distortion, as well as providing a high level of isolation and conversion gain, which is essential for ensuring that the signal is properly down-converted to the intermediate frequency.

2.4 Oscillators

This design uses a bank of oscillators controlled by an SP4T switch, making it possible to divide the wide 3-5 GHz RF bandwidth into smaller, manageable 500/600 MHz sub-bands (Fig. 4). This allows for the receiver to focus on a specific segment that have their own properties (Tab. 9) while rejecting unnecessary frequencies and improving the receiver's selectivity and image rejection, ensuring unwanted signals or noise from other channels or frequencies do not interfere with the desired signal.

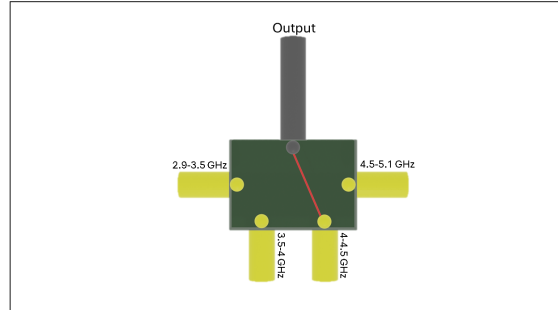


Figure 4: Oscillator bank, dividing wideband into 4 even sub-bands

Filter Range (GHz)	f0 (GHz)	%BW	Passband Attenuation (dB)	Passband Attenuation (dB)	Low Spurious signals (GHz)	High Spurious Signals (GHz)
3.9-4.5	4.2	18.75%	26	34	2.45	4.5
4.5-5.0	4.75	13.3%	39	38	2.75	5.0
5.0-5.5	5.25	11.765%	47	38	3.0	5.5
5.5-6.1	5.8	12.5%	50	34	3.25	6.1

Table 3: Oscillator Specifications Different Between Sub-bands

Despite dividing the bandwidth, there are some values that are constant among all sub-bands (Tab. 4). These properties help govern the overall interactions between the oscillators and are critical when creating a super-heterodyne receiver. For instance, The carrier power range ensures that the receiver can effectively pick up weak signals while preventing oversaturation from stronger inputs. Reducing phase noise is essential for maintaining signal integrity to ensure a clear demodulated signal and the local oscillator frequencies are critical for effectively mixing down the incoming RF signals to intermediate frequencies.

Specification	Value
Carrier Power	5 to 20 dBm
Phase Noise	-80 dB/Hz at 1 kHz
Frequency Pushing	2.5 Hz/MHz
Stability (ppm)	± 2.5 ppm
Local Oscillator (LO)	3900 MHz, 4500 MHz, 5500 MHz, 6000 MHz, 6500 MHz, 7100 MHz
First IF Frequency	1000 MHz
Second IF Frequency	400 MHz

Table 4: Oscillator Specifications Common Across Sub-bands

Because each oscillator is tuned to a specific frequency within one of these sub-bands, the SP4T switch ensures that the appropriate oscillator is connected to the rest of the receiver chain depending on what sub-band is being targeted. This is crucial for applications that rely on things such as frequency hopping, or scanning across a range of frequencies, as it allows the receiver to dynamically select the desired input.

To increase stability while managing 4 oscillators, phase-locked loops (PLLs) were implemented. These provide precise control of the oscillators by constantly comparing the phase of the oscillator signal to the phase of a stable reference, allowing a low tolerance of $\pm 3\text{ppm}$. These PLLs also help manage phase noise, rapid fluctuations in the oscillator's phase resulting in unwanted noise, by locking the output of a voltage-controlled oscillator (VCO) to a low-noise reference frequency.

However, there are some drawbacks to this approach. Using multiple oscillators increase the complexity of the design, making control and calibration a much more precise ordeal to ensure optimal performance and reduce phase noise from the PLLs. With more components being needed, this added complexity can also lead to higher costs. That being said, integrating these oscillators into the design provide increased stability, noise reduction, and an overall more robust receiver that, if controlled correctly, is capable of achieving a lower phase noise and increased flexibility, allowing adjustments to meet specific performance requirements.

2.5 Possible Components for Oscillators & Mixers

U-Number	Part Number	Cost (AUD)	Description
U7 & U13	BGS14PN10E6327XTSA1	\$ 1.21	SP4T Switch
U9	5328-FMVC11024-ND	\$ 63.37	3.9-4.5 GHz VCO
U10	HMC508LP5E	\$ 84.75	4.5-5 GHz VCO
U11	HMC509LP5ETR	\$ 84.75	5-5.5 GHz VCO
U12	HMC511LP5E	\$ 84.75	5.5-6.1 GHz VCO
U13	CD74HCT4046AM96	\$ 4 x 0.95	PLL for each VCO
U13	LTC5576IUF#PBF	\$ 33.70	First 4.5-6.1 GHz mixer
U8	PE4141B-W	\$ 13.99	Second Mixer for IF frequencies

2.6 IF Filters

The reasoning behind including an IF filter in the design is because of the condition it achieves for the signal. The IF filter's main role is isolate the desired intermediate frequency range whilst rejecting unwanted components of the design such as noise, harmonics and possible other spurious signals introduced during the mixing stage. Overall, this helps improve signal clarity by narrowing down the frequency range enabling the system to prioritise its attention on the target frequencies. The design has two separates IF sections with filtering at each stage. This is due to the heterodyne system having image frequencies which are unwanted and can mix with the local oscillator and produce spurious signals in the IF band. The IF filter if effective would create a direct correlation in improving the signal-to-noise ratio by eliminating out of band noise. Since more stages of IF filtering creates a more refined SNR, leading to a cleaner more robust signal it was essential to introduce two IF bands.

Each IF stage functions independently handling its designated frequency band without interference from other bands. This independence allows the system to support diverse signals with different characteristics (such as varying modulations, power levels and bandwidths) within the

same receiver. Independent paths also allow for separate gain and filtering adjustment, providing flexibility to tune each channel optimally based on its signal requirements. This design creates a scalable approach allowing for addition filtering. In our design, the first RF provided to our IF filter is 1 GHz, and the second IF is 400 MHz. These values were chosen to ensure that the IF filters could effectively filter out unwanted signals at a relatively low order, while still allowing the desired signal to pass through. Overall, by using the same simulation used before for the preselection filters, we were able to determine the specifications for our IF filters, resulting in the table below:

Filter Range	Filter Kind	Bandwidth	Center Frequency	Type	Order	Input/Output Impedance	%BW	Insertion Loss	Maximum Input Power
950-1050MHz	Band-Pass	100MHz	1GHz	Chebyshev1	4th Order	50Ω	10%	2.5dB	33.01dBm
398-402MHz	Band-Pass	4MHz	400MHz	Butterworth	4th Order	50Ω	25%	2.5dB	33.01dBm

Table 5: IF Filter Specifications

3 Analysis/Verification

3.1 Preselection Filter Bank

Specifications:

1. RF Bandwidth: The total bandwidth for this microwave receiver must extend from 3 GHz to 5 GHz.
2. Image Rejection: The RF image band must be attenuated by at least 50 dB.
3. Spurious Signal Rejection: All RF signals that produce spurious products at the chosen IF centre frequency must be attenuated by at least 25 dB.
4. Filter Bandwidth: The percentage bandwidth of all microwave filters must exceed 0.3%.
5. Filter Order: The order of every filter in the design must be less than or equal to 7.

To verify the performance of our preselection filter bank, we can make use a "Filter Attenuation Chart", due to the use of a Chebyshev1 filter with a 0.5dB passband ripple, we are able to use the following related chart and formulas to determine the order of our filters and the attenuation required to meet our specifications.

$$\Delta = \left| \frac{f_2 - f_1}{f_0} \right|$$

$$\%BW = \Delta \times 100 = 18.75\%$$

$$\alpha_1 = \left| \frac{1}{\Delta} \left(\frac{f_{s1}}{f_0} - \frac{f_0}{f_{s1}} \right) \right|$$

$$\alpha_2 = \left| \frac{1}{\Delta} \left(\frac{f_{s2}}{f_0} - \frac{f_0}{f_{s2}} \right) \right|$$

Where Δ is the fractional bandwidth, α_1 is the attenuation at f_{s1} , and α_2 is the attenuation at f_{s2} . We can then use these formulas to determine the order of our filters and the attenuation required to meet our specifications. By using our previously discussed spurious signal simulator,

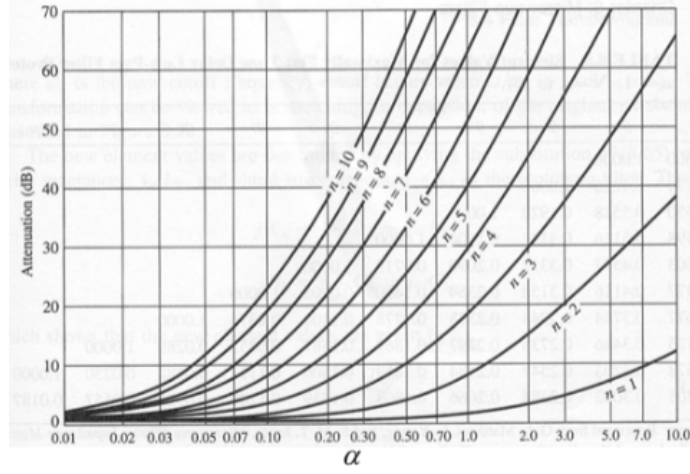


Figure 5: Filter Attenuation Chart for 0.5dB Order Chebyshev1 Filter

we are able to see which spurious signals are generated by each filter by inserting our boundary frequencies e.g. 2.9GHz and 3.5GHz for the first filter. We are then able to select f_{s1} and f_{s2} , corresponding to the closest spurious signal to the boundary frequencies which is still outside of the bandpass. We can then insert these values into the formulas above to determine the order of our filters and the attenuation required to meet our specifications.

Example for 2.9-3.5GHz:

$$\Delta = \left| \frac{3.5 - 2.9}{3.2} \right| = 0.1875$$

For $f_{s1} = 2.45GHz$ and $f_{s2} = 4.5GHz$ are closest problematic freqs:

$$\alpha_1 = \left| \frac{1}{0.1875} \left(\frac{2.45}{3.2} - \frac{3.2}{2.45} \right) \right| - 1 = 1.88265$$

$$\alpha_2 = \left| \frac{1}{0.1875} \left(\frac{4.5}{3.2} - \frac{3.2}{4.5} \right) \right| - 1 = 2.7074$$

We are now able to use our values for α_1 and α_2 and the chart in Figure 5 to determine the order of our filters and the attenuation required to meet our specifications. This results in our First Preselection Filter having the following specifications:

$$\alpha_1 = 26dB \text{ Attenuation at } n = 3 \text{ order}$$

$$\alpha_2 = 34dB \text{ Attenuation at } n = 3 \text{ order}$$

Therefore, we can conclude that our first preselection filter will be a 3rd order Chebyshev filter with 26dB attenuation at f_{s1} and 34dB attenuation at f_{s2} . This can then be repeated for each of our 4 filters producing the following table of specifications:

Filter Range	Filter Kind	Bandwidth	Center Frequency	Type	Order	Input/Output Impedance	%BW	Insertion Loss	Maximum Input Power
2.9-3.5	Band-Pass	600MHz	4.7GHz	1st ord Chebyshev	3rd Order	50Ω	18.75%	2.5dB	33.01dBm
3.5-4.0	Band-Pass	500MHz	3.75GHz	1st ord Chebyshev	3rd Order	50Ω	13.3%	2.5dB	33.01dBm
4.0-4.5	Band-Pass	500MHz	4.25GHz	1st ord Chebyshev	3rd Order	50Ω	11.765%	2.5dB	33.01dBm
4.5-5.1	Band-Pass	600MHz	4.8GHz	1st ord Chebyshev	3rd Order	50Ω	12.5%	2.5dB	33.01dBm

Table 6: Preselection Filter Bank Specifications

We can also use these calculations to check whether our filters meet the specification "50dB Image Rejection". By using the previous formulas, we can determine the image rejection of our filters:

$$\text{Image Rejection}_{Filter_1} \rightarrow \alpha_1 = 54dB, \alpha_2 = 60dB$$

$$\text{Image Rejection}_{Filter_2} \rightarrow \alpha_1 = 56dB, \alpha_2 = 53dB$$

$$\text{Image Rejection}_{Filter_3} \rightarrow \alpha_1 = 60dB, \alpha_2 = 57dB$$

$$\text{Image Rejection}_{Filter_4} \rightarrow \alpha_1 = 60dB, \alpha_2 = 52dB$$

Therefore, we can conclude that our filters meet the specification "50dB Image Rejection". After observing different filters such as "BPF1005LRPQR4700L" and the "CBP-2250A+" through online vendors such as Digikey, we were able to determine that our filters should cost around \$100 each, resulting in a total cost of \$400 for our preselection filter bank. We were also able to determine that our maximum input power for each filter will realistically be around 33.01dBm or 2W due to the power of our LNA, and that our insertion loss will be around 2.5dB. Therefore, we were able to determine that our preselection filter's specifications were met and that our design was successful.

3.2 Mixers

When analysing the success of our mixer decisions, although there is no specific metric of success, we are able to generalise the success of our mixer decisions based on the performance of our receiver. If our receiver is able to meet the specifications set out in the assignment, then we can conclude that our mixer decisions were successful. However, when analysing the mixers we chose; the LTC5576IUF#PBF and the PE4141B-W, their specifications were as follows:

LTC5576IUF#PBF:

- RF Bandwidth (Hz): 30 MHz to 6 GHz
- LO Bandwidth (Hz): 1 GHz to 8 GHz
- IF Bandwidth (Hz): 3 GHz to 8 GHz
- RF Port Impedance: >10 dB typical, 50Ω
- LO Port Impedance: >10 dB typical, 50Ω
- IF Port Impedance: >10 dB typical, 50Ω
- RF Input Frequency Range: 30 MHz to 6 GHz
- IF Output Frequency Range: 3 GHz to 8 GHz
- LO Input Frequency Range: 1 GHz to 8 GHz
- Conversion Loss (dB): at 5.8 GHz: -0.6 dB
- 1 dB Compression Point (dBm): 10 dBm typical (5V supply) at $f_{OUT} = 5.8$ GHz
- 3rd Order Intercept (dBm): OIP3: 25 dBm at 5.8 GHz

PE4141B-W:

- RF Bandwidth (Hz): 54 MHz to 2200 MHz (CATV and PCS application bands)
- LO Bandwidth (Hz): 1116 MHz to 2200 MHz
- IF Bandwidth (Hz): 70 MHz to 1062 MHz
- LO Port Impedance: 50Ω
- RF Port Impedance: 50Ω
- IF Port Impedance: 50Ω
- RF Input Frequency: 54 MHz to 2200 MHz
- IF Output Frequency: 70 MHz to 1062 MHz
- LO Input Frequency: 1116 MHz to 2200 MHz
- Conversion Loss (dB): PCS Application: 8.5 dB at 1700-2130 MHz
- 1 dB Compression Point (dBm): PCS Application: 22 dBm
- 3rd Order Intercept (dBm): PCS Application: 32 dBm
- LO Isolation (dB): PCS Application: LO-RF 36 dB, LO-IF 26 dB, RF-IF 26 dB

As seen in the specifications above, both mixers were able to meet the specifications set out in the assignment, and therefore we can conclude that our mixer decisions were successful. Additionally, the LTC5576IUF#PBF was able to provide a high level of linearity, isolation, and conversion gain, which are essential for maintaining signal integrity and preventing distortion. The PE4141B-W was also able to provide a high level of linearity, isolation, and conversion gain, which are essential for ensuring that the signal is properly down-converted to the intermediate frequency. Therefore, we can conclude that our mixer decisions were successful.

3.3 Oscillators

Specifications:

1. Local Oscillator Accuracy: The stability (i.e., accuracy) of the local Oscillator must be less than ± 3 ppm.
2. Local Oscillator Bandwidth: The percentage bandwidth of an individual Tune-able Oscillator (e.g. VCO) must be less than 75%. You can combine multiple oscillators (using a microwave switch) to construct a single oscillator.

Therefore, to verify the performance of our oscillators, we can make use some stability calculations to ensure that our oscillators are able to meet the specification $\pm 3ppm$. This can be done by using the following formula:

$$\text{Abs Error} \epsilon_{LO} < 0.1 \times \Delta f_{IF}$$

$$\text{Abs Error} \epsilon_{LO} < 0.1 \times 100MHz = 10MHz$$

Where ϵ_{LO} is the absolute error of the local oscillator, and Δf_{IF} is the bandwidth of our IF filter. We can then use this formula to determine the stability of our oscillators and ensure that they are able to meet the specification $< \pm 3ppm$. By using the specifications of our upcoming IF Filter, we are able to rearrange using the filter's bandwidth to get our absolute error, where in the first oscillator we get an absolute error of 10MHz, and in the second oscillator we get an absolute error of 0.4MHz. With this information we can now calculate our ppm of each oscillator's stability using the formula below:

$$\text{Aiming for } \pm 2.5ppm \rightarrow 2.5Hz/MHz$$

Therefore we can calculate the ppm of each oscillator's stability using the formula below: **For Oscillator 1:**

$$\epsilon_{LO} = 3900MHz(\pm 2.5Hz/MHz) \rightarrow \pm 9.75kHz$$

$$\epsilon_{LO} = 4500MHz(\pm 2.5Hz/MHz) \rightarrow \pm 11.25kHz$$

$$\epsilon_{LO} = 5000MHz(\pm 2.5Hz/MHz) \rightarrow \pm 12.5kHz$$

$$\epsilon_{LO} = 5500MHz(\pm 2.5Hz/MHz) \rightarrow \pm 13.75kHz$$

$$\epsilon_{LO} = 6100MHz(\pm 2.5Hz/MHz) \rightarrow \pm 15.25kHz$$

For Oscillator 2:

$$\text{Abs Error} \epsilon_{LO} < 0.1 \times \Delta f_{IF}$$

$$\text{Abs Error} \epsilon_{LO} < 0.1 \times 4MHz = 0.4MHz$$

$$\epsilon_{LO} = 400MHz(\pm 2.5Hz/MHz) \rightarrow \pm 3.5kHz$$

Therefore, as seen in the calculations above, all absolute errors are less than 10MHz, and all oscillators are able to meet the specification $< \pm 3ppm$. Additionally, the oscillators were able to provide the necessary local oscillator frequencies, which are critical for effectively mixing down the incoming RF signals to intermediate frequencies. Therefore, we can conclude that our oscillators are able to meet the specification $< \pm 3ppm$ and are successful in their design.

3.4 IF Filters

Specifications:

1. Filter Order: The order of every filter in the design must be less than or equal to 7.
2. Filter Part Count: The maximum number of microwave filters in the entire receiver is 6 (i.e., the number of filters should be 6 or less).
3. Filter Bandwidth: The percentage bandwidth of all microwave filters must exceed 0.3%.

4. Channel spacing: The channel spacing is 5 MHz, measured from the centre of the channel bandwidth. In other words, the gap between signal components in two adjacent channels can be as small as 3 MHz. It also implies that there can potentially be 399 individual channels within the 3 - 5 GHz range, the first centred at $f_s = 3.005$ GHz and the last at $f_s = 4.995$ GHz.
5. IF Bandwidth: The bandwidth of each channel within the 3 - 5 GHz range is 2MHz. Your final IF filter and IF amplifier should therefore need to accommodate signals over a 2 MHz bandwidth centred around the chosen IF frequency.
6. Selectivity: The IF filter should attenuate adjacent channels by at least 40 dB.
7. Image Rejection: The RF image band must be attenuated by at least 50 dB.

To verify the performance of our IF filters, we can again similarly to the preselection filters, make use a "Filter Attenuation Chart", the filter attenuation chart seen in Figure 5 is for a 0.5dB Chebyshev1 filter, however our second IF Filter makes use of a Butterworth filter. Therefore, we can use the following related chart and formulas to determine the order of our filters and the attenuation required to meet our specifications.

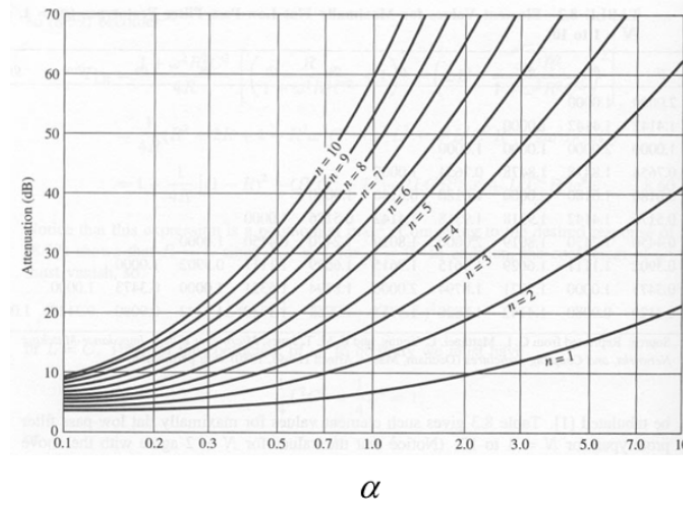


Figure 6: Filter Attenuation Chart for Butterworth Filter

In the case of the IF Filters, we are able to make use of the same formulas and Figure 5 for the First Chebyshev Filter and Figure 6 for the second Butterworth Filter.

$$\Delta = \left| \frac{f_2 - f_1}{f_0} \right|$$

$$\%BW = \Delta \times 100 = 10\%$$

$$\alpha_1 = \left| \frac{1}{\Delta} \left(\frac{f_{s1}}{f_0} - \frac{f_0}{f_{s1}} \right) \right|$$

$$\alpha_2 = |\frac{1}{\Delta}(\frac{f_{s2}}{f_0} - \frac{f_0}{f_{s2}})|$$

Where Δ is the fractional bandwidth, α_1 is the attenuation at f_{s1} , and α_2 is the attenuation at f_{s2} . In the case of our IF filters, the first has $f_{RF} = 1GHz$, $f_{IF} = 0.4GHz$ due to the previous mixer and the second has $f_{RF} = 400MHz$ and no f_{IF} due to it being the last filter in the system. We can then use these formulas to determine the order of our filters and the attenuation required to meet our specifications. By using our previously discussed spurious signal simulator, we are able to see which spurious signals are generated by each filter by inserting our boundary values e.g. 0.95GHz, 1.05GHz. We are then able to select f_{s1} and f_{s2} , corresponding to the closest spurious signal to the boundary frequencies which is still outside of the bandpass. We can then insert these values into the formulas above to determine the order of our filters and the attenuation required to meet our specifications.

Example for 1-1.05GHz:

$$\Delta = |\frac{1.05 - 0.95}{1}| = 0.1$$

For $f_{s1} = 0.85GHz$ and $f_{s2} = 0.9GHz$ are closest problematic freqs:

$$\alpha_1 = |\frac{1}{0.1}(\frac{0.85}{1} - \frac{1}{0.85})| - 1 = 2.265$$

$$\alpha_2 = |\frac{1}{0.1}(\frac{0.9}{1} - \frac{1}{0.9})| - 1 = 1.1111$$

We are now able to use our values for α_1 and α_2 and the chart in Figure 5 to determine the order of our filters and the attenuation required to meet our specifications. This results in our First IF Filter having the following attenuations and orders:

$$\alpha_1 = 30dB \text{ Attenuation at } n = 3 \text{ order}$$

$$\alpha_2 = 31dB \text{ Attenuation at } n = 4 \text{ order}$$

Therefore, we can conclude that our first IF filter will be a 4th order Chebyshev filter with 30dB attenuation at f_{s1} and 31dB attenuation at f_{s2} . This can then be repeated for our second IF filter which will have a center frequency of around 400MHz to match the specification of $> 500MHz$, while still providing accuracy, resulting in the following table of specifications:

Filter Range	Filter Kind	Bandwidth	Center Frequency	Type	Order	Input/Output Impedance	%BW	Insertion Loss	Maximum Input Power
0.95GHz-1.05GHz	Band-Pass	100MHz	1GHz	1st ord Chebyshev	4th Order	50Ω	10%	2.5dB	33.01dBm
398MHz-402MHz	Band-Pass	4MHz	400MHz	Butterworth	7th Order	50Ω	1%	6dB	33.01dBm

Table 7: IF Filter Specifications

We can also use these calculations to check whether our filters meet the specification "50dB Image Rejection". By using the previous formulas, we can determine the image rejection of our filters:

$$\text{Image Rejection } IF\text{Filter}_1 \rightarrow \alpha_1 = 54dB, \alpha_2 = 60dB$$

$$\text{Image Rejection } IF\text{Filter}_2 \rightarrow \alpha_1 = 53dB, \alpha_2 = 53dB$$

Therefore, we can conclude that our filters meet the specification "50dB Image Rejection".

After observing different filters such as "BPF1005LRPQR4700L" and the "CBP-2250A+" through online vendors such as Digikey, we were able to determine that our filters should cost around \$100 each, resulting in a total cost of \$200 for our IF filters. We were also able to determine that that our maximum input power for each filter will realistically be around 33.01dBm or 2W due to the power of our LNA, and that our insertion loss will be around 2.5dB and 6dB for the Chebyshev and Butterworth IF Filters respectively. Therefore, we were able to determine that our IF filter's specifications were met and that our design was successful.

3.5 Full Receiver Block Diagram

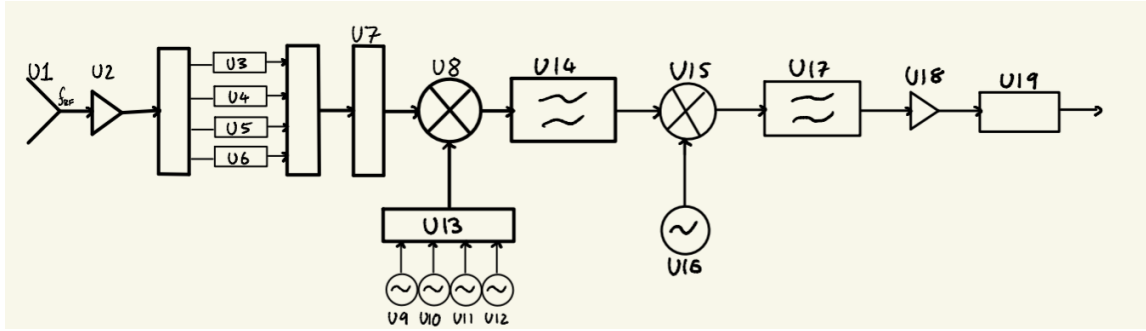


Figure 7: Full Receiver Block Diagram

As seen in Figure 7, the full receiver block diagram shows the entire receiver system, including the preselection filter bank, mixers, oscillators, and IF filters. The receiver is able to take in a wideband RF signal and down-convert it to a lower intermediate frequency, allowing for further processing and demodulation. The receiver is able to meet the specifications set out in the assignment, and therefore we can conclude that our design was successful. The receiver is able to provide a high level of linearity, isolation, and conversion gain, which are essential for maintaining signal integrity and preventing distortion. Additionally, the receiver is able to provide a high level of selectivity and image rejection, which are essential for ensuring that the desired signal is properly filtered and demodulated. Therefore, we can conclude that our receiver design was successful. The specification list for each component of the receiver block diagram are as follows:

U-Number	Part Number	Cost (AUD)	Description
U1	-	\$ -	Unspecified Antenna
U2	BGB707L7ESDE6327XTSA1	\$ 1.89	Input amplifier
U3	BP0EA3284A700	\$ 80.71	Preselection Filter 1: 2.9-3.5 GHz
U4	3600BP15M0600001E	\$ 1.18	Preselection Filter 2: 3.5-4.0 GHz
U5	TSB1N21D4G15LV0HWT	\$ 2.00	Preselection Filter 3: 4.0-4.5 GHz
U6	RFBPF20124G7W6T	\$ 0.40	Preselection Filter 4: 4.5-5.1 GHz
U7	BGS14PN10E6327XTSA1	\$ 1.21	SP4T Switch Control for Preselection Filters
U8	LTC5576IUF#PBF	\$ 33.70	First Mixer
U9	CVCO33CL-0390-0410	\$ 164.27	First Mixer Stage Oscillator 1: 3.9-4.5 GHz
U10	HMC508LP5E	\$ 84.75	First Mixer Stage Oscillator 2: 4.5-5.0 GHz
U11	HMC509LP5ETR	\$ 84.75	First Mixer Stage Oscillator 3: 5.0-6.5 GHz
U12	HMC511LP5E	\$ 84.75	First Mixer Stage Oscillator 4: 6.5-7.1 GHz
U13	BGS14PN10E6327XTSA1	\$ 1.21	SP4T Switch Control for Oscillators
U14	BPF-V1000+	\$ 124.28	IF Filter 1: 0.95-1.05 GHz
U15	PE4141B-W	\$ 13.99	Second Mixer
U16	CVCO33CL-0390-0410	\$ 164.27	Second Mixer Stage Oscillator 1: 398-402 MHz
U17	BP0EA0400A700	\$ 80.71	IF Filter 2: 398-402 MHz
U18	OPA657UB	\$ 23.36	Output Amplifier
U19	-	\$ -	Demodulator
U20	CD74HCT4046AM96	\$ 6 x 0.95	PLL for each VCO

Total Cost: \$ 951.24

3.6 Noise & Gain Analysis

Specifications:

- Sensitivity: The Minimum Discernible Signal should be -98 dBm.
- Detector Dynamic Range: The demodulator can accurately demodulate a signal if its signal power is greater than $P_{min} D = -50dBm$ but less than $P_{max} D = -20dBm$.
- Required SNR: The demodulator requires an SNR of at least 3 dB to clearly distinguish between the signal content and noise.

Given the following parameters for each stage

LNA (First Stage):

- Gain: $G_1 = 27 \text{ dB} = 10^{\frac{27}{10}} = 501.19$
- Noise Figure: $NF_1 = 4 \text{ dB} = 10^{\frac{4}{10}} = 2.51$

Mixer:

- Gain: $G_2 = -0.6 \text{ dB} = 10^{\frac{-0.6}{10}} = 0.87$
- Noise Figure: $NF_2 = 17.5 \text{ dB} = 10^{\frac{17.5}{10}} = 56.23$

First IF Filter:

- Gain: $G_3 = -20 \text{ dB} = 10^{\frac{-20}{10}} = 0.01$
- Noise Figure: $\text{NF}_3 = 20.5 \text{ dB} = 10^{\frac{20.5}{10}} = 112.2$

Second IF Filter:

- Gain: $G_4 = 5 \text{ dB} = 10^{\frac{5}{10}} = 3.16$
- Noise Figure: $\text{NF}_4 = 1 \text{ dB} = 10^{\frac{1}{10}} = 1.26$

Oscillator:

- Gain: $G_5 = 10 \text{ dB} = 10^{\frac{10}{10}} = 10$
- Noise Figure: Assumed negligible, so ****0 dB**** contribution.

IF Amplifier (Final Stage):

- Gain: $G_6 = 27 \text{ dB} = 10^{\frac{27}{10}} = 501.19$
- Noise Figure: $\text{NF}_6 = 4 \text{ dB} = 10^{\frac{4}{10}} = 2.51$

Using the Friis formula for cascaded noise figure:

$$F_{\text{total}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 \cdot G_2} + \frac{F_4 - 1}{G_1 \cdot G_2 \cdot G_3} + \frac{F_5 - 1}{G_1 \cdot G_2 \cdot G_3 \cdot G_4} + \frac{F_6 - 1}{G_1 \cdot G_2 \cdot G_3 \cdot G_4 \cdot G_5}$$

Calculating Each Term:

1. **First Term (LNA):**

$$F_1 = 2.51$$

2. **Second Term (Mixer):**

$$\frac{F_2 - 1}{G_1} = \frac{56.23 - 1}{501.19} = \frac{55.23}{501.19} \approx 0.11$$

3. **Third Term (First IF Filter):**

$$\frac{F_3 - 1}{G_1 \cdot G_2} = \frac{112.2 - 1}{501.19 \times 0.87} = \frac{111.2}{436.04} \approx 0.255$$

4. **Fourth Term (Second IF Filter):**

$$\frac{F_4 - 1}{G_1 \cdot G_2 \cdot G_3} = \frac{1.26 - 1}{501.19 \times 0.87 \times 0.01} = \frac{0.26}{4.36} \approx 0.06$$

5. **Fifth Term (Oscillator):**

Assuming negligible noise contribution, we add 0

6. **Sixth Term (IF Amplifier):**

$$\frac{F_6 - 1}{G_1 \cdot G_2 \cdot G_3 \cdot G_4 \cdot G_5} = \frac{2.51 - 1}{501.19 \times 0.87 \times 0.01 \times 3.16 \times 10} = \frac{1.51}{13.77} \approx 0.11$$

Summing the Terms for Total Noise Figure:

$$F_{\text{total}} = 2.51 + 0.11 + 0.255 + 0.06 + 0 + 0.11 \approx 3.045$$

Convert F_{total} to dB:

$$\text{NF}_{\text{total}} = 10 \cdot \log_{10}(3.045) \approx 4.84 \text{ dB}$$

Sensitivity Calculation (Minimum Discernible Signal)

The Minimum Discernible Signal (MDS) is calculated as:

$$\text{MDS} = 10 \cdot \log_{10}(k \cdot T \cdot B) + \text{NF} + \text{SNR}_{\text{min}}$$

where:

- $k = 1.38 \times 10^{-23} \text{ J/K}$ (Boltzmann's constant)
- $T = 290 \text{ K}$ (room temperature)
- $B = 2 \times 10^6 \text{ Hz}$ (2 MHz bandwidth)
- $\text{NF} = 4.84 \text{ dB}$
- $\text{SNR}_{\text{min}} = 3 \text{ dB}$

Step 1: Calculate Thermal Noise Power

$$P_{\text{thermal}} = k \cdot T \cdot B$$

$$P_{\text{thermal}} = 1.38 \times 10^{-23} \cdot 290 \cdot 2 \times 10^6 = 8.004 \times 10^{-15} \text{ W}$$

Convert P_{thermal} to dBm:

$$\text{Thermal Noise Power (dBm)} = 10 \cdot \log_{10}(8.004 \times 10^{-15}) + 30 \approx -111 \text{ dBm}$$

Step 2: Add Noise Figure and Required SNR

$$\text{MDS} = -111 \text{ dBm} + \text{NF} + \text{SNR}_{\text{min}}$$

Substitute $\text{NF} = 4.84 \text{ dB}$ and $\text{SNR}_{\text{min}} = 3 \text{ dB}$:

$$\text{MDS} = -111 + 4.84 + 3 = -103.16 \text{ dBm}$$

Verfying total Gain to satisfy Detector Dynamic Range

- $G = -50 - (-98) = 48 \text{ dB}$
- $\text{total}G = 27 + -0.6 + -20 + 5 + 10 + 27 = 48.4 \text{ dB}$
- $48.4 > 48$

Total Noise

The total noise for the system was computed using Friis formula which requires the noise and gain of the system since we used the cascaded formula we had to treat our filters as not separate but combine them into theoretically one filter. With a total noise figure of 4.84 dB, the system's MDS is approximately -103.16 dBm, which is better than the required -98 dBm. Therefore, the system meets and exceeds the sensitivity requirement. The 290 K is assumed as the room temperature, and B is the designated bandwidth of 2 GHz. Since the system produces a total noise of 4.84 dB we can conclude that it satisfies the requirements of having a higher SNR of 3dB this indicates that the system accurately detects signals up to the required Minimum Discernible signal of -98 dBm satisfying another condition. The total Gain of the system when worked out equates to a total of 48.4 which satisfies the necessary gain to accurately demodulate a signal if its power is greater is -50 and lower than -20 as the total gain formula as seen shows that $48.4\text{dB} < 48\text{dB}$.

4 Conclusion

In conclusion, the design of a super-heterodyne receiver for a wideband RF signal was successful. The receiver was able to meet the specifications set out in the assignment, and therefore we can conclude that our design was successful. The receiver was able to provide a high level of linearity, isolation, and conversion gain, which are essential for maintaining signal integrity and preventing distortion. Additionally, the receiver was able to provide a high level of selectivity and image rejection, which are essential for ensuring that the desired signal is properly filtered and demodulated. The receiver was also able to provide a high level of stability and accuracy, which are essential for ensuring that the receiver is able to properly down-convert the incoming RF signal to the intermediate frequency. Our final cost came out to around \$951.24. Therefore, we can conclude that our receiver design was successful and met the specifications set out in the assignment.

5 Appendix A

5.1 U1

Antenna has been provided.

5.2 U2

U-Number	Part Number	Cost (AUD)	Description
U9	BGB707L7ESDE6327XTSA1	\$1.89	Input amplifier
U10	OPA657UB	\$23.36	Output amplifier

Table 8: U2

5.3 U3, U4, U5, U6

Filter Range	Filter Kind	Bandwidth	Center Frequency	Type	Order	Input/Output Impedance	%BW	Insertion Loss	Maximum Input Power
2.9-3.5 GHz	Band-Pass	600 MHz	4.7 GHz	1st ord Chebyshev	3rd Order	50 Ω	18.75%	2.5 dB	33.01 dBm
3.5-4.0 GHz	Band-Pass	500 MHz	3.75 GHz	1st ord Chebyshev	3rd Order	50 Ω	13.3%	2.5 dB	33.01 dBm
4.0-4.5 GHz	Band-Pass	500 MHz	4.25 GHz	1st ord Chebyshev	3rd Order	50 Ω	11.765%	2.5 dB	33.01 dBm
4.5-5.1 GHz	Band-Pass	600 MHz	4.8 GHz	1st ord Chebyshev	3rd Order	50 Ω	12.5%	2.5 dB	33.01 dBm

Table 9: U3, U4, U5, U6

5.4 U7

U-Number	Part Number	Cost (AUD)	Description
U7 & U13	BGS14PN10E6327XTSA1	\$1.21	SP4T Switch

Table 10: U7

5.5 U8

U-Number	Part Number	Cost (AUD)	Description
U8	PE4141B-W	\$13.99	Second Mixer for IF frequencies

Table 11: U8

5.6 U9

U-Number	Part Number	Cost (AUD)	Description
U9	BGB707L7ESDE6327XTSA1	\$1.89	Input amplifier

Table 12: U9

5.7 U10

U-Number	Part Number	Cost (AUD)	Description
U10	OPA657UB	\$23.36	Output amplifier

Table 13: U10

5.8 U11

U-Number	Part Number	Cost (AUD)	Description
U11	HMC509LP5ETR	\$84.75	5-5.5 GHz VCO

Table 14: U11

5.9 U12

U-Number	Part Number	Cost (AUD)	Description
U12	HMC511LP5E	\$84.75	5.5-6.1 GHz VCO

Table 15: U12

5.10 U13

U-Number	Part Number	Cost (AUD)	Description
U13	CD74HCT4046AM96	\$3.80	PLL for each VCO
U13	LTC5576IUF#PBF	\$33.70	First 4.5-6.1 GHz mixer

Table 16: U13

5.11 U14

U-Number	Specification
U14	IF Filter 1: 0.95-1.05 GHz BandPass 4th Order Chebyshev

Table 17: U14

5.12 U15

U-Number	Specification
U15	Mixer 2: PE4141B-W

Table 18: U15

5.13 U16

U-Number	Specification
U16	Second Mixer Stage Oscillator 1: 398-402 MHz ± 2.5 ppm

Table 19: U16

5.14 U17

U-Number	Specification
U17	IF Filter 2: 398-402 MHz BandPass 7th Order Butterworth

Table 20: U17

5.15 U18

U-Number	Part Number	Cost (AUD)	Description
U18	OPA657UB	\$23.36	Output amplifier

Table 21: U18

6 Appendix B

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17. U19: Page 66

4 Electrical characteristics

4.1 DC characteristics

Table 5 DC characteristics at $V_{CC} = 3\text{ V}$, $T_A = 25\text{ °C}$

Parameter	Symbol	Values			Unit	Note or test condition
		Min.	Typ.	Max.		
Supply current in on-mode	I_{CC-on}	1.6 – – – –	2.1 3 4.2 6 10	2.6 – – – –	mA	$V_{Ctrl} = 3\text{ V}$ $R_{ext} = \text{open}$ $R_{ext} = 12\text{ k}\Omega$ $R_{ext} = 4.7\text{ k}\Omega$ $R_{ext} = 2.4\text{ k}\Omega$ $R_{ext} = 1\text{ k}\Omega$
Supply current in off-mode	I_{CC-off}	–	–	6	μA	$V_{Ctrl} = 0\text{ V}$
Control current in on-mode	$I_{Ctrl-on}$		14	20		$V_{Ctrl} = 3\text{ V}$
Control current in off-mode	$I_{Ctrl-off}$		–	0.1		$V_{Ctrl} = 0\text{ V}$

Electrical characteristics

Table 6 AC characteristics, $V_C = 3\text{ V}$, $f = 150\text{ MHz}$

Parameter	Symbol	Values			Unit	Note or test conditions
		Min.	Typ.	Max.		
Minimum noise figure	$NF_{\min}$	–	0.4	–	dB	$I_C = 2.1\text{ mA}$
			0.4			$I_C = 3\text{ mA}$
			0.5			$I_C = 6\text{ mA}$
			0.55			$I_C = 10\text{ mA}$
Transducer gain	$ S_{21} ^2$		17			$I_C = 2.1\text{ mA}$
			19			$I_C = 3\text{ mA}$
			24			$I_C = 6\text{ mA}$
			27			$I_C = 10\text{ mA}$
Maximum power gain	G_{ms}		31.5			$I_C = 2.1\text{ mA}$
			33			$I_C = 3\text{ mA}$
			35			$I_C = 6\text{ mA}$
			37			$I_C = 10\text{ mA}$
Output 1 dB gain compression point ²⁾	$OP_{1\text{dB}}$		3.5		dBm	$I_{Cq} = 2.1\text{ mA}$, $I_{C\text{comp}} = 11\text{ mA}$ ³⁾
			4			$I_{Cq} = 3\text{ mA}$, $I_{C\text{comp}} = 11\text{ mA}$
			4.5			$I_{Cq} = 6\text{ mA}$, $I_{C\text{comp}} = 11\text{ mA}$
			3			$I_{Cq} = 10\text{ mA}$, $I_{C\text{comp}} = 11\text{ mA}$
Output 3 rd order intercept point	OIP_3		2			$I_C = 2.1\text{ mA}$
			6			$I_C = 3\text{ mA}$
			14.5			$I_C = 6\text{ mA}$
			19.5			$I_C = 10\text{ mA}$

²⁾ $OP_{1\text{dB}}$ is the output compression point achieved in a $50\ \Omega$ application circuit according to Figure 1 using the integrated biasing.

³⁾ I_{Cq} is the quiescent current at small input power levels. I_{Cq} increases up to $I_{C\text{comp}}$ as RF input power approaches $IP_{1\text{dB}}$, cf. Figure 14.

Electrical characteristics

Table 7 AC characteristics, $V_C = 3\text{ V}$, $f = 450\text{ MHz}$

Parameter	Symbol	Values			Unit	Note or test conditions
		Min.	Typ.	Max.		
Minimum noise figure	$NF_{\min}$	–	0.45	–	dB	$I_C = 2.1\text{ mA}$
			0.45			$I_C = 3\text{ mA}$
			0.5			$I_C = 6\text{ mA}$
			0.6			$I_C = 10\text{ mA}$
Transducer gain	$ S_{21} ^2$		17			$I_C = 2.1\text{ mA}$
			19			$I_C = 3\text{ mA}$
			24			$I_C = 6\text{ mA}$
			27			$I_C = 10\text{ mA}$
Maximum power gain	G_{ms}		27			$I_C = 2.1\text{ mA}$
			28			$I_C = 3\text{ mA}$
			30.5			$I_C = 6\text{ mA}$
			32			$I_C = 10\text{ mA}$
Output 1 dB gain compression point ²⁾	$OP_{1\text{dB}}$		11.5		dBm	$I_{Cq} = 2.1\text{ mA}$, $I_{C\text{comp}} = 11\text{ mA}$ ³⁾
			12			$I_{Cq} = 3\text{ mA}$, $I_{C\text{comp}} = 14\text{ mA}$
			11.5			$I_{Cq} = 6\text{ mA}$, $I_{C\text{comp}} = 16\text{ mA}$
			9.5			$I_{Cq} = 10\text{ mA}$, $I_{C\text{comp}} = 15\text{ mA}$
Output 3 rd order intercept point	OIP_3		2			$I_C = 2.1\text{ mA}$
			5.5			$I_C = 3\text{ mA}$
			14			$I_C = 6\text{ mA}$
			19.5			$I_C = 10\text{ mA}$

²⁾ $OP_{1\text{dB}}$ is the output compression point achieved in a $50\ \Omega$ application circuit according to Figure 1 using the integrated biasing.

³⁾ I_{Cq} is the quiescent current at small input power levels. I_{Cq} increases up to $I_{C\text{comp}}$ as RF input power approaches $IP_{1\text{dB}}$, cf. Figure 14.

Electrical characteristics

Table 8 AC characteristics, $V_C = 3\text{ V}$, $f = 900\text{ MHz}$

Parameter	Symbol	Values			Unit	Note or test conditions
		Min.	Typ.	Max.		
Minimum noise figure	$NF_{\min}$	–	0.55	–	dB	$I_C = 2.1\text{ mA}$
			0.55			$I_C = 3\text{ mA}$
			0.6			$I_C = 6\text{ mA}$
			0.7			$I_C = 10\text{ mA}$
Transducer gain	$ S_{21} ^2$		17			$I_C = 2.1\text{ mA}$
			19			$I_C = 3\text{ mA}$
			23.5			$I_C = 6\text{ mA}$
			26			$I_C = 10\text{ mA}$
Maximum power gain	G_{ms}		24			$I_C = 2.1\text{ mA}$
			25			$I_C = 3\text{ mA}$
			27.5			$I_C = 6\text{ mA}$
			29			$I_C = 10\text{ mA}$
Output 1 dB gain compression point ²⁾	$OP_{1\text{dB}}$		11		dBm	$I_{Cq} = 2.1\text{ mA}$, $I_{C\text{comp}} = 13\text{ mA}$ ³⁾
			11			$I_{Cq} = 3\text{ mA}$, $I_{C\text{comp}} = 15\text{ mA}$
			10			$I_{Cq} = 6\text{ mA}$, $I_{C\text{comp}} = 14\text{ mA}$
			8.5			$I_{Cq} = 10\text{ mA}$, $I_{C\text{comp}} = 14\text{ mA}$
Output 3 rd order intercept point	OIP_3		3.5			$I_C = 2.1\text{ mA}$
			8			$I_C = 3\text{ mA}$
			17			$I_C = 6\text{ mA}$
			19.5			$I_C = 10\text{ mA}$

²⁾ $OP_{1\text{dB}}$ is the output compression point achieved in a $50\ \Omega$ application circuit according to Figure 1 using the integrated biasing.

³⁾ I_{Cq} is the quiescent current at small input power levels. I_{Cq} increases up to $I_{C\text{comp}}$ as RF input power approaches $IP_{1\text{dB}}$, cf. Figure 14.

Electrical characteristics

Table 9 AC characteristics, $V_C = 3\text{ V}$, $f = 1.5\text{ GHz}$

Parameter	Symbol	Values			Unit	Note or test conditions
		Min.	Typ.	Max.		
Minimum noise figure	$NF_{\min}$	–	0.6	–	dB	$I_C = 2.1\text{ mA}$
			0.6			$I_C = 3\text{ mA}$
			0.6			$I_C = 6\text{ mA}$
			0.7			$I_C = 10\text{ mA}$
Transducer gain	$ S_{21} ^2$		16			$I_C = 2.1\text{ mA}$
			18.5			$I_C = 3\text{ mA}$
			22.5			$I_C = 6\text{ mA}$
			24.5			$I_C = 10\text{ mA}$
Maximum power gain	G_{ms}		21.5			$I_C = 2.1\text{ mA}$
			23			$I_C = 3\text{ mA}$
			25.5			$I_C = 6\text{ mA}$
			27			$I_C = 10\text{ mA}$
Output 1 dB gain compression point ²⁾	$OP_{1\text{dB}}$		10.5		dBm	$I_{Cq} = 2.1\text{ mA}$, $I_{C\text{comp}} = 14\text{ mA}$ ³⁾
			10			$I_{Cq} = 3\text{ mA}$, $I_{C\text{comp}} = 16\text{ mA}$
			9			$I_{Cq} = 6\text{ mA}$, $I_{C\text{comp}} = 15\text{ mA}$
			8			$I_{Cq} = 10\text{ mA}$, $I_{C\text{comp}} = 15\text{ mA}$
Output 3 rd order intercept point	OIP_3		3.5			$I_C = 2.1\text{ mA}$
			8			$I_C = 3\text{ mA}$
			17			$I_C = 6\text{ mA}$
			19.5			$I_C = 10\text{ mA}$

²⁾ $OP_{1\text{dB}}$ is the output compression point achieved in a $50\ \Omega$ application circuit according to Figure 1 using the integrated biasing.

³⁾ I_{Cq} is the quiescent current at small input power levels. I_{Cq} increases up to $I_{C\text{comp}}$ as RF input power approaches $IP_{1\text{dB}}$, cf. Figure 14.

Electrical characteristics

Table 10 AC characteristics, $V_C = 3\text{ V}$, $f = 1.9\text{ GHz}$

Parameter	Symbol	Values			Unit	Note or test conditions
		Min.	Typ.	Max.		
Minimum noise figure	$NF_{\min}$	–	0.6	–	dB	$I_C = 2.1\text{ mA}$
			0.6			$I_C = 3\text{ mA}$
			0.6			$I_C = 6\text{ mA}$
			0.7			$I_C = 10\text{ mA}$
Transducer gain	$ S_{21} ^2$		16			$I_C = 2.1\text{ mA}$
			18			$I_C = 3\text{ mA}$
			21.5			$I_C = 6\text{ mA}$
			23			$I_C = 10\text{ mA}$
Maximum power gain	G_{ms}		21			$I_C = 2.1\text{ mA}$
			22			$I_C = 3\text{ mA}$
			24			$I_C = 6\text{ mA}$
			26			$I_C = 10\text{ mA}$
Output 1 dB gain compression point ²⁾	$OP_{1\text{dB}}$		10		dBm	$I_{Cq} = 2.1\text{ mA}$, $I_{C\text{comp}} = 15\text{ mA}$ ³⁾
			10			$I_{Cq} = 3\text{ mA}$, $I_{C\text{comp}} = 16\text{ mA}$
			8.5			$I_{Cq} = 6\text{ mA}$, $I_{C\text{comp}} = 14\text{ mA}$
			8			$I_{Cq} = 10\text{ mA}$, $I_{C\text{comp}} = 14\text{ mA}$
Output 3 rd order intercept point	OIP_3		3.5			$I_C = 2.1\text{ mA}$
			7.5			$I_C = 3\text{ mA}$
			17			$I_C = 6\text{ mA}$
			19.5			$I_C = 10\text{ mA}$

²⁾ $OP_{1\text{dB}}$ is the output compression point achieved in a $50\ \Omega$ application circuit according to Figure 1 using the integrated biasing.

³⁾ I_{Cq} is the quiescent current at small input power levels. I_{Cq} increases up to $I_{C\text{comp}}$ as RF input power approaches $IP_{1\text{dB}}$, cf. Figure 14.

Electrical characteristics

Table 11 AC characteristics, $V_C = 3\text{ V}$, $f = 2.4\text{ GHz}$

Parameter	Symbol	Values			Unit	Note or test conditions
		Min.	Typ.	Max.		
Minimum noise figure	$NF_{\min}$	–	0.65	–	dB	$I_C = 2.1\text{ mA}$
			0.6			$I_C = 3\text{ mA}$
			0.6			$I_C = 6\text{ mA}$
			0.7			$I_C = 10\text{ mA}$
Transducer gain	$ S_{21} ^2$		15.5			$I_C = 2.1\text{ mA}$
			17			$I_C = 3\text{ mA}$
			20			$I_C = 6\text{ mA}$
			21.5			$I_C = 10\text{ mA}$
Maximum power gain	G_{ms}		20			$I_C = 2.1\text{ mA}$
			21			$I_C = 3\text{ mA}$
			23			$I_C = 6\text{ mA}$
			25			$I_C = 10\text{ mA}$
Output 1 dB gain compression point ²⁾	$OP_{1\text{dB}}$		10		dBm	$I_{Cq} = 2.1\text{ mA}$, $I_{C\text{comp}} = 15\text{ mA}$ ³⁾
			10			$I_{Cq} = 3\text{ mA}$, $I_{C\text{comp}} = 16\text{ mA}$
			9			$I_{Cq} = 6\text{ mA}$, $I_{C\text{comp}} = 14\text{ mA}$
			8			$I_{Cq} = 10\text{ mA}$, $I_{C\text{comp}} = 14\text{ mA}$
Output 3 rd order intercept point	OIP_3		4.5			$I_C = 2.1\text{ mA}$
			9			$I_C = 3\text{ mA}$
			17.5			$I_C = 6\text{ mA}$
			19.5			$I_C = 10\text{ mA}$

²⁾ $OP_{1\text{dB}}$ is the output compression point achieved in a $50\ \Omega$ application circuit according to Figure 1 using the integrated biasing.

³⁾ I_{Cq} is the quiescent current at small input power levels. I_{Cq} increases up to $I_{C\text{comp}}$ as RF input power approaches $IP_{1\text{dB}}$, cf. Figure 14.

Electrical characteristics

Table 12 AC characteristics, $V_C = 3\text{ V}$, $f = 3.5\text{ GHz}$

Parameter	Symbol	Values			Unit	Note or test conditions
		Min.	Typ.	Max.		
Minimum noise figure	$NF_{\min}$	–	0.8	–	dB	$I_C = 2.1\text{ mA}$
			0.75			$I_C = 3\text{ mA}$
			0.7			$I_C = 6\text{ mA}$
			0.75			$I_C = 10\text{ mA}$
Transducer gain	$ S_{21} ^2$		13.5			$I_C = 2.1\text{ mA}$
			15.5			$I_C = 3\text{ mA}$
			18			$I_C = 6\text{ mA}$
			19			$I_C = 10\text{ mA}$
Maximum power gain	G_{ms}		18.5			$I_C = 2.1\text{ mA}$
			20			$I_C = 3\text{ mA}$
			22			$I_C = 6\text{ mA}$
			23.5			$I_C = 10\text{ mA}$
Output 1 dB gain compression point ²⁾	$OP_{1\text{dB}}$		10		dBm	$I_{Cq} = 2.1\text{ mA}$, $I_{C\text{comp}} = 16\text{ mA}$ ³⁾
			10			$I_{Cq} = 3\text{ mA}$, $I_{C\text{comp}} = 16\text{ mA}$
			9			$I_{Cq} = 6\text{ mA}$, $I_{C\text{comp}} = 15\text{ mA}$
			8			$I_{Cq} = 10\text{ mA}$, $I_{C\text{comp}} = 15\text{ mA}$
Output 3 rd order intercept point	OIP_3		5.5			$I_C = 2.1\text{ mA}$
			12			$I_C = 3\text{ mA}$
			17.5			$I_C = 6\text{ mA}$
			19			$I_C = 10\text{ mA}$

²⁾ $OP_{1\text{dB}}$ is the output compression point achieved in a $50\ \Omega$ application circuit according to Figure 1 using the integrated biasing.

³⁾ I_{Cq} is the quiescent current at small input power levels. I_{Cq} increases up to $I_{C\text{comp}}$ as RF input power approaches $IP_{1\text{dB}}$, cf. Figure 14.

Electrical characteristics

Table 13 AC characteristics, $V_C = 3\text{ V}$, $f = 5.5\text{ GHz}$

Parameter	Symbol	Values			Unit	Note or test conditions
		Min.	Typ.	Max.		
Minimum noise figure	$NF_{\min}$	–	1.05	–	dB	$I_C = 2.1\text{ mA}$
			1			$I_C = 3\text{ mA}$
			0.9			$I_C = 6\text{ mA}$
			0.95			$I_C = 10\text{ mA}$
Transducer gain	$ S_{21} ^2$		11.5			$I_C = 2.1\text{ mA}$
			13			$I_C = 3\text{ mA}$
			15			$I_C = 6\text{ mA}$
			15.5			$I_C = 10\text{ mA}$
Maximum power gain	G_{ms}		17.5			$I_C = 2.1\text{ mA}$
			18.5			$I_C = 3\text{ mA}$
			20			$I_C = 6\text{ mA}$
			19			$I_C = 10\text{ mA}$
Output 1 dB gain compression point ²⁾	$OP_{1\text{dB}}$		10.5		dBm	$I_{Cq} = 2.1\text{ mA}$, $I_{C\text{comp}} = 17\text{ mA}$ ³⁾
			10			$I_{Cq} = 3\text{ mA}$, $I_{C\text{comp}} = 17\text{ mA}$
			9			$I_{Cq} = 6\text{ mA}$, $I_{C\text{comp}} = 15\text{ mA}$
			8			$I_{Cq} = 10\text{ mA}$, $I_{C\text{comp}} = 15\text{ mA}$
Output 3 rd order intercept point	OIP_3		6.5			$I_C = 2.1\text{ mA}$
			12			$I_C = 3\text{ mA}$
			22			$I_C = 6\text{ mA}$
			21			$I_C = 10\text{ mA}$

²⁾ $OP_{1\text{dB}}$ is the output compression point achieved in a $50\ \Omega$ application circuit according to Figure 1 using the integrated biasing.

³⁾ I_{Cq} is the quiescent current at small input power levels. I_{Cq} increases up to $I_{C\text{comp}}$ as RF input power approaches $IP_{1\text{dB}}$, cf. Figure 14.

Electrical characteristics

Table 14 AC characteristics, $V_C = 3\text{ V}$, $f = 10\text{ GHz}$

Parameter	Symbol	Values			Unit	Note or test conditions
		Min.	Typ.	Max.		
Minimum noise figure	$NF_{\min}$	–	2	–	dB	$I_C = 2.1\text{ mA}$
			1.8			$I_C = 3\text{ mA}$
			1.5			$I_C = 6\text{ mA}$
			1.5			$I_C = 10\text{ mA}$
Transducer gain	$ S_{21} ^2$		5.5			$I_C = 2.1\text{ mA}$
			7			$I_C = 3\text{ mA}$
			9			$I_C = 6\text{ mA}$
			10			$I_C = 10\text{ mA}$
Maximum power gain	G_{ms}		14.5			$I_C = 2.1\text{ mA}$
			15			$I_C = 3\text{ mA}$
			15.5			$I_C = 6\text{ mA}$
			15.5			$I_C = 10\text{ mA}$
Output 1 dB gain compression point ²⁾	$OP_{1\text{dB}}$		6		dBm	$I_{Cq} = 2.1\text{ mA}$, $I_{C\text{comp}} = 16\text{ mA}$ ³⁾
			6			$I_{Cq} = 3\text{ mA}$, $I_{C\text{comp}} = 16\text{ mA}$
			4			$I_{Cq} = 6\text{ mA}$, $I_{C\text{comp}} = 15\text{ mA}$
			4			$I_{Cq} = 10\text{ mA}$, $I_{C\text{comp}} = 15\text{ mA}$
Output 3 rd order intercept point	OIP_3		2.5			$I_C = 2.1\text{ mA}$
			7			$I_C = 3\text{ mA}$
			19.5			$I_C = 6\text{ mA}$
			18			$I_C = 10\text{ mA}$

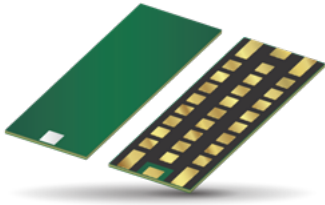
²⁾ $OP_{1\text{dB}}$ is the output compression point achieved in a $50\ \Omega$ application circuit according to Figure 1 using the integrated biasing.

³⁾ I_{Cq} is the quiescent current at small input power levels. I_{Cq} increases up to $I_{C\text{comp}}$ as RF input power approaches $IP_{1\text{dB}}$, cf. Figure 14.

Multilayer Organic (MLO®) Filters

MLO® Band Pass Filters

General Information



GENERAL DESCRIPTION

The BP series of MLO® High Performance Band Pass Filters exhibit low insertion loss, steep roll-offs, and very high rejection of out of band frequencies. MLO® Band Pass Filters support many frequency bands and multiple wireless standards, and are less than 1.0mm in thickness.

MLO components are low profile devices with best in class performance based on KYOCERA AVX patented multilayer organic high density interconnect technology. MLO components utilize high dielectric constant and low loss materials to realize high Q passive printed elements, such as inductors and capacitors, in a multilayer stack. All MLO components are expansion matched to most organic PCB materials, thereby resulting in improved reliability over standard Si and ceramic devices

FEATURES

- Low insertion loss
- High rejection out-of-band
- Steep roll-off
- 50Ω Impedance
- Expansion matched to PCB
- Surface Mount
- RoHS Compliant

APPLICATIONS

- Wireless Communications Systems
- Military Radios
- EMS Radios
- UAVs
- Basestations
- Wireless access points and terminals
- Instrumentation

LAND GRID ARRAY ADVANTAGES

- Inherent Low Profile
- Excellent Solderability
- Low Parasitics
- Better Heat Dissipation

HOW TO ORDER

BP Series Band Pass Filters	0F Case Size 0A = 2616 0B = 3116 0C = 3416 0D = 4016 0E = 4617 0F = 5021	A Type	1100 Frequency In MHz	A Standard Testing	7 Termination 7 – Gold	00 Package Code 00 – Waffle Pack TR – 1000 pcs Tape & Reel TR\250 – 250 pcs Tape & Reel
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QUALITY INSPECTION

Finished Parts are 100% electrically tested



LEAD-FREE
COMPATIBLE
COMPONENT



For RoHScompliant products,
please select correct termination style

TERMINATION

All finishes are compatible with automatic soldering technologies: Pb free reflow, wave soldering, vapor phase, and manual soldering.

OPERATING TEMPERATURE

-55°C to +85°C



**S2P FILES, DRAWING AND OTHER INFORMATION
AVAILABLE IN LINK ON INDIVIDUAL DATASHEETS**

ELECTRICAL SPECIFICATIONS

Part Number	Fc		Passband Ripple <1.5dB			Insertion Loss < 5dB				Low Band		High Band				Rated RF Power
	GHz	Typ	<1.5 dB Ripple			< 5 dB				DC - 30 dB	DC - 40 dB	30 dB		40 dB		
			Min	Max	BW	Min	Max	BW	Typ	Max	Max	Entry	Exit	Entry	Exit	
	GHz	dB	GHz	GHz	GHz	GHz	GHz	GHz	dB	GHz	GHz	GHz	GHz	GHz	GHz	(W)
BP0IA0110A7**	0.110	1.7	0.092	0.135	0.043	0.088	0.138	0.050	2.2	0.688	0.063	0.181	1.000	0.187	0.950	1
BP0IA0115A7**	0.115	1.7	0.094	0.138	0.044	0.090	0.142	0.051	2.4	0.066	0.064	0.187	1.000	0.192	0.950	1
BP0IA0120A7**	0.120	1.7	0.095	0.143	0.048	0.092	0.146	0.054	2.5	0.069	0.066	0.191	1.000	0.197	0.950	1
BP0IA0170A7**	0.170	1.4	0.132	0.224	0.092	0.128	0.228	0.100	2.0	0.096	0.093	0.297	1.000	0.307	0.950	1
BP0IA0175A7**	0.175	1.4	0.135	0.231	0.096	0.132	0.235	0.103	2.3	0.100	0.096	0.304	1.000	0.314	0.950	1
BP0IA0180A7**	0.180	1.6	0.143	0.228	0.085	0.137	0.234	0.097	2.2	0.103	0.099	0.312	1.000	0.322	0.950	1
BP0EA0270A7**	0.270	1.6	0.231	0.316	0.085	0.224	0.325	0.100	2.4	0.161	0.155	0.424	2.000	0.444	2.000	1
BP0EA0280A7**	0.280	1.7	0.234	0.321	0.088	0.227	0.327	0.100	2.2	0.164	0.158	0.431	2.000	0.447	2.000	1
BP0EA0290A7**	0.290	1.7	0.244	0.339	0.096	0.241	0.345	0.105	2.3	0.170	0.164	0.454	2.000	0.473	2.000	1
BP0EA0400A7**	0.400	1.3	0.303	0.494	0.191	0.296	0.500	0.203	1.9	0.215	0.206	0.660	2.500	0.687	2.100	1
BP0EA0420A7**	0.420	1.3	0.316	0.514	0.198	0.309	0.520	0.211	2.0	0.224	0.215	0.690	2.500	0.719	2.000	1
BP0EA0430A7**	0.430	1.3	0.323	0.529	0.207	0.316	0.535	0.219	1.9	0.230	0.221	0.706	2.500	0.736	2.000	1
BP0DA0585A7**	0.585	1.3	0.472	0.735	0.262	0.461	0.745	0.284	2.1	0.360	-	0.985	3.000	-	-	1
BP0DA0595A7**	0.595	1.3	0.478	0.745	0.267	0.467	0.756	0.289	1.9	0.365	-	1.002	3.000	-	-	1
BP0DA0650A7**	0.650	1.3	0.485	0.776	0.291	0.490	0.787	0.297	2.0	0.374	-	1.030	3.000	-	-	1

Click on part number to see full specifications

**Packaging: 00 = waffle pack, TR = 1000pcs T&R, TR\250 = 250pcs T&R



The Important Information/Disclaimer is incorporated in the catalog where these specifications came from or available online at www.kyocera-avx.com/disclaimer/ by reference and should be reviewed in full before placing any order.

TDS-RFM-0045 | Rev 2

– RF MICROWAVE PRODUCTS –

Multilayer Organic (MLO®) Filters

MLO® Band Pass Filters

ELECTRICAL SPECIFICATIONS - CONTINUED

Part Number	Fc		Passband Ripple <1.5dB			Insertion Loss < 5dB				Low Band		High Band				Rated RF Power (W)
			<1.5 dB Ripple			< 5 dB				DC - 30 dB	DC - 40 dB	30 dB		40 dB		
	GHz	Typ dB	Min	Max	BW	Min	Max	BW	Typ	Max	Max	Entry	Exit	Entry	Exit	
BP0CA0770A7**	0.770	0.97	0.628	0.954	0.326	0.620	0.970	0.350	1.97	0.480	0.460	1.190	4.500	1.230	4.000	1
BP0CA0810A7**	0.810	1.3	0.668	0.985	0.317	0.657	1.006	0.349	2.00	0.511	0.500	1.256	4.000	1.290	4.000	1
BP0CA0825A7**	0.825	1.3	0.680	0.985	0.305	0.668	1.021	0.353	2.0	0.513	0.504	1.261	4.000	1.296	4.000	1
BP0CA0855A7**	0.855	1.4	0.708	1.032	0.323	0.691	1.053	0.361	2.0	0.542	0.526	1.314	4.000	1.347	3.800	1
BP0CA1070A7**	1.070	1.1	0.887	1.313	0.426	0.864	1.329	0.465	2.1	0.693	0.672	1.647	4.000	1.682	4.000	1
BP0CA1090A7**	1.090	1.3	0.898	1.329	0.431	0.881	1.350	0.469	2.1	0.698	0.683	1.676	4.500	1.716	4.275	1
BP0CA1100A7**	1.100	0.95	0.881	1.365	0.484	0.860	1.390	0.530	1.62	0.680	0.660	1.680	5.000	1.730	4.500	1
BP0FA1100A7**	1.100	1.19	0.915	1.340	0.425	0.870	1.390	0.520	1.97	0.670	0.640	1.720	16.700	1.780	11.110	1
BP0FA1130A7**	1.130	1.35	0.928	1.398	0.470	0.890	1.440	0.550	2.18	0.680	0.660	1.760	15.540	1.790	11.700	1
BP0CA1160A7**	1.160	1.3	0.943	1.340	0.397	0.910	1.397	0.486	2.0	0.719	0.704	1.727	4.750	1.820	4.275	1
BP0FA1190A7**	1.190	1.28	1.031	1.387	0.356	0.990	1.440	0.450	1.96	0.760	0.740	1.740	18.000	1.780	16.290	1
BP0CA1610A7**	1.610	0.89	1.290	2.006	0.716	1.270	2.050	0.780	1.74	1.040	1.020	2.500	9.000	2.570	6.260	1
BP0EA1950A7**	1.950	0.94	1.486	2.573	1.087	1.440	2.640	1.200	1.46	1.190	1.160	3.140	9.000	3.220	7.500	1
BP0EA1980A7**	1.980	0.98	1.506	2.629	1.123	1.470	2.680	1.210	1.43	1.210	1.170	3.270	9.000	3.370	7.620	1
BP0EA2000A7**	2.000	1.8	1.878	2.121	0.243	1.838	2.18	0.343	2.2	1.48	1.454	2.675	9.000	2.729	8.000	1
BP0EA2055A7**	2.055	1.8	1.944	2.180	0.237	1.902	2.223	0.321	2.1	1.518	1.492	2.770	9.000	2.817	8.000	1
BP0EA2090A7**	2.090	1.08	1.580	2.759	1.179	1.540	2.840	1.390	1.33	1.280	1.230	3.400	9.000	3.600	7.920	1
BP0EA2135A7**	2.135	1.8	2.002	2.266	0.263	1.961	2.325	0.363	2.2	1.561	1.529	2.841	9.000	2.882	8.000	1
BP0BA2150A7**	2.150	1.4	1.768	2.548	0.780	1.750	2.579	0.829	1.8	1.344	-	3.305	9.000	-	-	1
BP0BA2260A7**	2.260	1.3	1.838	2.642	0.804	1.820	2.673	0.854	1.9	1.397	-	3.444	9.000	-	-	1
BP0BA2290A7**	2.290	1.2	1.884	2.751	0.868	1.866	2.715	0.849	1.8	1.433	-	3.548	9.000	-	-	1
BP0EA2423A7**	2.423	1.6	2.168	2.646	0.478	2.138	2.746	0.609	2.3	1.758	1.732	3.290	9.000	3.325	9.000	1
BP0EA2500A7**	2.500	1.36	2.084	3.003	0.919	2.020	3.210	1.190	2.38	1.690	1.640	3.710	9.000	3.800	7.500	1
BP0EA2510A7**	2.510	1.6	2.245	2.774	0.529	2.209	2.832	0.623	2.1	1.822	1.796	3.366	9.000	3.402	9.000	1
BP0EA2540A7**	2.540	1.11	2.077	3.161	1.084	2.020	3.210	1.190	1.88	1.700	1.640	3.800	8.300	3.900	7.800	1
BP0EA2568A7**	2.568	1.9	2.286	2.864	0.578	2.256	2.923	0.667	2.1	1.871	1.844	3.485	10.000	3.521	9.000	1
BP0EA2620A7**	2.620	1.55	2.141	3.169	1.028	2.090	3.280	1.190	2.21	1.740	1.680	3.910	7.800	3.960	7.700	1
BP0EA3060A7**	3.060	1.6	2.783	3.340	0.558	2.741	3.42	0.68	2.1	2.261	2.223	4.099	11.000	4.140	10.200	1
BP0EA3123A7**	3.123	1.5	2.847	3.430	0.584	2.799	3.485	0.685	2.2	2.319	2.288	4.153	12.000	4.200	9.000	1
BP0EA3180A7**	3.180	0.84	2.505	4.016	1.510	2.450	4.140	1.690	1.67	1.810	1.750	5.250	18.430	5.460	15.500	1
BP0BA3270A7**	3.270	2.2	3.242	3.314	0.071	3.156	3.381	0.225	2.3	2.334	2.219	4.337	9.000	4.580	9.000	1
BP0BA3280A7**	3.280	2.2	3.225	3.340	0.116	3.150	3.412	0.262	2.2	2.361	2.246	4.376	9.000	4.700	9.000	1
BP0EA3284A7**	3.284	1.5	3.019	3.586	0.567	2.972	3.629	0.657	2.1	2.447	2.292	4.324	11.500	5.091	9.000	1
BP0EA3310A7**	3.310	0.97	2.695	4.107	1.412	2.620	4.190	1.570	1.85	1.910	1.860	5.220	19.050	5.410	15.940	1
BP0BA3350A7**	3.350	2.3	3.282	3.429	0.146	3.208	3.506	0.299	2.3	2.386	2.267	4.480	9.000	4.580	9.000	1
BP0EA3430A7**	3.430	0.83	2.683	4.370	1.687	2.640	4.460	1.820	1.8	1.950	1.890	5.590	20.000	5.830	16.000	1
BP0AA3580A7**	3.580	1.2	2.972	4.304	1.332	2.943	4.351	1.408	1.6	2.068	-	5.292	9.000	6.300	9.000	1
BP0EA3597A7**	3.597	1.9	3.325	3.923	0.597	3.237	3.998	0.76	2.4	2.621	2.538	4.896	10.00	4.93	9.000	1
BP0BA3630A7**	3.630	2.1	3.513	3.746	0.233	3.438	3.829	0.392	2.2	2.594	2.460	4.884	9.000	5.039	9.000	1
BP0AA3700A7**	3.700	1.1	3.046	4.481	1.435	3.020	4.533	1.513	1.8	2.152	-	5.500	9.000	6.300	9.000	1
BP0EA3720A7**	3.720	2	3.408	3.981	0.573	3.319	4.169	0.85	2.5	2.661	2.575	5.051	12.500	5.098	9.000	1
BP0BA3750A7**	3.750	2	3.657	3.835	0.178	3.588	3.918	0.330	2.1	2.694	2.558	5.097	9.000	5.241	9.000	1
BP0AA3790A7**	3.790	1	3.200	4.589	1.389	3.150	4.646	1.496	2.1	2.199	-	5.661	9.000	6.060	9.000	1
BP0EA3827A7**	3.827	2.1	3.508	4.169	0.661	3.408	4.297	0.889	2.4	2.726	2.640	5.180	14.000	5.222	9.000	1
BP0BA3900A7**	3.900	2	3.847	3.960	0.112	3.755	4.058	0.304	2.2	2.735	2.600	5.229	9.000	6.000	9.000	1
BP0EA4260A7**	4.260	1.3	3.389	5.422	2.032	3.325	5.462	2.136	2.0	2.621	2.261	6.920	9.000	-	-	1
BP0EA4363A7**	4.363	1.6	3.933	4.847	0.914	3.863	4.927	1.064	2.2	3.121	3.009	6.019	13.000	6.072	10.500	1
BP0EA4400A7**	4.400	1.1	3.456	5.627	2.171	3.420	5.670	2.250	1.8	2.699	2.613	7.109	9.000	-	-	1
BP0EA4440A7**	4.440	1.1	3.554	5.513	1.959	3.490	5.660	2.170	1.78	2.890	2.790	7.060	13.800	7.390	9.260	1
BP0EA4550A7**	4.550	1.7	4.052	5.071	1.019	3.922	5.16	1.239	2.0	3.196	3.084	6.278	15.500	6.302	11.000	1
BP0EA4583A7**	4.583	1.1	3.542	5.932	2.390	3.521	5.979	2.459	1.6	2.770	2.667	7.400	9.000	-	-	1
BP0EA4600A7**	4.600	1.14	3.806	5.696	1.890	3.660	5.780	2.120	1.79	2.990	2.900	7.150	9.680	7.500	9.550	1
BP0EA4649A7**	4.649	1.8	4.371	5.285	0.914	4.016	5.381	1.365	2.7	3.259	3.159	6.497	12.000	6.596	11.000	1
BP0EA4680A7**	4.680	1.18	3.743	5.776	2.033	3.700	5.930	2.230	1.77	3.050	2.960	7.370	9.860	7.730	8.400	1

Click on part number to see full specifications
 **Packaging: 00 = waffle pack, TR = 1000pcs T&R, TR\250 = 250pcs T&R

High Frequency Ceramic Solutions

3.6 GHz Band Pass Filter

P/N 3600BP15M600

Detail Specification: 07/14/05

Page 1 of 2

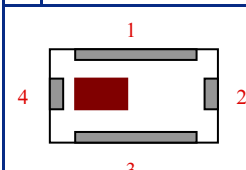
General Specifications

Part Number	3600BP15M600	
Frequency (MHz)	3300 - 3900	
Insertion Loss	1.8 dB max.	
Return Loss	9.5 dB min.	
Attenuation (min.)	0.1 - 2.6 Ghz	15 dB
Attenuation (min.)	4.4 Ghz	9 dB
Attenuation (min.)	6.0 - 9.9 Ghz	20 dB

Input Power	500 mW max.
Impedance	50 Ω
Operating Temperature	-40 to +85°C
Storage Temperature	-40 to +85°C
Reel Quantity	4,000

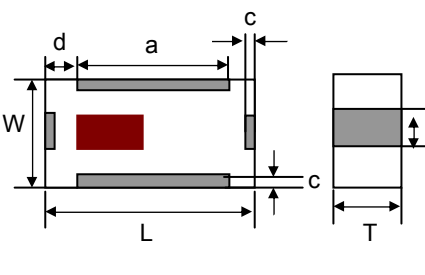
Terminal Configuration

No.	Function
1	GND
2	OUT
3	GND
4	IN



Mechanical Dimensions

	In	mm
L	0.079 $\pm$ 0.006	2.00 $\pm$ 0.15
W	0.049 $\pm$ 0.004	1.25 $\pm$ 0.10
T	0.037 $\pm$ 0.004	0.95 $\pm$ 0.10
a	0.049 $\pm$ 0.006	1.25 $\pm$ 0.15
b	0.012 $\pm$ 0.006	0.30 $\pm$ 0.15
c	0.010 $\pm$ 0.006	0.25 $\pm$ 0.15
g	0.014 $\pm$ 0.006	0.35 $\pm$ 0.15

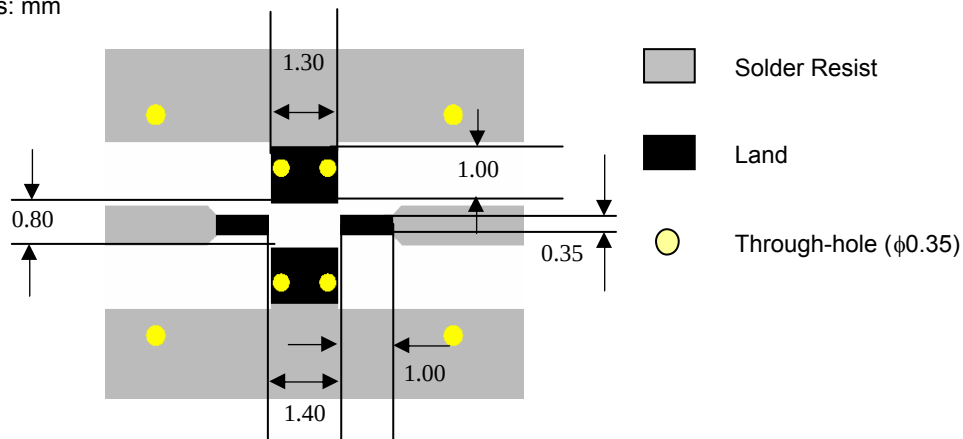


Mounting Considerations

Mount these devices with brown mark facing up.

* Line width should be designed to provide 50 Ω impedance matching characteristics.

Units: mm



Johanson Technology, Inc. reserves the right to make design changes without notice.

All sales are subject to Johanson Technology, Inc. terms and conditions.



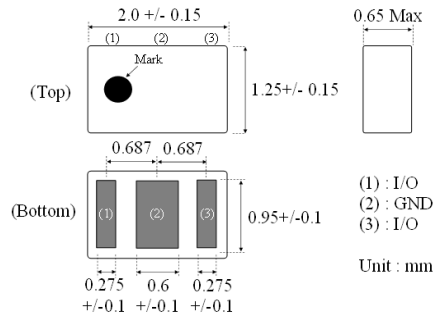
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4001 Calle Tecate • Camarillo, CA 93012 • TEL 805.389.1166 FAX 805.389.1821

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Multilayer Ceramic Devices (Band Pass Type)for 5G

TSB1N21D4G15LV0HWT (Previous PN: FI212B4150HW-T)



■ Features

- Item Summary
Band Pass Type, 2.0x1.25x0.65mm
for 5G
- Lifecycle Stage
Mass Production (Preferred)
- Standard packaging quantity (minimum)
Taping Embossed 5000pcs

■ Products characteristics table

System	5G
Passband Frequency1 (f1)	3300 to 5000 MHz
Input Impedance	50 Ω
Output Impedance	50 Ω
Operating Temp. Range	-40 to +85 °C
RoHS Compliance (10 subst.)	Yes
REACH Compliance (233 subst.)	Yes
Halogen Free	Yes
Soldering	Reflow

■ External Dimensions

Dimension L	2.0 ±0.15 mm
Dimension W	1.25 ±0.15 mm
Dimension T	Max 0.65 mm

The data is reference only. Electrical characteristics vary depending on environment or measurement condition.
 TAIYO YUDEN reserves the right to make change to the data at any time without notice.
 Before making final selection, please check product specification.

2023.03.01

ELECTRICAL CHARACTERISTICS

RFBPF20124G7W6T	Specification
Frequency Range	4400 ~ 5000 MHz
Insertion Loss (at 25°C)	2.00 dB max. @ 4400 ~ 4600 MHz 1.60 dB max. @ 4600 ~ 4800 MHz 1.80 dB max. @ 4800 ~ 5000 MHz
Insertion Loss (at -40 ~ +85°C)	2.30 dB max. @ 4400 ~ 4600 MHz 1.80 dB max. @ 4600 ~ 4800 MHz 2.30 dB max. @ 4800 ~ 5000 MHz
Attenuation	37 dB min. @ 450 ~ 2200 MHz 37 dB min. @ 2300 ~ 2483 MHz 33 dB min. @ 2496 ~ 2690 MHz 15 dB min. @ 5490 ~ 5670 MHz 25 dB min. @ 5670 ~ 5950 MHz 25 dB min. @ 6200 ~ 8000 MHz 25 dB min. @ 8800 ~ 10000 MHz 15 dB min. @ 13200 ~ 15000 MHz 15 dB min. @ 17600 ~ 20000 MHz
VSWR	2.0 max.
Characteristics impedance	50 Ω
Power Capacity	2W max.
Moisture sensitivity levels	MSL is LEVEL 1 (Refer to : IPC/JEDEC J-STD-020)
HBM ESD	Pass 1KV on all pins (Base on AEC-Q200-002)
HBM ESD	Pass 200V (Base on EIA/JESD22-A115)

Operating & Storage Condition (Component)

Operation Temperature Range: -40 ~ +85 °C

Storage Temperature Range: -40 ~ +85 °C

Storage Condition before Soldering (Included packaging material)

Storage Temperature Range: +5 ~ +40 °C

Humidity: 30 to 70% relative humidity

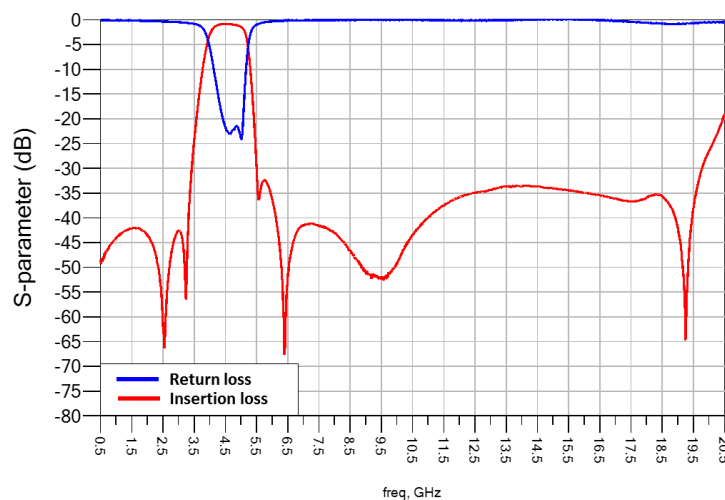
Typical Electrical Chart

Table 3: Maximum Ratings, Table II at $T_A = 25\text{ °C}$, unless otherwise specified

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Maximum DC-voltage on RF-Ports and RF-Ground	V_{RFDC}	0	–	0	V	No DC voltages allowed on RF-Ports
Control Voltage Levels	V_{CTRL}	-0.7	–	3.3	V	–

4 Operation Ranges

Table 4: Operation Ranges

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Supply voltage	V_{DD}	1.8	2.85	3.6	V	–
Supply current ¹⁾	I_{DD}	–	75	120	μA	–
Control voltage low	$V_{Ctrl,low}$	0		0.45	V	–
Control voltage high	$V_{Ctrl,high}$	1.2	1.8	2.85	V	$V_{Ctrl,high} \ll V_{DD}$
Control current low	$I_{Ctrl,low}$	-1	0	1	μA	–
Control current high	$I_{Ctrl,high}$	-1	0	1	μA	$V_{Ctrl,high} \ll V_{DD}$
Ambient temperature	T_A	-40	25	85	°C	–
RF switching time ²⁾	t_{sw}	1	2	5	μs	–
Startup time ²⁾	t_{sw}		10	30	μs	–

¹⁾ $T_A = -30\text{ °C} - +85\text{ °C}$, $V_{BATT} = 1.8 - 3.6\text{ V}$
²⁾ Represents actual alpha status. To be updated.

5 Logic Table

Table 5: Logic Table

CTRL 1	CTRL 2	Mode
0	0	RF1 connected to ANT
0	1	RF2 connected to ANT
1	0	RF3 connected to ANT
1	1	RF4 connected to ANT

6 RF Characteristics for RF1 and RF3

Table 6: RF Specifications

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Insertion Loss						
698 - 960 MHz	IL	–	0.18	0.26	dB	V _{DD} = 1.8 – 3.6 V, T _A = -30 ... +85 °C, Z ₀ = 50 Ω, P _{IN} up to 38 dBm
1710 - 1980 MHz		–	0.29	0.36	dB	
1981 - 2170 MHz		–	0.30	0.41	dB	
2171 - 2690 MHz		–	0.51	0.68	dB	
3400 - 3800 MHz		–	1.20	1.40	dB	
5150 - 5850 MHz		–	1.90	2.35	dB	
Return Loss						
All Ports @ 698 - 915 MHz	RL	23	30	–	dB	
All Ports @ 1710 - 1980 MHz		16	19	–	dB	
All Ports @ 1981 - 2170 MHz		14	17	–	dB	
All Ports @ 2171 - 2690 MHz		11	12	–	dB	
All Ports @ 3400 - 3800 MHz		7	8	–	dB	
All Ports @ 5150 - 5850 MHz		6	7	–	dB	
Isolation RFC						
698 - 915 MHz	ISO	34	41	–	dB	
1710 - 1980 MHz		27	32	–	dB	
1981 - 2170 MHz		26	30	–	dB	
2171 - 2690 MHz		24	28	–	dB	
3400 - 3800 MHz		20	24	–	dB	
5150 - 5850 MHz		15	18	–	dB	
Isolation RF1,2,3,4 - RF4,3,2,1						
698 - 915 MHz	ISO	43	50	–	dB	
1710 - 1980 MHz		34	38	–	dB	
1981 - 2170 MHz		33	36	–	dB	
2170 - 2690 MHz		30	33	–	dB	
3400 - 3800 MHz		24	28	–	dB	
5150 - 5850 MHz		18	21	–	dB	

7 RF Characteristics for RF2 and RF4

Table 7: RF Specifications

Parameter	Symbol	Values			Unit	Note / Test Condition	
		Min.	Typ.	Max.			
Insertion Loss							
698 - 960 MHz	IL	–	0.30	0.40	dB	V _{DD} = 1.8 – 3.6 V, T _A = -30 ... +85 °C, Z ₀ = 50 Ω, P _{IN} up to 38 dBm	
1710 - 1980 MHz		–	0.40	0.50	dB		
1981 - 2170 MHz		–	0.41	0.54	dB		
2171 - 2690 MHz		–	0.64	0.80	dB		
3400 - 3800 MHz		–	1.19	1.45	dB		
5150 - 5850 MHz		–	1.78	2.09	dB		
Return Loss							
All Ports @ 698 - 915 MHz	RL	23	27	–	dB		
All Ports @ 1710 - 1980 MHz		17	20	–	dB		
All Ports @ 1981 - 2170 MHz		14	18	–	dB		
All Ports @ 2171 - 2690 MHz		11	15	–	dB		
All Ports @ 3400 - 3800 MHz		7	9	–	dB		
All Ports @ 5150 - 5850 MHz		6	8	–	dB		
Isolation RFC							
698 - 915 MHz	ISO	34	41	–	dB		
1710 - 1980 MHz		27	32	–	dB		
1981 - 2170 MHz		26	30	–	dB		
2171 - 2690 MHz		24	28	–	dB		
3400 - 3800 MHz		20	24	–	dB		
5150 - 5850 MHz		14	18	–	dB		
Isolation RF1,2,3 - RF3,2,1							
698 - 915 MHz	ISO	43	50	–	dB		
1710 - 1980 MHz		34	38	–	dB		
1981 - 2170 MHz		33	36	–	dB		
2170 - 2690 MHz		30	33	–	dB		
3400 - 3800 MHz		24	28	–	dB		
5150 - 5850 MHz		18	21	–	dB		

8 RF large signal parameter

Table 8: RF large signal specifications for RF1 and RF3

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Max. RF input power	–	–	–	38	dBm	for typical H_x behavior
Harmonic Generation up to 12.75 GHz ^(1,2,3)						
Second Order Harmonics	P_{H2}	–	-100	–	dBc	25 dBm, 50Ω, CW mode
Third Order Harmonics	P_{H3}	–	-115	–	dBc	25 dBm, 50Ω, CW mode
All RF Ports	P_{Hx}	–	-100	–	dBc	25 dBm, 50Ω, CW mode
Intermodulation Distortion IMD2 ^(1,2,3)						
IIP2, low	IIP2,l	–	110	–	dBm	IIP2 conditions table 8
IIP2, high	IIP2,h	–	125	–	dBm	
Intermodulation Distortion IMD3 ^(1,2,3)						
IIP3	IIP3	–	72	–	dBm	IIP3 conditions table 9
SV LTE Intermodulation ^(1,2,3)						
IIP3,SVLTE	IIP3,SV	–	73	–	dBm	SV-LTE conditions table 10

¹⁾Terminating Port Impedance: $Z_0 = 50 \Omega$ ²⁾Supply Voltage: $V_{DD} = 1.8 - 3.6 V$ ³⁾On application board without any matching components

Table 9: RF large signal specifications for RF2 and RF4

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Max. RF input power	–	–	–	38	dBm	for typical H_x behavior
Harmonic Generation up to 12.75 GHz ^(1,2,3)						
Second Order Harmonics	P_{H2}	–	-105	–	dBc	25 dBm, 50Ω, CW mode
Third Order Harmonics	P_{H3}	–	-105	–	dBc	25 dBm, 50Ω, CW mode
All RF Ports	P_{Hx}	–	-105	–	dBc	25 dBm, 50Ω, CW mode
Intermodulation Distortion IMD2 ^(1,2,3)						
IIP2, low	IIP2,l	–	110	–	dBm	IIP2 conditions table 8
IIP2, high	IIP2,h	–	130	–	dBm	
Intermodulation Distortion IMD3 ^(1,2,3)						
IIP3	IIP3	–	74	–	dBm	IIP3 conditions table 9
SV LTE Intermodulation ^(1,2,3)						
IIP3,SVLTE	IIP3,SV	–	74	–	dBm	SV-LTE conditions table 10

¹⁾Terminating Port Impedance: $Z_0 = 50 \Omega$ ²⁾Supply Voltage: $V_{DD} = 1.8 - 3.6 V$ ³⁾On application board without any matching components

Table 10: IIP2 conditions table

Band	In-Band Frequency [MHz]	Blocker Frequency 1 [MHz]	Blocker Power 1 [dBm]	Blocker Frequency 2 [MHz]	Blocker Power 2 [dBm]
Band 1 Low	2140	1950	20	190	-15
Band 1 High	2140	1950	20	4090	-15
Band 5 Low	881.5	836.5	20	45	-15
Band 5 High	881.5	836.5	20	1718	-15

Table 11: IIP3 conditions table

Band	In-Band Frequency [MHz]	Blocker Frequency 1 [MHz]	Blocker Power 1 [dBm]	Blocker Frequency 2 [MHz]	Blocker Power 2 [dBm]
Band 1	2140	1950	20	1760	-15
Band 5	881.5	836.5	20	791.5	-15

Table 12: SV-LTE conditions table

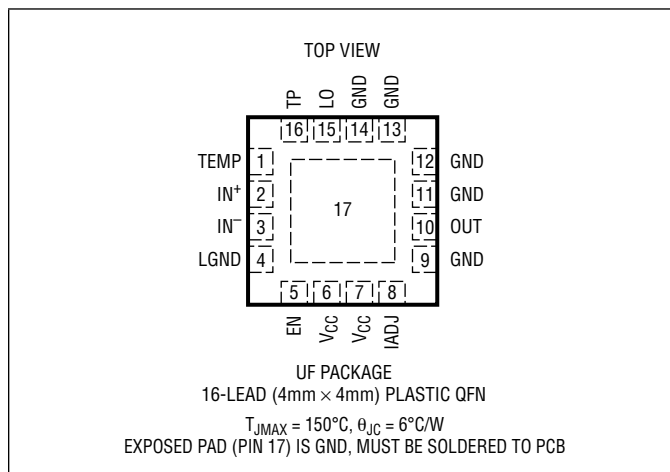
Band	In-Band Frequency [MHz]	Blocker Frequency 1 [MHz]	Blocker Power 1 [dBm]	Blocker Frequency 2 [MHz]	Blocker Power 2 [dBm]
Band 5	872	827	23	872	14
Band 13	747	786	23	747	14
Band 20	878	833	23	2544	14

ABSOLUTE MAXIMUM RATINGS

(Note 1)

Supply Voltage (V_{CC})	6V
Enable Voltage	-0.3V to $V_{CC} + 0.3V$
IADJ Pin Voltage	-0.3 to 2.7V
LO Input Power (1GHz to 8GHz)	+10dBm
IN ⁺ , IN ⁻ Input Power (30MHz to 6GHz)	+15dBm
TEMP Input Current	10mA
Operating Temperature Range (T_C)	-40°C to 105°C
Junction Temperature (T_J)	150°C
Storage Temperature Range	-65°C to 150°C

PIN CONFIGURATION



ORDER INFORMATION

<http://www.linear.com/product/LTC5576#orderinfo>

LEAD FREE FINISH	TAPE AND REEL	PART MARKING	PACKAGE DESCRIPTION	CASE TEMPERATURE RANGE
LTC5576IUF#PBF	LTC5576IUF#TRPBF	5576	16-Lead (4mm × 4mm) Plastic QFN	-40°C to 105°C

Consult LTC Marketing for parts specified with wider operating temperature ranges.

For more information on lead free part marking, go to: <http://www.linear.com/leadfree/>For more information on tape and reel specifications, go to: <http://www.linear.com/tapeandreel/>. Some packages are available in 500 unit reels through designated sales channels with #TRMPBF suffix.

DC ELECTRICAL CHARACTERISTICS

The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at $T_C = 25^{\circ}\text{C}$, $V_{CC} = 5V$. Test circuit shown in Figure 1. (Note 2)

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
Supply Voltage (V_{CC})	5V Supply ●	4.5	5	5.3	V
	3.3V Supply ●	3.1	3.3	3.5	V
Supply Current	5V, $R_1 = 2.61k\Omega$		99	112	mA
	3.3V, $R_1 = 649\Omega$		85		mA
	Shutdown ($EN = \text{Low}$)		1.3		mA
Enable Logic Input (EN)					
EN Input High Voltage (On)		1.8			V
EN Input Low Voltage (Off)				0.5	V
EN Input Current	-0.3V to $V_{CC} + 0.3V$	-20		200	μA
Turn-On Time			0.6		μs
Turn-Off Time			0.6		μs
Current Adjust Pin (IADJ)					
Open Circuit DC Voltage			1.8		V
Short Circuit DC Current			1.9		mA
Temperature Sensing Diode (TEMP)					
DC Voltage at $T_J = 25^{\circ}\text{C}$	$I_{IN} = 10\mu\text{A}$		697		mV
	$I_{IN} = 80\mu\text{A}$		755		mV
Voltage Temperature Coefficient	$I_{IN} = 10\mu\text{A}$	-1.80			mV/°C
	$I_{IN} = 80\mu\text{A}$	-1.61			mV/°C

5576fa

AC ELECTRICAL CHARACTERISTICS The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at $T_C = 25^\circ\text{C}$. $V_{CC} = 5\text{V}$, $\text{EN} = \text{High}$, $P_{LO} = 0\text{dBm}$. Test circuit shown in Figure 1. (Notes 2, 3)

PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
LO Input Frequency Range	External Matching Required	●		1 to 8		GHz
Input (IN) Frequency Range	External Matching Required	●		0.03 to 6		GHz
Output (OUT) Frequency Range	External Matching Required	●		3 to 8		GHz
Input Return Loss	$Z_0 = 50\Omega$			>10		dB
LO Input Return Loss	$Z_0 = 50\Omega$			>10		dB
LO Input Power	Single-Ended	●	-6	0	6	dBm
LO to IN Leakage	$f_{LO} = 1\text{GHz to } 8\text{GHz}$			≤ -30		dBm
IN to LO Isolation	$f_{IN} = 0.1\text{GHz to } 6\text{GHz}$			>35		dB

AC ELECTRICAL CHARACTERISTICS The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at $T_C = 25^\circ\text{C}$. $V_{CC} = 5\text{V}$, $\text{EN} = \text{High}$, $P_{IN} = -10\text{dBm}$ (-10dBm/Tone for 2-tone tests, $\Delta f = 2\text{MHz}$), $P_{LO} = 0\text{dBm}$, unless otherwise noted. Test circuit shown in Figure 1. (Notes 2, 3 and 4)

5V Upmixer Application: Low Side LO, $P_{LO} = 0\text{dBm}$, $P_{IN} = -10\text{dBm}$

PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
Conversion Gain	$f_{IN} = 456\text{MHz}$, $f_{OUT} = 3.5\text{GHz}$, $f_{LO} = 3.044\text{GHz}$			-0.6		dB
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 5.8\text{GHz}$, $f_{LO} = 4.9\text{GHz}$		-1.5	-0.6		dB
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 8\text{GHz}$, $f_{LO} = 7.1\text{GHz}$			-2.0		dB
Conversion Gain vs Temperature	$T_C = -40^\circ\text{C to } 105^\circ\text{C}$, $f_{OUT} = 5.8\text{GHz}$	●		-0.009		dB/ $^\circ\text{C}$
Output 3rd Order Intercept	$f_{IN} = 456\text{MHz}$, $f_{OUT} = 3.5\text{GHz}$, $f_{LO} = 3.044\text{GHz}$			25		dBm
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 5.8\text{GHz}$, $f_{LO} = 4.9\text{GHz}$			25		dBm
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 8\text{GHz}$, $f_{LO} = 7.1\text{GHz}$			25		dBm
SSB Noise Figure	$f_{IN} = 456\text{MHz}$, $f_{OUT} = 3.5\text{GHz}$, $f_{LO} = 3.044\text{GHz}$			12.4		dB
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 5.8\text{GHz}$, $f_{LO} = 4.9\text{GHz}$			14.1		dB
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 8\text{GHz}$, $f_{LO} = 7.1\text{GHz}$			17.5		dB
SSB Noise Floor at $P_{IN} = 5\text{dBm}$	$f_{IN} = 1\text{GHz}$, $f_{OUT} = 5801\text{MHz}$, $f_{LO} = 4899\text{MHz}$			-154		dBm/Hz
Input 1dB Compression	$f_{IN} = 456\text{MHz}$, $f_{OUT} = 3.5\text{GHz}$, $f_{LO} = 3.044\text{GHz}$			10.8		dBm
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 5.8\text{GHz}$, $f_{LO} = 4.9\text{GHz}$			10.4		dBm
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 8\text{GHz}$, $f_{LO} = 7.1\text{GHz}$			10.3		dBm
LO-OUT Leakage	$f_{IN} = 456\text{MHz}$, $f_{OUT} = 3.5\text{GHz}$, $f_{LO} = 3.044\text{GHz}$			-36		dBm
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 5.8\text{GHz}$, $f_{LO} = 4.9\text{GHz}$			-35		dBm
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 8\text{GHz}$, $f_{LO} = 7.1\text{GHz}$			-28		dBm
IN to OUT Isolation	$f_{IN} = 456\text{MHz}$, $f_{OUT} = 3.5\text{GHz}$, $f_{LO} = 3.044\text{GHz}$			70		dB
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 5.8\text{GHz}$, $f_{LO} = 4.9\text{GHz}$			38		dB
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 8\text{GHz}$, $f_{LO} = 7.1\text{GHz}$			35		dB

AC ELECTRICAL CHARACTERISTICS The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at $T_C = 25^\circ\text{C}$. $V_{CC} = 3.3\text{V}$, $EN = \text{High}$, $P_{IN} = -10\text{dBm}$ (-10dBm/Tone for 2-tone tests, $\Delta f = 2\text{MHz}$), $P_{LO} = 0\text{dBm}$, unless otherwise noted. Test circuit shown in Figure 1. (Notes 2, 3 and 4)

3.3V Upmixer Application: Low Side LO, $P_{LO} = 0\text{dBm}$, $P_{IN} = -10\text{dBm}$

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
Conversion Gain	$f_{IN} = 456\text{MHz}$, $f_{OUT} = 3.5\text{GHz}$, $f_{LO} = 3.044\text{GHz}$		-0.6		dB
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 5.8\text{GHz}$, $f_{LO} = 4.9\text{GHz}$		-0.6		dB
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 8\text{GHz}$, $f_{LO} = 7.1\text{GHz}$		-2.0		dB
Conversion Gain vs Temperature	$T_C = -40^\circ\text{C}$ to 105°C , $f_{OUT} = 5.8\text{GHz}$	●	-0.009		dB/ $^\circ\text{C}$
Output 3rd Order Intercept	$f_{IN} = 456\text{MHz}$, $f_{OUT} = 3.5\text{GHz}$, $f_{LO} = 3.044\text{GHz}$		21		dBm
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 5.8\text{GHz}$, $f_{LO} = 4.9\text{GHz}$		23		dBm
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 8\text{GHz}$, $f_{LO} = 7.1\text{GHz}$		19		dBm
SSB Noise Figure	$f_{IN} = 456\text{MHz}$, $f_{OUT} = 3.5\text{GHz}$, $f_{LO} = 3.044\text{GHz}$		11.5		dB
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 5.8\text{GHz}$, $f_{LO} = 4.9\text{GHz}$		12.8		dB
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 8\text{GHz}$, $f_{LO} = 7.1\text{GHz}$		17.8		dB
SSB Noise Floor at $P_{IN} = 5\text{dBm}$	$f_{IN} = 1\text{GHz}$, $f_{OUT} = 5801\text{MHz}$, $f_{LO} = 4899\text{MHz}$		-154		dBm/Hz
Input 1dB Compression	$f_{IN} = 456\text{MHz}$, $f_{OUT} = 3.5\text{GHz}$, $f_{LO} = 3.044\text{GHz}$		8.4		dBm
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 5.8\text{GHz}$, $f_{LO} = 4.9\text{GHz}$		8.5		dBm
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 8\text{GHz}$, $f_{LO} = 7.1\text{GHz}$		8.1		dBm
LO-OUT Leakage	$f_{IN} = 456\text{MHz}$, $f_{OUT} = 3.5\text{GHz}$, $f_{LO} = 3.044\text{GHz}$		-39		dBm
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 5.8\text{GHz}$, $f_{LO} = 4.9\text{GHz}$		-36		dBm
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 8\text{GHz}$, $f_{LO} = 7.1\text{GHz}$		-27		dBm
IN to OUT Isolation	$f_{IN} = 456\text{MHz}$, $f_{OUT} = 3.5\text{GHz}$, $f_{LO} = 3.044\text{GHz}$		70		dB
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 5.8\text{GHz}$, $f_{LO} = 4.9\text{GHz}$		38		dB
	$f_{IN} = 900\text{MHz}$, $f_{OUT} = 8\text{GHz}$, $f_{LO} = 7.1\text{GHz}$		33		dB

Note 1: Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. Exposure to any Absolute Maximum Rating condition for extended periods may affect device reliability and lifetime.

Note 2: The LTC5576 is guaranteed functional over the -40°C to 105°C case temperature range.

Note 3: SSB noise figure measured with a small-signal noise source, bandpass filter and 3dB matching pad on IN port, and bandpass filter on the LO input.

Note 4: Specified performance includes all external component and evaluation PCB losses.

**VCO (Voltage Controlled Oscillator) 0.5 inch
Commercial SMT (Surface Mount), Frequency Range
of 3 GHz to 3.5 GHz, Phase Noise -81 dBc/Hz**

FMVC11024 is a High Reliability Low Noise Voltage Controlled Oscillator (VCO) which covers a 3000 MHz to 3500 MHz frequency band with a voltage tuning range from 0.0V to 10V. This design features exceptional phase noise performance of -81 dBc/Hz @ 10 kHz offset. Supply Voltage is +11V with a generated output power level of +8.0 dBm and 2nd harmonic output of -16 dBc typical. The assembly is RoHS compliant and available in a compact 0.5 inch SMT package with an industry standard mounting footprint. The bottom surface is copper clad with 2-5 micro inches of immersion Gold over 150 - 250 micro inches of Electro less Nickel which makes it resistant to oxidation for ease of soldering. The VCO operates over a temperature range of -40°C to +85°C and is designed to meet a variety of MIL-STD-202 test conditions including Humidity, Shock, and Vibration.

Electrical Specifications

Description	Min	Typ	Max	Units
Frequency Range	3		3.5	GHz
Tuning Voltage	0		10	Vdc
Supply Voltage (DC)	10	11	12	Vdc
Supply Current (DC)		20	22	mA
Phase Noise @10kHz Offset		-81	-80	dBc/Hz
Phase Noise @100kHz Offset		-103	-102	dBc/Hz
Output Power	+8	+10	+11.5	dBm
Tuning Sensitivity (Kvco)	40		135	MHz/V
Pushing		2.5	5	MHz/V
Pulling (pk-pk)		12	20	MHz
Tuning Port Capacitance		18		pF
Load Impedance		50		Ohms
2nd Harmonics		-16	-13	dBc

Mechanical Specifications

Size	
Length	0.5 in [12.7 mm]
Width	0.5 in [12.7 mm]
Height	0.13 in [3.3 mm]
Body Material and Plating	Copper Clad, Nickel, Gold
Design	Commercial
Package Style	Surface Mount

Environmental Specifications

Temperature	
Operating Range	-40 to +85 deg C
Storage Range	-55 to +125 deg C
Humidity	MIL-STD-202, Method 103, 90% RH, +65 C
Shock	MIL-STD-202, Method 213I
Vibration	MIL-STD-202, Method 204D
Temperature Cycle	MIL-STD-202, Method 107B



Features:

- 3000 MHz to 3500 MHz Bandwidth
- -81 dBc/Hz @ 10kHz offset
- Tuning Voltage 0.0V to 10v
- Pout = +8.0 dBm
- Harmonics = -16 dBc
- RoHS Compliant Assembly
- 0.5" SMT package
- Industry Standard Mounting Footprint
- Designed to meet MIL-STD-202 Environmental Conditions

Applications:

- Phase Locked Loop
- Function Generators
- Frequency Synthesizers
- Receivers
- Electronic Jamming Equipment
- Local Oscillator
- Wireless Communications
- SATCOM
- Optical Communications
- Military Electronic Systems

Fairview Microwave
1130 Junction Dr. #100
Allen, TX 75013
Tel: 1-800-715-4396 / (972) 649-6678
Fax: (972) 649-6689
www.fairviewmicrowave.com
sales@fairviewmicrowave.com



MMIC VCO w/ HALF FREQUENCY OUTPUT 7.3 - 8.2 GHz

Typical Applications

Low noise MMIC VCO w/Half Frequency, for:

- VSAT Radio
- Point to Point/Multi-Point Radio
- Test Equipment & Industrial Controls
- Military End-Use

Features

Dual Output: $F_o = 7.3 - 8.2$ GHz
 $F_o/2 = 3.65 - 4.1$ GHz

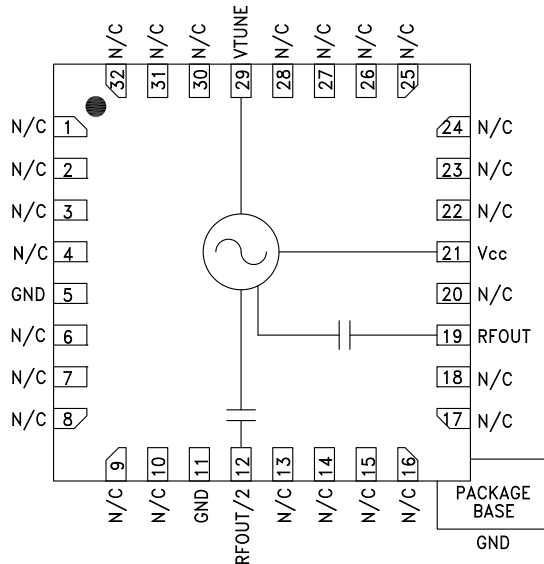
Pout: +15.0 dBm

Phase Noise: -116 dBc/Hz @100 kHz Typ.

No External Resonator Needed

32 Lead 5x5mm SMT Package: 25mm²

Functional Diagram



General Description

The HMC508LP5 & HMC508LP5E are GaAs InGaP Heterojunction Bipolar Transistor (HBT) MMIC VCOs. The HMC508LP5 & HMC508LP5E integrate resonators, negative resistance devices, varactor diodes and feature a half frequency output. The VCO's phase noise performance is excellent over temperature, shock, and process due to the oscillator's monolithic structure. Power output is +15 dBm typical from a +5V supply. The voltage controlled oscillator is packaged in a leadless QFN 5x5 mm surface mount package, and requires no external matching components.

Electrical Specifications, $T_A = +25^\circ\text{C}$, $V_{cc} = +5\text{V}$

Parameter	Min.	Typ.	Max.	Units
Frequency Range				
F_o		7.3 - 8.2		GHz
$F_o/2$		3.65 - 4.1		GHz
Power Output				
RFOUT	+12		+17	dBm
RFOUT/2	+4		+10	dBm
SSB Phase Noise @ 100 kHz Offset, $V_{tune} = +5\text{V}$ @ RFOUT		-116		dBc/Hz
Tune Voltage				
V_{tune}	2		13	V
Supply Current (I_{cc}) ($V_{cc} = +5.0\text{V}$)	200	240	280	mA
Tune Port Leakage Current ($V_{tune} = 13\text{V}$)			10	μA
Output Return Loss		2		dB
Harmonics/Subharmonics				
1/2		40		dBc
2nd		20		dBc
3rd		35		dBc
Pulling (into a 2.0:1 VSWR)		8		MHz pp
Pushing @ $V_{tune} = 5\text{V}$		10		MHz/V
Frequency Drift Rate		1.0		MHz/ $^\circ\text{C}$

MMIC VCO w/ HALF FREQUENCY OUTPUT 7.8 - 8.8 GHz

Typical Applications

Low noise MMIC VCO w/Half Frequency, for:

- VSAT Radio
- Point to Point/Multi-Point Radio
- Test Equipment & Industrial Controls
- Military End-Use

Features

Dual Output: $F_o = 7.8 - 8.8 \text{ GHz}$
 $F_o/2 = 3.9 - 4.4 \text{ GHz}$

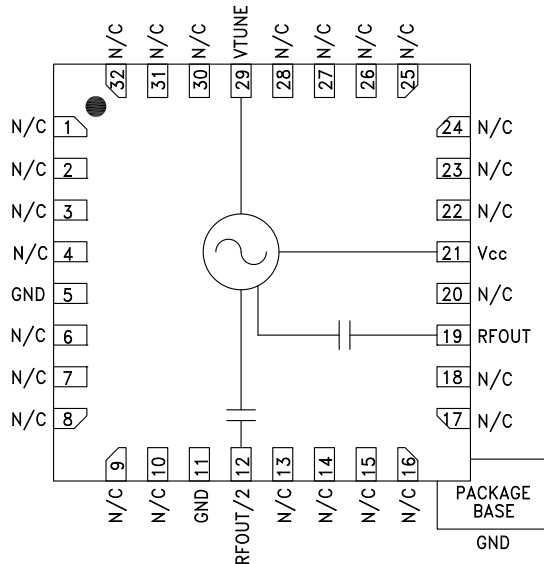
Pout: +13 dBm

Phase Noise: -115 dBc/Hz @100 kHz Typ.

No External Resonator Needed

32 Lead 5x5mm SMT Package: 25mm²

Functional Diagram



General Description

The HMC509LP5 & HMC509LP5E are GaAs InGaP Heterojunction Bipolar Transistor (HBT) MMIC VCOs. The HMC509LP5 & HMC509LP5E integrate resonators, negative resistance devices, varactor diodes and feature a half frequency output. The VCO's phase noise performance is excellent over temperature, shock, and process due to the oscillator's monolithic structure. Power output is +13 dBm typical from a +5V supply. The voltage controlled oscillator is packaged in a leadless QFN 5x5 mm surface mount package, and requires no external matching components.

Electrical Specifications, $T_A = +25^\circ \text{C}$, $V_{cc} = +5V$

Parameter	Min.	Typ.	Max.	Units
Frequency Range		7.8 - 8.8 3.9 - 4.4		GHz GHz
Power Output	+10 +5		+15 +10	dBm dBm
SSB Phase Noise @ 100 kHz Offset, $V_{tune} = +5V$ @ RFOUT		-115		dBc/Hz
Tune Voltage	2		13	V
Supply Current (I_{cc}) ($V_{cc} = +5.0V$)	200	250	270	mA
Tune Port Leakage Current ($V_{tune} = 13V$)			10	μA
Output Return Loss		2		dB
Harmonics/Subharmonics	1/2 2nd 3rd	35 10 32		dBc dBc dBc
Pulling (into a 2.0:1 VSWR)		5		MHz pp
Pushing @ $V_{tune} = 5V$		10		MHz/V
Frequency Drift Rate		0.9		MHz/ $^\circ C$

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Typical Applications

Low noise MMIC VCO w/Half Frequency, for:

- VSAT Radio
- Point to Point/Multi-Point Radio
- Test Equipment & Industrial Controls
- Military End-Use

Features

Dual Output: $F_o = 9.05 - 10.15$ GHz

$F_o/2 = 4.525 - 5.075$ GHz

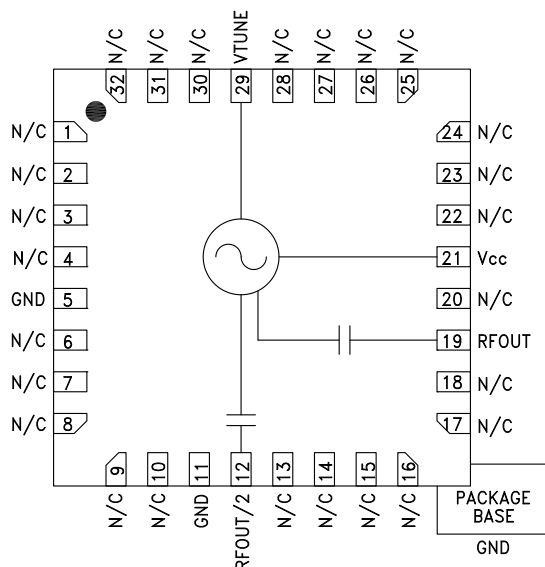
Pout: +13 dBm

Phase Noise: -115 dBc/Hz @100 kHz Typ.

No External Resonator Needed

32 Lead 5x5mm SMT Package: 25mm²

Functional Diagram



General Description

The HMC511LP5 & HMC511LP5E are GaAs InGaP Heterojunction Bipolar Transistor (HBT) MMIC VCOs. The HMC511LP5 & HMC511LP5E integrate resonators, negative resistance devices, varactor diodes and feature a half frequency output. The VCO's phase noise performance is excellent over temperature, shock, and process due to the oscillator's monolithic structure. Power output is +13 dBm typical from a +5V supply. The voltage controlled oscillator is packaged in a leadless QFN 5x5 mm surface mount package, and requires no external matching components.

Electrical Specifications, $T_A = +25^\circ\text{C}$, $V_{cc} = +5\text{V}$

Parameter	Min.	Typ.	Max.	Units
Frequency Range		9.05 - 10.15		GHz
		4.525 - 5.075		GHz
Power Output				
RFOUT	+9		+16	dBm
RFOUT/2	+5		+11	dBm
SSB Phase Noise @ 100 kHz Offset, $V_{tune} = +5\text{V}$ @ RFOUT		-115		dBc/Hz
Tune Voltage	2		13	V
Supply Current (I_{cc}) ($V_{cc} = +5.0\text{V}$)	200	265	300	mA
Tune Port Leakage Current ($V_{tune} = 13\text{V}$)			10	μA
Output Return Loss		2		dB
Harmonics/Subharmonics				
1/2		38		dBc
2nd		15		dBc
3rd		30		dBc
Pulling (into a 2.0:1 VSWR)		8		MHz pp
Pushing @ $V_{tune} = 5\text{V}$		15		MHz/V
Frequency Drift Rate		0.9		MHz/ $^\circ\text{C}$

Surface Mount Bandpass Filter

50Ω 940 to 1060 MHz

BPF-V1000+



Features

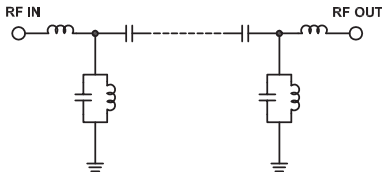
- Flat group delay
- Steep rejection up to 3000 MHz
- Thin package

Generic photo used for illustration purposes only
CASE STYLE: KV1974

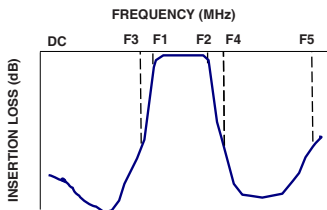
Applications

- Aviation / Aeronautical
- Radar and Navigation system

Functional Schematic



Typical Frequency Response



Electrical Specifications at 25°C

Parameter	F#	Frequency (MHz)	Min.	Typ.	Max.	Unit
Pass Band	Center Frequency	—	—	1000	—	MHz
	Insertion Loss	F1-F2	—	5	6	dB
	Group delay flatness	F1-F2	—	3	—	ns
	VSWR	F1-F2	—	1.5	2	:1
Stop Band, Lower	Insertion Loss	DC-F3	30	40	—	dB
	VSWR	DC-F3	—	10	—	:1
Stop Band, Upper	Insertion Loss	F4-F5	30	40	—	dB
	VSWR	F4-F5	—	10	—	:1

Maximum Ratings

Operating Temperature	-40°C to 85°C
Storage Temperature	-55°C to 100°C
RF Power Input	0.5 W

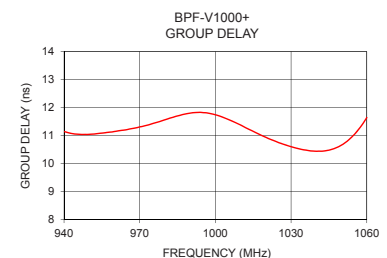
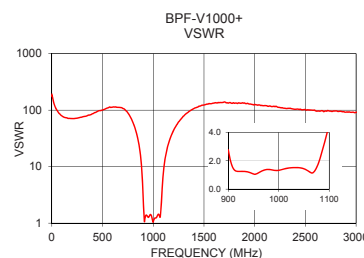
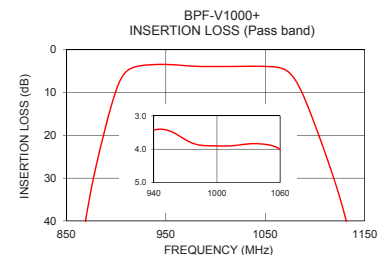
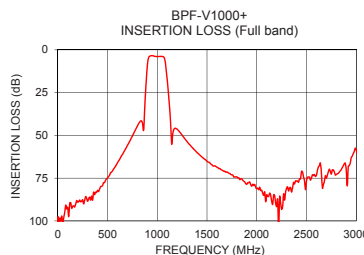
Permanent damage may occur if any of these limits are exceeded.

Typical Performance Data at 25°C

Frequency (MHz)	Insertion Loss (dB)	VSWR (:1)	Frequency (MHz)	Group Delay (nsec)
1	99.47	192.05	940	11.15
250	87.68	72.29	945	11.05
860	47.17	23.99	950	11.04
880	27.10	12.58	960	11.15
885	22.16	9.85	970	11.31
895	13.22	4.90	975	11.43
905	6.75	1.88	980	11.57
920	3.92	1.25	990	11.80
940	3.43	1.34	995	11.82
1000	3.90	1.05	1000	11.74
1060	4.01	1.36	1005	11.58
1085	9.06	3.30	1010	11.38
1100	17.59	7.90	1015	11.15
1105	20.75	9.66	1020	10.94
1120	30.84	14.93	1025	10.75
1125	34.60	16.70	1030	10.60
1140	49.83	22.00	1035	10.49
2000	81.15	126.29	1040	10.44
2500	76.64	102.68	1050	10.66
3000	58.51	91.68	1060	11.64

+RoHS Compliant

The +Suffix identifies RoHS Compliance. See our web site for RoHS Compliance methodologies and qualifications



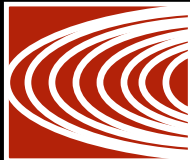
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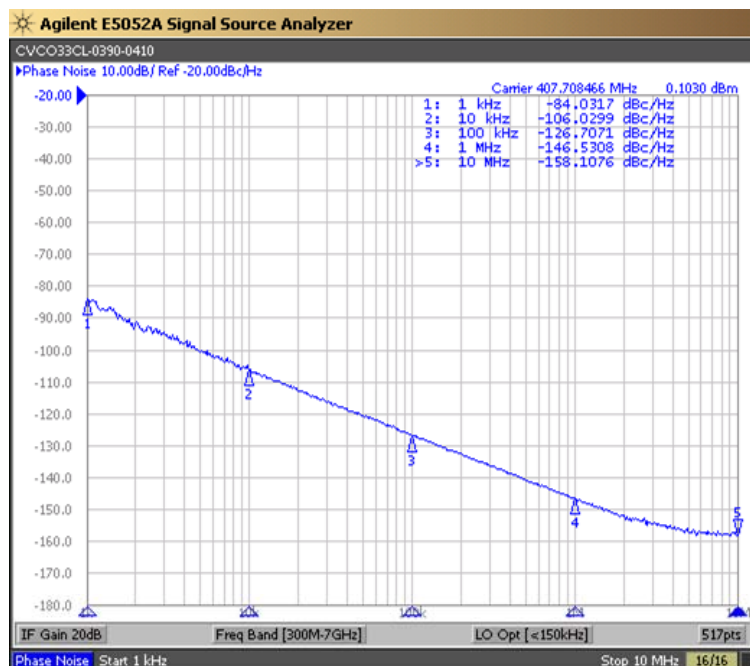
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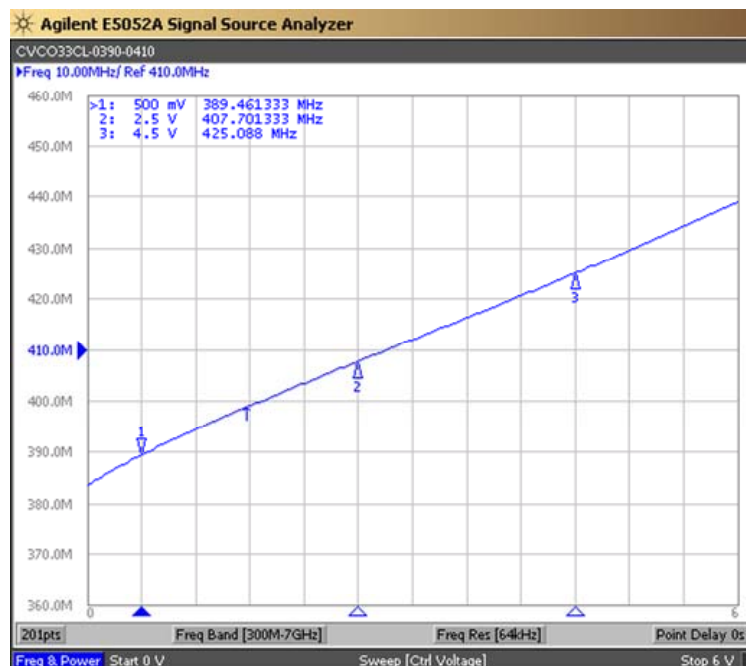


PERFORMANCE SPECIFICATION	MIN	TYP	MAX	UNITS
Lower Frequency:			390	MHz
Upper Frequency:	410			MHz
Tuning Voltage:	0.5		4.5	VDC
Supply Voltage:	4.75	5.0	5.25	VDC
Output Power:	-3.0	0	+3.0	dBm
Supply Current:		10	15	mA
Harmonic Suppression (2 nd Harmonic):		-12	-9	dBc
Pushing:		1.0	1.5	MHz/V
Pulling, all Phases:		1.0	1.5	MHz pk-pk
Tuning Sensitivity:		12		MHz/V
Phase Noise @ 10kHz offset:		-110		dBc/Hz
Phase Noise @ 100kHz offset:		-130		dBc/Hz
Load Impedance:		50		Ω
Input Capacitance:			100	pF
Operating Temperature Range:	-40		+85	$^{\circ}\text{C}$
Storage Temperature Range:	-45		+90	$^{\circ}\text{C}$

Phase Noise (1 Hz BW, Typical)



Tuning Curve (Typical)



Product Control:

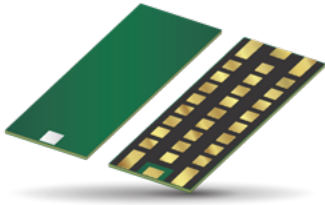
Crystek Part Number:	CVCO33CL-0390-0410	Release Date:	30-Oct-2017
Revision Level:	F	Responsible:	C. Vales



Multilayer Organic (MLO®) Filters

MLO® Band Pass Filters

General Information



GENERAL DESCRIPTION

The BP series of MLO® High Performance Band Pass Filters exhibit low insertion loss, steep roll-offs, and very high rejection of out of band frequencies. MLO® Band Pass Filters support many frequency bands and multiple wireless standards, and are less than 1.0mm in thickness.

MLO components are low profile devices with best in class performance based on KYOCERA AVX patented multilayer organic high density interconnect technology. MLO components utilize high dielectric constant and low loss materials to realize high Q passive printed elements, such as inductors and capacitors, in a multilayer stack. All MLO components are expansion matched to most organic PCB materials, thereby resulting in improved reliability over standard Si and ceramic devices

FEATURES

- Low insertion loss
- High rejection out-of-band
- Steep roll-off
- 50Ω Impedance
- Expansion matched to PCB
- Surface Mount
- RoHS Compliant

APPLICATIONS

- Wireless Communications Systems
- Military Radios
- EMS Radios
- UAVs
- Basestations
- Wireless access points and terminals
- Instrumentation

LAND GRID ARRAY ADVANTAGES

- Inherent Low Profile
- Excellent Solderability
- Low Parasitics
- Better Heat Dissipation

HOW TO ORDER

BP Series Band Pass Filters	0F Case Size 0A = 2616 0B = 3116 0C = 3416 0D = 4016 0E = 4617 0F = 5021	A Type	1100 Frequency In MHz	A Standard Testing	7 Termination 7 – Gold	00 Package Code 00 – Waffle Pack TR – 1000 pcs Tape & Reel TR\250 – 250 pcs Tape & Reel
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QUALITY INSPECTION

Finished Parts are 100% electrically tested



TERMINATION

All finishes are compatible with automatic soldering technologies: Pb free reflow, wave soldering, vapor phase, and manual soldering.

OPERATING TEMPERATURE

-55°C to +85°C

LEAD-FREE
COMPATIBLE
COMPONENT



For RoHSc compliant products,
please select correct termination style



**S2P FILES, DRAWING AND OTHER INFORMATION
AVAILABLE IN LINK ON INDIVIDUAL DATASHEETS**

ELECTRICAL SPECIFICATIONS

Part Number	Fc		Passband Ripple <1.5dB			Insertion Loss < 5dB				Low Band		High Band				Rated RF Power
			<1.5 dB Ripple			< 5 dB				DC - 30 dB	DC - 40 dB	30 dB		40 dB		
			Min	Max	BW	Min	Max	BW	Typ	Max	Max	Entry	Exit	Entry	Exit	
	GHz	dB	GHz	GHz	GHz	GHz	GHz	GHz	dB	GHz	GHz	GHz	GHz	GHz	GHz	(W)
BP0IA0110A7**	0.110	1.7	0.092	0.135	0.043	0.088	0.138	0.050	2.2	0.688	0.063	0.181	1.000	0.187	0.950	1
BP0IA0115A7**	0.115	1.7	0.094	0.138	0.044	0.090	0.142	0.051	2.4	0.066	0.064	0.187	1.000	0.192	0.950	1
BP0IA0120A7**	0.120	1.7	0.095	0.143	0.048	0.092	0.146	0.054	2.5	0.069	0.066	0.191	1.000	0.197	0.950	1
BP0IA0170A7**	0.170	1.4	0.132	0.224	0.092	0.128	0.228	0.100	2.0	0.096	0.093	0.297	1.000	0.307	0.950	1
BP0IA0175A7**	0.175	1.4	0.135	0.231	0.096	0.132	0.235	0.103	2.3	0.100	0.096	0.304	1.000	0.314	0.950	1
BP0IA0180A7**	0.180	1.6	0.143	0.228	0.085	0.137	0.234	0.097	2.2	0.103	0.099	0.312	1.000	0.322	0.950	1
BP0EA0270A7**	0.270	1.6	0.231	0.316	0.085	0.224	0.325	0.100	2.4	0.161	0.155	0.424	2.000	0.444	2.000	1
BP0EA0280A7**	0.280	1.7	0.234	0.321	0.088	0.227	0.327	0.100	2.2	0.164	0.158	0.431	2.000	0.447	2.000	1
BP0EA0290A7**	0.290	1.7	0.244	0.339	0.096	0.241	0.345	0.105	2.3	0.170	0.164	0.454	2.000	0.473	2.000	1
BP0EA0400A7**	0.400	1.3	0.303	0.494	0.191	0.296	0.500	0.203	1.9	0.215	0.206	0.660	2.500	0.687	2.100	1
BP0EA0420A7**	0.420	1.3	0.316	0.514	0.198	0.309	0.520	0.211	2.0	0.224	0.215	0.690	2.500	0.719	2.000	1
BP0EA0430A7**	0.430	1.3	0.323	0.529	0.207	0.316	0.535	0.219	1.9	0.230	0.221	0.706	2.500	0.736	2.000	1
BP0DA0585A7**	0.585	1.3	0.472	0.735	0.262	0.461	0.745	0.284	2.1	0.360	-	0.985	3.000	-	-	1
BP0DA0595A7**	0.595	1.3	0.478	0.745	0.267	0.467	0.756	0.289	1.9	0.365	-	1.002	3.000	-	-	1
BP0DA0650A7**	0.650	1.3	0.485	0.776	0.291	0.490	0.787	0.297	2.0	0.374	-	1.030	3.000	-	-	1

Click on part number to see full specifications

**Packaging: 00 = waffle pack, TR = 1000pcs T&R, TR\250 = 250pcs T&R



The Important Information/Disclaimer is incorporated in the catalog where these specifications came from or available online at www.kyocera-avx.com/disclaimer/ by reference and should be reviewed in full before placing any order.

TDS-RFM-0045 | Rev 2

– RF MICROWAVE PRODUCTS –

Multilayer Organic (MLO®) Filters

MLO® Band Pass Filters

ELECTRICAL SPECIFICATIONS - CONTINUED

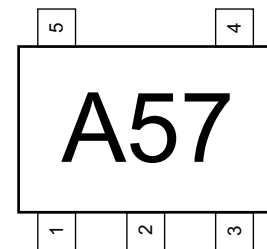
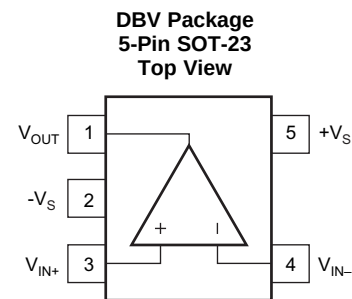
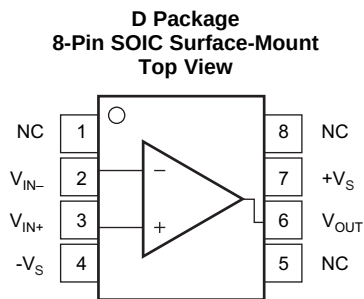
Part Number	Fc		Passband Ripple <1.5dB			Insertion Loss < 5dB				Low Band		High Band				Rated RF Power (W)
			<1.5 dB Ripple			< 5 dB				DC - 30 dB	DC - 40 dB	30 dB		40 dB		
	GHz	Typ dB	Min	Max	BW	Min	Max	BW	Typ	Max	Max	Entry	Exit	Entry	Exit	
BP0CA0770A7**	0.770	0.97	0.628	0.954	0.326	0.620	0.970	0.350	1.97	0.480	0.460	1.190	4.500	1.230	4.000	1
BP0CA0810A7**	0.810	1.3	0.668	0.985	0.317	0.657	1.006	0.349	2.00	0.511	0.500	1.256	4.000	1.290	4.000	1
BP0CA0825A7**	0.825	1.3	0.680	0.985	0.305	0.668	1.021	0.353	2.0	0.513	0.504	1.261	4.000	1.296	4.000	1
BP0CA0855A7**	0.855	1.4	0.708	1.032	0.323	0.691	1.053	0.361	2.0	0.542	0.526	1.314	4.000	1.347	3.800	1
BP0CA1070A7**	1.070	1.1	0.887	1.313	0.426	0.864	1.329	0.465	2.1	0.693	0.672	1.647	4.000	1.682	4.000	1
BP0CA1090A7**	1.090	1.3	0.898	1.329	0.431	0.881	1.350	0.469	2.1	0.698	0.683	1.676	4.500	1.716	4.275	1
BP0CA1100A7**	1.100	0.95	0.881	1.365	0.484	0.860	1.390	0.530	1.62	0.680	0.660	1.680	5.000	1.730	4.500	1
BP0FA1100A7**	1.100	1.19	0.915	1.340	0.425	0.870	1.390	0.520	1.97	0.670	0.640	1.720	16.700	1.780	11.110	1
BP0FA1130A7**	1.130	1.35	0.928	1.398	0.470	0.890	1.440	0.550	2.18	0.680	0.660	1.760	15.540	1.790	11.700	1
BP0CA1160A7**	1.160	1.3	0.943	1.340	0.397	0.910	1.397	0.486	2.0	0.719	0.704	1.727	4.750	1.820	4.275	1
BP0FA1190A7**	1.190	1.28	1.031	1.387	0.356	0.990	1.440	0.450	1.96	0.760	0.740	1.740	18.000	1.780	16.290	1
BP0CA1610A7**	1.610	0.89	1.290	2.006	0.716	1.270	2.050	0.780	1.74	1.040	1.020	2.500	9.000	2.570	6.260	1
BP0EA1950A7**	1.950	0.94	1.486	2.573	1.087	1.440	2.640	1.200	1.46	1.190	1.160	3.140	9.000	3.220	7.500	1
BP0EA1980A7**	1.980	0.98	1.506	2.629	1.123	1.470	2.680	1.210	1.43	1.210	1.170	3.270	9.000	3.370	7.620	1
BP0EA2000A7**	2.000	1.8	1.878	2.121	0.243	1.838	2.18	0.343	2.2	1.48	1.454	2.675	9.000	2.729	8.000	1
BP0EA2055A7**	2.055	1.8	1.944	2.180	0.237	1.902	2.223	0.321	2.1	1.518	1.492	2.770	9.000	2.817	8.000	1
BP0EA2090A7**	2.090	1.08	1.580	2.759	1.179	1.540	2.840	1.390	1.33	1.280	1.230	3.400	9.000	3.600	7.920	1
BP0EA2135A7**	2.135	1.8	2.002	2.266	0.263	1.961	2.325	0.363	2.2	1.561	1.529	2.841	9.000	2.882	8.000	1
BP0BA2150A7**	2.150	1.4	1.768	2.548	0.780	1.750	2.579	0.829	1.8	1.344	-	3.305	9.000	-	-	1
BP0BA2260A7**	2.260	1.3	1.838	2.642	0.804	1.820	2.673	0.854	1.9	1.397	-	3.444	9.000	-	-	1
BP0BA2290A7**	2.290	1.2	1.884	2.751	0.868	1.866	2.715	0.849	1.8	1.433	-	3.548	9.000	-	-	1
BP0EA2423A7**	2.423	1.6	2.168	2.646	0.478	2.138	2.746	0.609	2.3	1.758	1.732	3.290	9.000	3.325	9.000	1
BP0EA2500A7**	2.500	1.36	2.084	3.003	0.919	2.020	3.210	1.190	2.38	1.690	1.640	3.710	9.000	3.800	7.500	1
BP0EA2510A7**	2.510	1.6	2.245	2.774	0.529	2.209	2.832	0.623	2.1	1.822	1.796	3.366	9.000	3.402	9.000	1
BP0EA2540A7**	2.540	1.11	2.077	3.161	1.084	2.020	3.210	1.190	1.88	1.700	1.640	3.800	8.300	3.900	7.800	1
BP0EA2568A7**	2.568	1.9	2.286	2.864	0.578	2.256	2.923	0.667	2.1	1.871	1.844	3.485	10.000	3.521	9.000	1
BP0EA2620A7**	2.620	1.55	2.141	3.169	1.028	2.090	3.280	1.190	2.21	1.740	1.680	3.910	7.800	3.960	7.700	1
BP0EA3060A7**	3.060	1.6	2.783	3.340	0.558	2.741	3.42	0.68	2.1	2.261	2.223	4.099	11.000	4.140	10.200	1
BP0EA3123A7**	3.123	1.5	2.847	3.430	0.584	2.799	3.485	0.685	2.2	2.319	2.288	4.153	12.000	4.200	9.000	1
BP0EA3180A7**	3.180	0.84	2.505	4.016	1.510	2.450	4.140	1.690	1.67	1.810	1.750	5.250	18.430	5.460	15.500	1
BP0BA3270A7**	3.270	2.2	3.242	3.314	0.071	3.156	3.381	0.225	2.3	2.334	2.219	4.337	9.000	4.580	9.000	1
BP0BA3280A7**	3.280	2.2	3.225	3.340	0.116	3.150	3.412	0.262	2.2	2.361	2.246	4.376	9.000	4.700	9.000	1
BP0EA3284A7**	3.284	1.5	3.019	3.586	0.567	2.972	3.629	0.657	2.1	2.447	2.292	4.324	11.500	5.091	9.000	1
BP0EA3310A7**	3.310	0.97	2.695	4.107	1.412	2.620	4.190	1.570	1.85	1.910	1.860	5.220	19.050	5.410	15.940	1
BP0BA3350A7**	3.350	2.3	3.282	3.429	0.146	3.208	3.506	0.299	2.3	2.386	2.267	4.480	9.000	4.580	9.000	1
BP0EA3430A7**	3.430	0.83	2.683	4.370	1.687	2.640	4.460	1.820	1.8	1.950	1.890	5.590	20.000	5.830	16.000	1
BP0AA3580A7**	3.580	1.2	2.972	4.304	1.332	2.943	4.351	1.408	1.6	2.068	-	5.292	9.000	6.300	9.000	1
BP0EA3597A7**	3.597	1.9	3.325	3.923	0.597	3.237	3.998	0.76	2.4	2.621	2.538	4.896	10.00	4.93	9.000	1
BP0BA3630A7**	3.630	2.1	3.513	3.746	0.233	3.438	3.829	0.392	2.2	2.594	2.460	4.884	9.000	5.039	9.000	1
BP0AA3700A7**	3.700	1.1	3.046	4.481	1.435	3.020	4.533	1.513	1.8	2.152	-	5.500	9.000	6.300	9.000	1
BP0EA3720A7**	3.720	2	3.408	3.981	0.573	3.319	4.169	0.85	2.5	2.661	2.575	5.051	12.500	5.098	9.000	1
BP0BA3750A7**	3.750	2	3.657	3.835	0.178	3.588	3.918	0.330	2.1	2.694	2.558	5.097	9.000	5.241	9.000	1
BP0AA3790A7**	3.790	1	3.200	4.589	1.389	3.150	4.646	1.496	2.1	2.199	-	5.661	9.000	6.060	9.000	1
BP0EA3827A7**	3.827	2.1	3.508	4.169	0.661	3.408	4.297	0.889	2.4	2.726	2.640	5.180	14.000	5.222	9.000	1
BP0BA3900A7**	3.900	2	3.847	3.960	0.112	3.755	4.058	0.304	2.2	2.735	2.600	5.229	9.000	6.000	9.000	1
BP0EA4260A7**	4.260	1.3	3.389	5.422	2.032	3.325	5.462	2.136	2.0	2.621	2.261	6.920	9.000	-	-	1
BP0EA4363A7**	4.363	1.6	3.933	4.847	0.914	3.863	4.927	1.064	2.2	3.121	3.009	6.019	13.000	6.072	10.500	1
BP0EA4400A7**	4.400	1.1	3.456	5.627	2.171	3.420	5.670	2.250	1.8	2.699	2.613	7.109	9.000	-	-	1
BP0EA4440A7**	4.440	1.1	3.554	5.513	1.959	3.490	5.660	2.170	1.78	2.890	2.790	7.060	13.800	7.390	9.260	1
BP0EA4550A7**	4.550	1.7	4.052	5.071	1.019	3.922	5.16	1.239	2.0	3.196	3.084	6.278	15.500	6.302	11.000	1
BP0EA4583A7**	4.583	1.1	3.542	5.932	2.390	3.521	5.979	2.459	1.6	2.770	2.667	7.400	9.000	-	-	1
BP0EA4600A7**	4.600	1.14	3.806	5.696	1.890	3.660	5.780	2.120	1.79	2.990	2.900	7.150	9.680	7.500	9.550	1
BP0EA4649A7**	4.649	1.8	4.371	5.285	0.914	4.016	5.381	1.365	2.7	3.259	3.159	6.497	12.000	6.596	11.000	1
BP0EA4680A7**	4.680	1.18	3.743	5.776	2.033	3.700	5.930	2.230	1.77	3.050	2.960	7.370	9.860	7.730	8.400	1

Click on part number to see full specifications
 **Packaging: 00 = waffle pack, TR = 1000pcs T&R, TR\250 = 250pcs T&R

5 Related Operational Amplifier Products

DEVICE	V _S (V)	BW (MHz)	SLEW RATE (V/μs)	VOLTAGE NOISE (nV/√Hz)	AMPLIFIER DESCRIPTION
OPA657	±5	1600	700	4.8	Gain of +7 stable FET Input
OPA656	±5	230	290	7	Unity-Gain Stable FET-Input
OPA659	±6	350	2550	8.9	Unity-Gain Stable FET-Input
LMH6629	5	4000	1600	0.69	Gain of +10 stable Bipolar Input
THS4631	±15	210	1000	7	Unity-Gain Stable FET-Input
OPA857	5	4750	220	—	Programmable Gain (5 kΩ / 20 kΩ) Transimpedance Amplifier

6 Pin Configuration and Functions



Pin Orientation/Package Marking

Pin Functions

NAME	PIN		I/O	DESCRIPTION
	SOIC	SOT-23		
NC	1	—	—	No Connection
	5			
	8			
V _{IN-}	2	4	I	Inverting Input
V _{IN+}	3	3	I	Noninverting Input
-V _S	4	2	POW	Negative Power Supply
V _{OUT}	6	1	O	Output of amplifier
+V _S	7	5	POW	Positive Power Supply

7 Specifications

7.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)⁽¹⁾

	MIN	MAX	UNIT
Supply voltage (Total Bipolar Supplies)		±6.5	V
Internal power dissipation	See Thermal Information		
Differential input voltage	−V _S	+V _S	V
Input voltage	−V _S	+V _S	V
Junction temperature (T _J)		175	°C
Storage temperature	−65	125	°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

7.2 ESD Ratings

		VALUE	UNIT
V _(ESD) Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±2000	V
	Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±500	
	Machine Model (MM)	±200	

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

7.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

	MIN	NOM	MAX	UNIT
V _S Total supply voltage	8	10	12	V
T _A Ambient temperature	−40	25	85	°C

7.4 Thermal Information

THERMAL METRIC ⁽¹⁾		OPA657		UNIT
		D (SOIC)	DBV (SOT-23)	
		8 PINS	5 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	125	150	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	85.2	140.8	°C/W
R _{θJB}	Junction-to-board thermal resistance	75.9	62.8	°C/W
ψ _{JT}	Junction-to-top characterization parameter	26.2	24.4	°C/W
ψ _{JB}	Junction-to-board characterization parameter	75.4	61.8	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	—	—	°C/W

- (1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).

7.5 Electrical Characteristics: $V_S = \pm 5\text{ V}$

At $R_F = +453\ \Omega$, $R_L = +100\ \Omega$, and $G = +10\text{ V/V}$, unless otherwise noted. See Figure 29 for AC performance.

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT	TEST LEVEL ⁽¹⁾
AC PERFORMANCE (see Figure 29)							
Small-signal bandwidth	G = +7 V/V, V _O = 200 mV _{PP}	T _J = 25°C	350		MHz	C	
	G = +10 V/V, V _O = 200 mV _{PP}	T _J = 25°C	275				
	G = +20 V/V, V _O = 200 mV _{PP}	T _J = 25°C	90				
Gain-bandwidth product	G > +40 V/V	T _J = 25°C	1600		dB	C	
Bandwidth for 0.1dB Flatness	G = +10 V/V, 2 V _{PP}	T _J = 25°C	30		MHz	C	
Peaking at a Gain of +7		T _J = 25°C	7		dB	C	
Large-Signal Bandwidth	G = +10 V/V, 2 V _{PP}	T _J = 25°C	180		MHz	C	
Slew Rate	G = +10 V/V, 1-V Step	T _J = 25°C	700		V/μs	C	
Rise-and-Fall Time	0.2-V Step	T _J = 25°C	1		ns	C	
Settling Time to 0.02%	G = +10 V/V, V _O = 2-V Step	T _J = 25°C	20		ns	C	
Harmonic Distortion	G = +10 V/V, f = 5 MHz, V _O = 2 V _{PP}					C	
2nd-Harmonic	R _L = 200 Ω	T _J = 25°C	−70		dBc	C	
	R _L > 500 Ω	T _J = 25°C	−74		dBc	C	
3rd-Harmonic	R _L = 200 Ω	T _J = 25°C	−99		dBc	C	
	R _L > 500 Ω	T _J = 25°C	−106		dBc	C	
Input Voltage Noise	f > 100 kHz	T _J = 25°C	4.8		nV/√Hz	C	
Input Current Noise	f > 100 kHz	T _J = 25°C	1.3		fA/√Hz	C	
DC PERFORMANCE ⁽²⁾							
Open-Loop Voltage Gain (A _{OL})	V _{CM} = 0 V, R _L = 100 Ω	T _J = 25°C	65	70	dB	A	
		T _J = 0°C to 70°C ⁽³⁾	64				
		T _J = −40°C to 85°C ⁽³⁾	63				
Input Offset Voltage	V _{CM} = 0 V	T _J = 25°C	±0.25	±1.8	mV	A	
		T _J = 0°C to 70°C ⁽³⁾		±2.2			
		T _J = −40°C to 85°C ⁽³⁾		±2.6			
Average Offset Voltage Drift	V _{CM} = 0 V	T _J = 25°C	±12	±2	μV/°C	A	
		T _J = 0°C to 70°C ⁽³⁾	±12				
		T _J = −40°C to 85°C ⁽³⁾	±12				
Input Bias Current	V _{CM} = 0 V	T _J = 25°C	±2	±20	pA	A	
		T _J = 0°C to 70°C ⁽³⁾		±1800			
		T _J = −40°C to 85°C ⁽³⁾		±5000			
Input Offset Current	V _{CM} = 0 V	T _J = 25°C	±1	±10	pA	A	
		T _J = 0°C to 70°C ⁽³⁾		±900			
		T _J = −40°C to 85°C ⁽³⁾		±2500			
INPUT							
Most Positive Input Voltage ⁽⁴⁾		T _J = 25°C	2	2.5	V	A	
		T _J = 0°C to 70°C ⁽³⁾	1.9				
		T _J = −40°C to 85°C ⁽³⁾	1.8				
Most Negative Input Voltage ⁽⁴⁾		T _J = 25°C	−3.5	−4	V	A	
		T _J = 0°C to 70°C ⁽³⁾	−3.4				
		T _J = −40°C to 85°C ⁽³⁾	−3.3				
Common-Mode Rejection Ratio (CMRR)	V _{CM} = ±0.5 V	T _J = 25°C	83	89	dB	A	
		T _J = 0°C to 70°C ⁽³⁾	81				
		T _J = −40°C to 85°C ⁽³⁾	79				
Input Impedance							

(1) Test Levels: (A) 100% tested at $+25^\circ\text{C}$. Over-temperature limits by characterization and simulation. (B) Limits set by characterization and simulation. (C) Typical value only for information.

(2) Current is considered positive out-of-node. V_{CM} is the input common-mode voltage.

(3) Junction temperature = ambient at low temperature limit; junction temperature = ambient $+20^\circ\text{C}$ at high temperature limit for over temperature specifications.

(4) Tested $< 3\text{ dB}$ below minimum specified CMRR at $\pm\text{CMIR}$ limits.

Electrical Characteristics: $V_S = \pm 5\text{ V}$ (continued)

At $R_F = +453\ \Omega$, $R_L = +100\ \Omega$, and $G = +10\text{ V/V}$, unless otherwise noted. See Figure 29 for AC performance.

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT	TEST LEVEL ⁽¹⁾
Differential		T _J = 25°C	10 ¹² 0.7			Ω pF	C
Common-Mode		T _J = 25°C	10 ¹² 4.5			Ω pF	C
OUTPUT							
Voltage Output Swing	No load	T _J = 25°C	±3.9		V	A	
		T _J = 0°C to 70°C ⁽³⁾	±3.7				
	R _L = 100 Ω	T _J = 25°C	±3.3	±3.5	V	B	
		T _J = 0°C to 70°C ⁽³⁾	±3.2				
		T _J = −40°C to 85°C ⁽³⁾	±3.1				
Current Output, Sourcing		T _J = 25°C	50	70	mA	A	
		T _J = 0°C to 70°C ⁽³⁾	48				
		T _J = −40°C to 85°C ⁽³⁾	46				
Current Output, Sinking		T _J = 25°C	−50	−70	mA	A	
		T _J = 0°C to 70°C ⁽³⁾	−48				
		T _J = −40°C to 85°C ⁽³⁾	−46				
Closed-Loop Output Impedance	G = +10 V/V, f = 0.1 MHz	T _J = 25°C	0.02			Ω	A
POWER SUPPLY							
Specified Operating Voltage		T _J = 25°C	±5			V	A
Minimum Operating Voltage		T _J = 25°C	±4			V	C
Maximum Operating Voltage Range		T _J = 25°C	±6		V	A	
		T _J = 0°C to 70°C ⁽³⁾	±6				
		T _J = −40°C to 85°C ⁽³⁾	±6				
Maximum Quiescent Current		T _J = 25°C	14	16	mA	A	
		T _J = 0°C to 70°C ⁽³⁾	16.2				
		T _J = −40°C to 85°C ⁽³⁾	16.3				
Minimum Quiescent Current		T _J = 25°C	11.7	14	mA	A	
		T _J = 0°C to 70°C ⁽³⁾	11.4				
		T _J = −40°C to 85°C ⁽³⁾	11.1				
Power-Supply Rejection Ratio (+PSRR)	+VS = 4.5 V to 5.5 V	T _J = 25°C	76	80	dB	A	
		T _J = 0°C to 70°C ⁽³⁾	74				
		T _J = −40°C to 85°C ⁽³⁾	72				
Power-Supply Rejection Ratio (−PSRR)	−VS = 4.5 V to 5.5 V	T _J = 25°C	62	68	dB	A	
		T _J = 0°C to 70°C ⁽³⁾	60				
		T _J = −40°C to 85°C ⁽³⁾	58				
TEMPERATURE RANGE							
Specified Operating Range: U, N Package		T _J = 25°C	−40 to +85			°C	
Thermal Resistance, R _{θJA}	Junction-to-Ambient						
U: SO-8		T _J = 25°C	125			°C/W	
N: SOT23-5		T _J = 25°C	150			°C/W	

7.6 Electrical Characteristics: $V_S = \pm 5\text{ V}$, High-Grade DC Specifications

At $R_F = 453\ \Omega$, $R_L = 100\ \Omega$, and $G = +10\text{ V/V}$, unless otherwise noted. ⁽¹⁾

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT	TEST LEVEL (2)
Input Offset Voltage	$V_{CM} = 0\text{ V}$	$T_J = 25^\circ\text{C}$		± 0.1	$\pm 0.6^{(3)}$	mV	A
		$T_J = 0^\circ\text{C to } 70^\circ\text{C}^{(4)}$			± 0.85		
		$T_J = -40^\circ\text{C to } 85^\circ\text{C}^{(4)}$			± 0.9		
Input Offset Voltage Drift	$V_{CM} = 0\text{ V}$	$T_J = 25^\circ\text{C}$		± 2	$\pm 6^{(3)}$	$\mu\text{V}/^\circ\text{C}$	A
		$T_J = 0^\circ\text{C to } 70^\circ\text{C}^{(4)}$			± 6		
		$T_J = -40^\circ\text{C to } 85^\circ\text{C}^{(4)}$			± 6		
Input Bias Current	$V_{CM} = 0\text{ V}$	$T_J = 25^\circ\text{C}$		± 1	$\pm 5^{(3)}$	pA	A
		$T_J = 0^\circ\text{C to } 70^\circ\text{C}^{(4)}$			± 450		
		$T_J = -40^\circ\text{C to } 85^\circ\text{C}^{(4)}$			± 1250		
Input Offset Current	$V_{CM} = 0\text{ V}$	$T_J = 25^\circ\text{C}$		± 0.5	$\pm 5^{(3)}$	pA	A
		$T_J = 0^\circ\text{C to } 70^\circ\text{C}^{(4)}$			± 450		
		$T_J = -40^\circ\text{C to } 85^\circ\text{C}^{(4)}$			± 1250		
Common-Mode Rejection Ratio (CMRR)	$V_{CM} = \pm 0.5\text{ V}$	$T_J = 25^\circ\text{C}$	91 ⁽³⁾	98		dB	A
		$T_J = 0^\circ\text{C to } 70^\circ\text{C}^{(4)}$	89				
		$T_J = -40^\circ\text{C to } 85^\circ\text{C}^{(4)}$	87				
Power-Supply Rejection Ratio (+PSRR)	$+V_S = 4.5\text{ V to } 5.5\text{ V}$	$T_J = 25^\circ\text{C}$	78 ⁽³⁾	82		dB	A
		$T_J = 0^\circ\text{C to } 70^\circ\text{C}^{(4)}$	76				
		$T_J = -40^\circ\text{C to } 85^\circ\text{C}^{(4)}$	74				
Power-Supply Rejection Ratio (–PSRR)	$-V_S = -4.5\text{ V to } -5.5\text{ V}$	$T_J = 25^\circ\text{C}$	68 ⁽³⁾	74		dB	A
		$T_J = 0^\circ\text{C to } 70^\circ\text{C}^{(4)}$	66				
		$T_J = -40^\circ\text{C to } 85^\circ\text{C}^{(4)}$	64				

(1) All other specifications are the same as the standard-grade.

(2) Test Levels: (A) 100% tested at $+25^\circ\text{C}$. Over temperature limits by characterization and simulation.

(3) Junction temperature = ambient for $+25^\circ\text{C}$ specifications.

(4) Junction temperature = ambient at low temperature limit; junction temperature = ambient $+20^\circ\text{C}$ at high temperature limit for over temperature specifications.

CD54HC4046A, CD74HC4046A, CD54HCT4046A, CD74HCT4046A

Absolute Maximum Ratings

DC Supply Voltage, V_{CC} -0.5V to 7V
 DC Input Diode Current, I_{IK}
 For $V_I < -0.5V$ or $V_I > V_{CC} + 0.5V$ $\pm 20mA$
 DC Output Diode Current, I_{OK}
 For $V_O < -0.5V$ or $V_O > V_{CC} + 0.5V$ $\pm 20mA$
 DC Drain Current, per Output, I_O
 For $-0.5V < V_O < V_{CC} + 0.5V$ $\pm 25mA$
 DC Output Source or Sink Current per Output Pin, I_O
 For $V_O > -0.5V$ or $V_O < V_{CC} + 0.5V$ $\pm 25mA$
 DC V_{CC} or Ground Current, I_{CC} $\pm 50mA$

Thermal Information

Package Thermal Impedance, θ_{JA} (see Note 1):
 E (PDIP) Package $67^{\circ}C/W$
 M (SOIC) Package $73^{\circ}C/W$
 NS (SOP) Package $64^{\circ}C/W$
 PW (TSSOP) Package $108^{\circ}C/W$
 Maximum Junction Temperature $150^{\circ}C$
 Maximum Storage Temperature Range $-65^{\circ}C$ to $150^{\circ}C$
 Maximum Lead Temperature (Soldering 10s) $300^{\circ}C$
 (SOIC - Lead Tips Only)

Operating Conditions

Temperature Range, T_A $-55^{\circ}C$ to $125^{\circ}C$
 Supply Voltage Range, V_{CC}
 HC Types $2V$ to $6V$
 HCT Types $4.5V$ to $5.5V$
 DC Input or Output Voltage, V_I , V_O $0V$ to V_{CC}
 Input Rise and Fall Time
 $2V$ $1000ns$ (Max)
 $4.5V$ $500ns$ (Max)
 $6V$ $400ns$ (Max)

CAUTION: Stresses above those listed in "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied.

NOTE:

1. The package thermal impedance is calculated in accordance with JESD 51-7.

DC Electrical Specifications

PARAMETER	SYMBOL	TEST CONDITIONS		V _{CC} (V)	25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS
		V _I (V)	I _O (mA)		MIN	TYP	MAX	MIN	MAX	MIN	MAX	
HC TYPES												
VCO SECTION												
INH High Level Input Voltage	V _{IH}	-	-	3	2.1	-	-	2.1	-	2.1	-	V
				4.5	3.15	-	-	3.15	-	3.15	-	V
				6	4.2	-	-	4.2	-	4.2	-	V
INH Low Level Input Voltage	V _{IL}	-	-	3	-	-	0.9	-	0.9	-	0.9	V
				4.5	-	-	1.35	-	1.35	-	1.35	V
				6	-	-	1.8	-	1.8	-	1.8	V
VCO _{OUT} High Level Output Voltage CMOS Loads	V _{OH}	V _{IH} or V _{IL}	-0.02	3	2.9	-	-	2.9	-	2.9	-	V
			-0.02	4.5	4.4	-	-	4.4	-	4.4	-	V
			-0.02	6	5.9	-	-	5.9	-	5.9	-	V
VCO _{OUT} High Level Output Voltage TTL Loads	V _{OH}	V _{IH} or V _{IL}	-	-	-	-	-	-	-	-	-	V
			-4	4.5	3.98	-	-	3.84	-	3.7	-	V
			-5.2	6	5.48	-	-	5.34	-	5.2	-	V
VCO _{OUT} Low Level Output Voltage CMOS Loads	V _{OL}	V _{IH} or V _{IL}	0.02	2	-	-	0.1	-	0.1	-	0.1	V
			0.02	4.5	-	-	0.1	-	0.1	-	0.1	V
			0.02	6	-	-	0.1	-	0.1	-	0.1	V
VCO _{OUT} Low Level Output Voltage TTL Loads	V _{OL}	V _{IH} or V _{IL}	-	-	-	-	-	-	-	-	-	V
			4	4.5	-	-	0.26	-	0.33	-	0.4	V
			5.2	6	-	-	0.26	-	0.33	-	0.4	V
C1A, C1B Low Level Output Voltage (Test Purposes Only)	V _{OL}	V _{IL} or V _{IH}	4	4.5	-	-	0.40	-	0.47	-	0.54	V
			5.2	6	-	-	0.40	-	0.47	-	0.54	V

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DC Electrical Specifications (Continued)

PARAMETER	SYMBOL	TEST CONDITIONS		V _{CC} (V)	25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS
		V _I (V)	I _O (mA)		MIN	TYP	MAX	MIN	MAX	MIN	MAX	
INH VCO _{IN} Input Leakage Current	I _I	V _{CC} or GND	-	6	-	-	±0.1	-	±1	-	±1	μA
R1 Range (Note 2)	-	-	-	4.5	3	-	300	-	-	-	-	kΩ
R2 Range (Note 2)	-	-	-	4.5	3	-	300	-	-	-	-	kΩ
C1 Capacitance Range	-	-	-	3	-	-	No Limit	-	-	-	-	pF
				4.5	-	-		-	-	-	-	pF
				6	-	-		-	-	-	-	pF
VCO _{IN} Operating Voltage Range	-	Over the range specified for R1 for Linearity See Figure 10, and 34 - 37 (Note 3)		3	1.1	-	1.9	-	-	-	-	V
				4.5	1.1	-	3.2	-	-	-	-	V
				6	1.1	-	4.6	-	-	-	-	V
PHASE COMPARATOR SECTION												
SIG _{IN} , COMP _{IN} DC Coupled High-Level Input Voltage	V _{IH}	-	-	2	1.5	-	-	1.5	-	1.5	-	V
				4.5	3.15	-	-	3.15	-	3.15	-	V
				6	4.2	-	-	4.2	-	4.2	-	V
SIG _{IN} , COMP _{IN} DC Coupled Low-Level Input Voltage	V _{IL}	-	-	2	-	-	0.5	-	0.5	-	0.5	V
				4.5	-	-	1.35	-	1.35	-	1.35	V
				6	-	-	1.8	-	1.8	-	1.8	V
PCP _{OUT} , PCn OUT High-Level Output Voltage CMOS Loads	V _{OH}	V _{IL} or V _{IH}	-0.02	2	1.9	-	-	1.9	-	1.9	-	V
				4.5	4.4	-	-	4.4	-	4.4	-	V
				6	5.9	-	-	5.9	-	5.9	-	V
PCP _{OUT} , PCn OUT High-Level Output Voltage TTL Loads	V _{OH}	V _{IL} or V _{IH}	-4	4.5	3.98	-	-	3.84	-	3.7	-	V
			-5.2	6	5.48	-	-	5.34	-	5.2	-	V
PCP _{OUT} , PCn OUT Low-Level Output Voltage CMOS Loads	V _{OL}	V _{IL} or V _{IH}	0.02	2	-	-	0.1	-	0.1	-	0.1	V
				4.5	-	-	0.1	-	0.1	-	0.1	V
				6	-	-	0.1	-	0.1	-	0.1	V
PCP _{OUT} , PCn OUT Low-Level Output Voltage TTL Loads	V _{OL}	V _{IL} or V _{IH}	4	4.5	-	-	0.26	-	0.33	-	0.4	V
			5.2	6	-	-	0.26	-	0.33	-	0.4	V
SIG _{IN} , COMP _{IN} Input Leakage Current	I _I	V _{CC} or GND	-	2	-	-	±3	-	±4	-	±5	μA
				3	-	-	±7	-	±9	-	±11	μA
				4.5	-	-	±18	-	±23	-	±29	μA
				6	-	-	±30	-	±38	-	±45	μA
PC2 _{OUT} Three-State Off-State Current	I _{OZ}	V _{IL} or V _{IH}	-	6	-	-	±0.5	-	±5	-	±10	μA
SIG _{IN} , COMP _{IN} Input Resistance	R _I	V _I at Self-Bias Operation Point: ΔV _I = 0.5V, See Figure 10		3	-	800	-	-	-	-	-	kΩ
				4.5	-	250	-	-	-	-	-	kΩ
				6	-	150	-	-	-	-	-	kΩ
DEMODULATOR SECTION												
Resistor Range	R _S	at R _S > 300kΩ Leakage Current Can Influence V _{DEMOUT}		3	50	-	300	-	-	-	-	kΩ
				4.5	50	-	300	-	-	-	-	kΩ
				6	50	-	300	-	-	-	-	kΩ

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DC Electrical Specifications (Continued)

PARAMETER	SYMBOL	TEST CONDITIONS		V _{CC} (V)	25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS
		V _I (V)	I _O (mA)		MIN	TYP	MAX	MIN	MAX	MIN	MAX	
Offset Voltage V _{COIN} to V _{DEM}	V _{OFF}	V _I = V _{VCO IN} = $\frac{V_{CC}}{2}$ Values Taken Over R _S Range See Figure 23		3	-	±30	-	-	-	-	-	mV
				4.5	-	±20	-	-	-	-	-	mV
				6	-	±10	-	-	-	-	-	mV
Dynamic Output Resistance at DEM _{OUT}	R _D	V _{DEMOUT} = $\frac{V_{CC}}{2}$		3	-	25	-	-	-	-	-	Ω
				4.5	-	25	-	-	-	-	-	Ω
				6	-	25	-	-	-	-	-	Ω
Quiescent Device Current	I _{CC}	Pins 3, 5 and 14 at V _{CC} Pin 9 at GND, I ₁ at Pins 3 and 14 to be excluded		6	-	-	8	-	80	-	160	μA
HCT TYPES												
VCO SECTION												
INH High Level Input Voltage	V _{IH}	-	-	4.5 to 5.5	2	-	-	2	-	2	-	V
INH Low Level Input Voltage	V _{IL}	-	-	4.5 to 5.5	-	-	0.8	-	0.8	-	0.8	V
VCO _{OUT} High Level Output Voltage CMOS Loads	V _{OH}	V _{IH} or V _{IL}	-0.02	4.5	4.4	-	-	4.4	-	4.4	-	V
VCO _{OUT} High Level Output Voltage TTL Loads			-4	4.5	3.98	-	-	3.84	-	3.7	-	V
VCO _{OUT} Low Level Output Voltage CMOS Loads	V _{OL}	V _{IH} or V _{IL}	0.02	4.5	-	-	0.1	-	0.1	-	0.1	V
VCO _{OUT} Low Level Output Voltage TTL Loads			4	4.5	-	-	0.26	-	0.33	-	0.4	V
C1A, C1B Low Level Output Voltage (Test Purposes Only)	V _{OL}	V _{IH} or V _{IL}	4	4.5	-	-	0.40	-	0.47	-	0.54	V
INH VCO _{IN} Input Leakage Current	I _I	Any Voltage Between V _{CC} and GND		5.5	-		±0.1	-	±1	-	±1	μA
R1 Range (Note 2)	-	-	-	4.5	3	-	300	-	-	-	-	kΩ
R2 Range (Note 2)	-	-	-	4.5	3	-	300	-	-	-	-	kΩ
C1 Capacitance Range	-	-	-	4.5	0	-	No Limit	-	-	-	-	pF
VCO _{IN} Operating Voltage Range	-	Over the range specified for R1 for Linearity See Figure 10, and 34 - 37 (Note 3)		4.5	1.1	-	3.2	-	-	-	-	V
PHASE COMPARATOR SECTION												
SIG _{IN} , COMP _{IN} DC Coupled High-Level Input Voltage	V _{IH}	-	-	4.5 to 5.5	2	-	-	2	-	2	-	V

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DC Electrical Specifications (Continued)

PARAMETER	SYMBOL	TEST CONDITIONS		V _{CC} (V)	25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS
		V _I (V)	I _O (mA)		MIN	TYP	MAX	MIN	MAX	MIN	MAX	
SIG _{IN} , COMP _{IN} DC Coupled Low-Level Input Voltage	V _{IL}	-	-	4.5 to 5.5	-	-	0.8	-	0.8	-	0.8	V
PCP _{OUT} , PCn OUT High-Level Output Voltage CMOS Loads	V _{OH}	V _{IL} or V _{IH}	-	4.5	4.4	-	-	4.4	-	4.4	-	V
PCP _{OUT} , PCn OUT High-Level Output Voltage TTL Loads	V _{OH}	V _{IL} or V _{IH}	-	4.5	3.98	-	-	3.84	-	3.7	-	V
PCP _{OUT} , PCn OUT Low-Level Output Voltage CMOS Loads	V _{OL}	V _{IL} or V _{IH}	-	4.5	-	-	0.1	-	0.1	-	0.1	V
PCP _{OUT} , PCn OUT Low-Level Output Voltage TTL Loads	V _{OL}	V _{IL} or V _{IH}	-	4.5	-	-	0.26	-	0.33	-	0.4	V
SIG _{IN} , COMP _{IN} Input Leakage Current	I _I	Any Voltage Between V _{CC} and GND	-	5.5	-	-	±30		±38		±45	μA
PC2 _{OUT} Three-State Off-State Current	I _{OZ}	V _{IL} or V _{IH}	-	5.5	-	-	±0.5	±5	-	-	±10	μA
SIG _{IN} , COMP _{IN} Input Resistance	R _I	V _I at Self-Bias Operation Point: ΔV _I = 0.5V, See Figure 10		4.5	-	250	-	-	-	-	-	kΩ
DEMODULATOR SECTION												
Resistor Range	R _S	at R _S > 300kΩ Leakage Current Can Influence V _{DEM OUT}		4.5	5	-	300	-	-	-	-	kΩ
Offset Voltage VCO _{IN} to V _{DEM}	V _{OFF}	V _I = V _{VCO IN} = $\frac{V_{CC}}{2}$ Values taken over R _S Range See Figure 23		4.5	-	±20	-	-	-	-	-	mV
Dynamic Output Resistance at DEM _{OUT}	R _D	V _{DEM OUT} = $\frac{V_{CC}}{2}$		4.5	-	25	-	-	-	-	-	Ω
Quiescent Device Current	I _{CC}	V _{CC} or GND	-	5.5	-	-	8	-	80	-	160	μA
Additional Quiescent Device Current Per Input Pin: 1 Unit Load	ΔI _{CC} (Note 4)	V _{CC} -2.1 Excluding Pin 5	-	4.5 to 5.5	-	100	360	-	450	-	490	μA

NOTES:

- The value for R1 and R2 in parallel should exceed 2.7kΩ.
- The maximum operating voltage can be as high as V_{CC} -0.9V, however, this may result in an increased offset voltage.
- For dual-supply systems theoretical worst case (V_I = 2.4V, V_{CC} = 5.5V) specification is 1.8mA.

CD54HC4046A, CD74HC4046A, CD54HCT4046A, CD74HCT4046A

HCT Input Loading Table

INPUT	UNIT LOADS
INH	1

NOTE: Unit load is ΔI_{CC} limit specific in DC Electrical Specifications Table, e.g., 360 μ A max. at 25°C.

Switching Specifications $C_L = 50$ pF, Input $t_r, t_f = 6$ ns

PARAMETER	SYMBOL	TEST CONDITIONS	V _{CC} (V)	25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS	
				MIN	TYP	MAX	MIN	MAX	MIN	MAX		
HC TYPES												
PHASE COMPARATOR SECTION												
Propagation Delay SIG _{IN} , COMP _{IN} to PC _I OUT	t _{PLH} , t _{PHL}		2	-	-	200	-	250	-	300	ns	
			4.5	-	-	40	-	50	-	60	ns	
			6	-	-	34	-	43	-	51	ns	
SIG _{IN} , COMP _{IN} to PCP _{OUT}			2	-	-	300	-	375	-	450	ns	
			4.5	-	-	60	-	75	-	90	ns	
			6	-	-	51	-	64	-	77	ns	
SIG _{IN} , COMP _{IN} to PC3 _{OUT}			2	-	-	245	-	305	-	307	ns	
			4.5	-	-	49	-	61	-	74	ns	
			6	-	-	42	-	52	-	63	ns	
Output Transition Time	t _{THL} , t _{TLH}		2	-	-	75	-	95	-	110	ns	
			4.5	-	-	15	-	19	-	22	ns	
			6	-	-	13	-	16	-	19	ns	
Output Enable Time, SIG _{IN} , COMP _{IN} to PC2 _{OUT}	t _{PZH} , t _{PZL}		2	-	-	265	-	330	-	400	ns	
			4.5	-	-	53	-	66	-	80	ns	
			6	-	-	45	-	56	-	68	ns	
Output Disable Time, SIG _{IN} , COMP _{IN} to PC2 _{OUT}	t _{PHZ} , t _{PLZ}		2	-	-	315	-	395	-	475	ns	
			4.5	-	-	63	-	79	-	95	ns	
			6	-	-	54	-	67	-	81	ns	
AC Coupled Input Sensitivity (p-p) at SIG _{IN} or COMP _{IN}		V _{I(P-P)}	3	-	11	-	-	-	-	-	mV	
			4.5	-	15	-	-	-	-	-	-	mV
			6	-	33	-	-	-	-	-	-	mV
VCO SECTION												
Frequency Stability with Temperature Change	$\frac{\Delta f}{\Delta T}$	R ₁ = 100kΩ, R ₂ = ∞	3	-	0.11	-	-	-	-	-	%/°C	
			4.5	-	0.11	-	-	-	-	-	%/°C	
			6	-	0.11	-	-	-	-	-	%/°C	
Maximum Frequency	f _{MAX}	C ₁ = 50pF R ₁ = 3.5kΩ R ₂ = ∞	3	-	24	-	-	-	-	-	MHz	
			4.5	-	24	-	-	-	-	-	MHz	
			6	-	24	-	-	-	-	-	MHz	
		C ₁ = 0pF R ₁ = 9.1kΩ R ₂ = ∞	3	-	38	-	-	-	-	-	MHz	
			4.5	-	38	-	-	-	-	-	MHz	
			6	-	38	-	-	-	-	-	MHz	

CD54HC4046A, CD74HC4046A, CD54HCT4046A, CD74HCT4046A

Switching Specifications $C_L = 50\text{pF}$, Input $t_r, t_f = 6\text{ns}$ (Continued)

PARAMETER	SYMBOL	TEST CONDITIONS	V _{CC} (V)	25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS			
				MIN	TYP	MAX	MIN	MAX	MIN	MAX				
Center Frequency		C ₁ = 40pF R ₁ = 3kΩ R ₂ = ∞ VCO _{IN} = VCC/2	3	7	10	-	-	-	-	-	MHz			
			4.5	12	17	-	-	-	-	-	MHz			
			6	14	21	-	-	-	-	-	MHz			
Frequency Linearity	Δf _{VCO}	R ₁ = 100kΩ R ₂ = ∞ C ₁ = 100pF	3	-	0.4	-	-	-	-	-	%			
			4.5	-	0.4	-	-	-	-	-	%			
			6	-	0.4	-	-	-	-	-	%			
Offset Frequency		R ₂ = 220kΩ C ₁ = 1nF	3	-	400	-	-	-	-	-	kHz			
			4.5	-	400	-	-	-	-	-	kHz			
			6	-	400	-	-	-	-	-	kHz			
DEMODULATOR SECTION														
V _{OUT} V _S f _{IN}		R ₁ = 100kΩ R ₂ = ∞ C ₁ = 100pF R _S = 10kΩ R ₃ = 100kΩ C ₂ = 100pF	3	-	-	-	-	-	-	-	mV/kHz			
			4.5	-	330	-	-	-	-	-	mV/kHz			
			6	-	-	-	-	-	-	-	mV/kHz			
HCT TYPES														
PHASE COMPARATOR SECTION														
Propagation Delay SIG _{IN} , COMP _{IN} to PC _{OUT}	t _{PHL} , t _{PLH}	C _L = 50pF	4.5	-	-	45	-	56	-	68	ns			
			SIG _{IN} , COMP _{IN} to PCP _{OUT}	t _{PHL} , t _{PLH}	C _L = 50pF	4.5	-	-	68	-	85	-	102	ns
			SIG _{IN} , COMP _{IN} to PC3 _{OUT}	t _{PHL} , t _{PLH}	C _L = 50pF	4.5	-	-	58	-	73	-	87	ns
Output Transition Time	t _{TLH} , t _{THL}	C _L = 50pF	4.5	-	-	15	-	19	-	22	ns			
Output Enable Time, SIG _{IN} , COMP _{IN} to PC2 _{OUT}	t _{PZH} , t _{PZL}	C _L = 50pF	4.5	-	-	60	-	75	-	90	pF			
Output Disable Time, SIG _{IN} , COMP _{IN} to PCZ _{OUT}	t _{PHZ} , t _{PLZ}	C _L = 50pF	4.5	-	-	68	-	85	-	102	pF			
AC Coupled Input Sensitivity (p-p) at SIG _{IN} or COMP _I		V _{I(P-P)}	4.5	-	15	-	-	-	-	-	mV			
VCO SECTION														
Frequency Stability with Temperature Change	Δf / ΔT	R ₁ = 100kΩ, R ₂ = ∞	4.5	-	0.11	-	-	-	-	-	%/°C			
Maximum Frequency	f _{MAX}	C ₁ = 50pF R ₁ = 3.5kΩ R ₂ = ∞	4.5	-	24	-	-	-	-	-	MHz			
		C ₁ = 0pF R ₁ = 9.1kΩ R ₂ = ∞	4.5	-	38	-	-	-	-	-	MHz			
Center Frequency		C ₁ = 40pF R ₁ = 3kΩ R ₂ = ∞ VCO _{IN} = VCC/2	4.5	12	17	-	-	-	-	-	MHz			
Frequency Linearity	Δf _{VCO}	R ₁ = 100kΩ R ₂ = ∞ C ₁ = 100pF	4.5	-	0.4	-	-	-	-	-	%			

Appendix C

Using filter bank due to its ability to provide cheap yet accurate filtering behaviour for 3-5GHz

∴ Filter 1) 2.9 → 3.5 GHz

$$f_1 = 2.9 \text{ GHz}, f_2 = 3.5 \text{ GHz}, f_0 = 3.2 \text{ GHz}$$

$$\Delta = \frac{f_2 - f_1}{f_0} = 0.1875$$

Closest spurious signals:

for 2.9: 2.45 GHz

for 3.5: 4.5 GHz

$$\alpha_1 = \left| \frac{1}{0.1875} \left(\frac{2.45}{3.2} - \frac{3.2}{2.45} \right) \right| - 1 = 1.882653$$

$$\% BW = 18.75\%$$

$$\alpha_2 = \left| \frac{1}{0.1875} \left(\frac{4.5}{3.2} - \frac{3.2}{4.5} \right) \right| - 1 = 2.7074$$

Using a 0.5 dB Cheby 1ord.

$\alpha_1 = 26 \text{ dB}$ attenuation, $n = 3$ order.

$\alpha_2 = 34 \text{ dB}$ attenuation, $n = 3$ order.

Filter 2) 3.5 → 4.0 GHz

$$f_1 = 3.5 \text{ GHz}, f_2 = 4.0 \text{ GHz}, f_0 = 3.75 \text{ GHz}$$

0.5 dB Cheby 1ord

$$\Delta = \frac{f_2 - f_1}{f_0} = 0.13$$

Closest Spurious Signals:

for 3.5: 2.75 GHz

for 4.0: 5 GHz

$$\alpha_1 = \left| \frac{1}{0.13} \left(\frac{2.75}{3.75} - \frac{3.75}{2.75} \right) \right| - 1 = 3.7285$$

$$\% BW = 13.3\%$$

$$\alpha_2 = \left| \frac{1}{0.13} \left(\frac{5}{3.75} - \frac{3.75}{5} \right) \right| - 1 = 3.3761$$

∴ $\alpha_1 = 39 \text{ dB}$ attenuation at $n = 3$

$\alpha_2 = 38 \text{ dB}$ attenuation at $n = 3$

Filter 3) 4.0 → 4.5 GHz

0.5 dB Cheby 1ord

$$f_1 = 4.0 \text{ GHz}, f_2 = 4.5 \text{ GHz}, f_0 = 4.25 \text{ GHz}$$

$$\Delta = \frac{f_2 - f_1}{f_0} = 0.11765$$

Closest Spurious Signals:

for 4.0 GHz: 3 GHz

for 4.5 GHz: 5.5 GHz

$$\alpha_1 = \left| \frac{1}{0.118} \left(\frac{3}{4.25} - \frac{4.25}{3} \right) \right| - 1 = 5.0236$$

$$\% BW = 11.765\%$$

$$\alpha_2 = \left| \frac{1}{0.118} \left(\frac{5.5}{4.25} - \frac{4.25}{5.5} \right) \right| - 1 = 3.4186$$

∴ $\alpha_1 = 27 \text{ dB}$ @ $n = 2$, 47 dB @ $n = 3$

$\alpha_2 = 38 \text{ dB}$ attenuation $n = 3$.

Filter 4) $4.5 \rightarrow 5.1 \text{ GHz}$

$$f_1 = 4.5 \text{ GHz}, f_2 = 5.1 \text{ GHz}, f_0 = 4.8 \text{ GHz}$$

$$\Delta = \frac{f_2 - f_1}{f_0} = 0.125$$

$$\alpha_1 = \left| \frac{1}{0.125} \left(\frac{3.25}{4.8} - \frac{4.8}{3.25} \right) \right| - 1$$
$$= 5.39872$$

Closest Spurious Signals:

for 4.5 GHz : 3.25 GHz

for 5.1 GHz : 6.1 GHz

0.5 dB Cheby Zord.

$$\% \text{BW} = 12.5\%$$

$$\alpha_2 = \left| \frac{1}{0.125} \left(\frac{6.1}{4.8} - \frac{4.8}{6.1} \right) \right| - 1$$
$$= 2.8716$$

$$\therefore \alpha_1 = 28 \text{ dB @ } n=2, 50 \text{ dB @ } n=3$$

$$\alpha_2 = 34 \text{ dB attenuation } n=3$$

First Oscillators:

$$f_{IF} = 1 \text{ GHz} \therefore f_{RF} = 2.9 \rightarrow 5.1 \text{ GHz}$$

$$f_{LO} = 3.9 \rightarrow 6.1 \text{ GHz}$$

First IF Filter:

Choosing 100 MHz due closest spurious signals being at 900 MHz for 1 GHz .

$$\therefore f_1 = 950 \text{ MHz}, f_2 = 1.05 \text{ GHz}, f_0 = 1 \text{ GHz}$$

$$\Delta = \frac{f_2 - f_1}{f_0} = 0.1$$

$$\alpha_1 = \left| \frac{1}{0.1} \left(\frac{0.85}{1} - \frac{1}{0.85} \right) \right| - 1 = 0$$
$$= 2.265$$

for 0.95 GHz : $0.875 \Rightarrow 0.9$ for 1 GHz Spurious

$$\% \text{BW} = 10\%$$

$$\alpha_2 = \left| \frac{1}{0.1} \left(\frac{0.9}{1} - \frac{1}{0.9} \right) \right| - 1$$
$$= 1.1$$

$$\therefore \alpha_1 = 30 \text{ dB attenuation at } n=3$$

$$\alpha_2 = 31 \text{ dB attenuation at } n=4.$$

Second IF Filter:

Due to criteria $\text{BW} = 4 \text{ MHz}$

$$f_1 = 398 \text{ MHz}, f_2 = 402 \text{ MHz}, f_0 = 400 \text{ MHz}$$

$$\Delta = \frac{f_2 - f_1}{f_0} = 0.01$$

$$\alpha_1 = \left| \frac{1}{0.01} \left(\frac{0.396}{0.4} - \frac{0.4}{0.396} \right) \right| - 1 = 1.0101$$

$398 \rightarrow 396$ as 396 is where adjacent channel starts.

$402 \rightarrow 404$ as 404 is where adjacent channel starts.

$$\alpha_2 = \left| \frac{1}{0.01} \left(\frac{0.402}{0.4} - \frac{0.4}{0.402} \right) \right| - 1 = 0.99$$

Oscillator:

$$\varepsilon_{L0} < 0.1 \Delta f_{IF}$$

$$\varepsilon_{L0} < 0.1 \cdot 100 \text{ MHz}, \quad \varepsilon_{L0} < 10 \text{ MHz}$$

$$\pm 2.5 \text{ ppm} = 2.5 \text{ Hz / MHz}$$

Oscillator:

1st Stage

3.9 $\rightarrow$ 4.5,

4.5 $\rightarrow$ 5,

5 $\rightarrow$ 5.5,

5.5 $\rightarrow$ 6.1

2nd Stage

1.4

$$\varepsilon_{L0} < 0.1 \cdot 4 \text{ MHz}$$

$$\varepsilon_{L0} < 0.4 \text{ MHz}$$

$$\varepsilon_{L0} = 1400 \text{ MHz} \Rightarrow \pm 3.5 \text{ kHz}$$

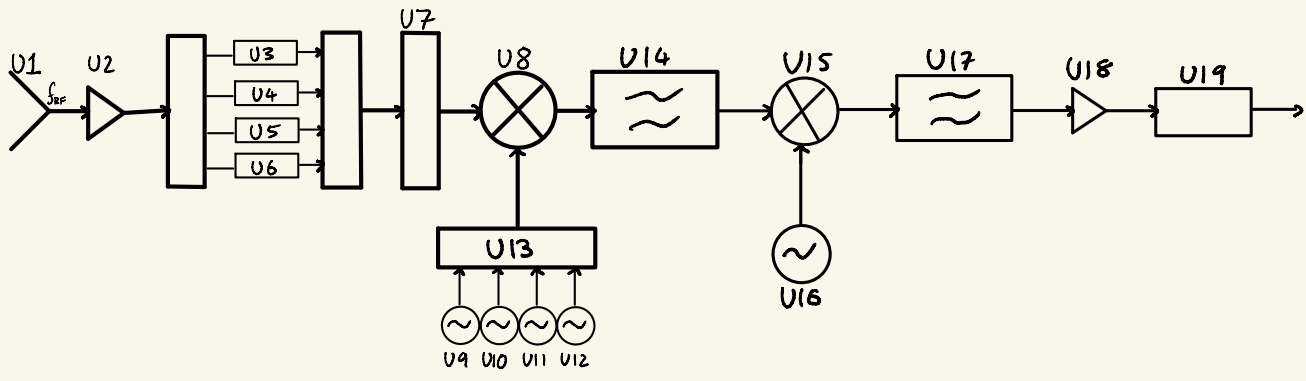
$$\varepsilon_{L0} = 3400 \text{ MHz} \Rightarrow \pm 9.75 \text{ kHz}$$

$$\varepsilon_{L0} = 4500 \text{ MHz} \Rightarrow \pm 11.25 \text{ kHz}$$

$$\varepsilon_{L0} = 5000 \text{ MHz} \Rightarrow \pm 12.5 \text{ kHz}$$

$$\varepsilon_{L0} = 5500 \text{ MHz} \Rightarrow \pm 13.75 \text{ kHz}$$

$$\varepsilon_{L0} = 6100 \text{ MHz} \Rightarrow \pm 15.25 \text{ kHz}$$



```

%% Comparison Between use of Butterworth vs Chebyshev
% MATLAB Script for Comparing Chebyshev and Butterworth Filters

% Clear workspace and command window
clear; clc; close all;

% Define filter parameters
Fs = 10e9;           % Sampling frequency (10 GHz)
Fc = 4e9;            % Center frequency (4 GHz)
BW = 2e6;            % Bandwidth (2 MHz)
N = 4;              % Filter order

% Define the normalized cutoff frequencies
Wn = [Fc - BW/2, Fc + BW/2] / (Fs / 2); % Normalized cutoff for bandpass filter

% Design Chebyshev Filter
[cheby_b, cheby_a] = cheby1(N, 1, Wn, 'bandpass');

% Design Butterworth Filter
[butter_b, butter_a] = butter(N, Wn, 'bandpass');

% Frequency response
[cheby_h, cheby_f] = freqz(cheby_b, cheby_a, 2048, 'half');
[butter_h, butter_f] = freqz(butter_b, butter_a, 2048, 'half');

% Convert frequency from normalized to actual frequency
cheby_f = cheby_f * Fs / (2 * pi); % Convert to frequency in Hz
butter_f = butter_f * Fs / (2 * pi); % Convert to frequency in Hz

% Plot frequency response
figure;
subplot(2,1,1);
plot(cheby_f / 1e9, 20*log10(abs(cheby_h)), 'b', 'LineWidth', 1.5); hold on;
plot(butter_f / 1e9, 20*log10(abs(butter_h)), 'r', 'LineWidth', 1.5);
title('Frequency Response of Chebyshev and Butterworth Filters');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
grid on;
legend('Chebyshev', 'Butterworth');
xlim([3.8 4.2]); % Limit x-axis to focus on the center frequency
ylim([-60 5]); % Set y-axis limits for better visualization

% Plot phase response
subplot(2,1,2);
plot(cheby_f / 1e9, unwrap(angle(cheby_h)) * (180/pi), 'b', 'LineWidth', 1.5); hold on;
plot(butter_f / 1e9, unwrap(angle(butter_h)) * (180/pi), 'r', 'LineWidth', 1.5);
title('Phase Response of Chebyshev and Butterworth Filters');
xlabel('Frequency (GHz)');
ylabel('Phase (degrees)');

```

```

grid on;
legend('Chebyshev', 'Butterworth');
xlim([3.8 4.2]); % Limit x-axis to focus on the center frequency

% Insertion loss comparison
cheby_insertion_loss = -20*log10(abs(cheby_h));
butter_insertion_loss = -20*log10(abs(butter_h));

% Display insertion loss at the center frequency
cheby_IL_at_Fc = cheby_insertion_loss(find(cheby_f >= Fc, 1));
butter_IL_at_Fc = butter_insertion_loss(find(butter_f >= Fc, 1));
fprintf('Insertion Loss at Center Frequency (Chebyshev): %.2f dB\n', cheby_IL_at_Fc);
fprintf('Insertion Loss at Center Frequency (Butterworth): %.2f dB\n', butter_IL_at_Fc);

% Show stopband attenuation
cheby_stopband_attenuation = 20 * log10(abs(cheby_h(find(cheby_f >= (Fc + BW), 1))));
butter_stopband_attenuation = 20 * log10(abs(butter_h(find(butter_f >= (Fc + BW), 1))));
fprintf('Stopband Attenuation (Chebyshev): %.2f dB\n', cheby_stopband_attenuation);
fprintf('Stopband Attenuation (Butterworth): %.2f dB\n', butter_stopband_attenuation);
%% Testing Filter Decided Upon
% MATLAB Script to Plot Filter Characteristics
% Define Filter Parameters
fs = 10e9; % Sampling frequency (example: 10 GHz, set this appropriately)
f = linspace(1e9, 5e9, 1000); % Frequency range from 1 GHz to 5 GHz

% Design Chebyshev Filter
order = 5; % Filter order
rp = 3; % Passband ripple (dB)
wc = 4e9; % Center frequency (4 GHz)

% Normalize cutoff frequency to the Nyquist frequency
wc_normalized = wc / (fs / 2); % Normalized cutoff frequency

% Design the Chebyshev filter
[b, a] = cheby1(order, rp, wc_normalized, 'low'); % Chebyshev filter coefficients

% Frequency Response
[H, f] = freqz(b, a, f, fs);

% Magnitude Response (in dB)
magResponse = 20*log10(abs(H));

% Plotting
figure;
subplot(2,1,1);
plot(f, magResponse);
hold on;
grid on;

% Mark Passband and Stopband

```

```

xline(3e9, 'r--', '3 GHz'); % Low end of the bandwidth
xline(5e9, 'r--', '5 GHz'); % High end of the bandwidth
title('Magnitude Response of Chebyshev Filter');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-80 10]); % Set y-axis limits
xlim([2.5e9 5.5e9]); % Focus on the relevant frequency range

% Mark -3 dB point
yline(-3, 'g--', '-3 dB Point');
yline(-20, 'k--', '-20 dB Attenuation');
text(4e9, -5, 'Passband', 'Color', 'g');
text(4.8e9, -25, 'Stopband', 'Color', 'k');

% Phase Response
subplot(2,1,2);
plot(f, angle(H));
grid on;
title('Phase Response of Chebyshev Filter');
xlabel('Frequency (Hz)');
ylabel('Phase (radians)');
ylim([-pi pi]);

% % Save Figure
% saveas(gcf, 'Filter_Response.png');

%%

% Define Filter Parameters
fs = 12e9; % Sampling frequency
order = 5; % Filter order
rp = 3; % Passband ripple (dB)

% Define cutoff frequencies
wc_low = 3e9; % Low-pass cutoff frequency
wc_high = 5e9; % High-pass cutoff frequency

% Normalize cutoff frequencies
wc_low_normalized = wc_low / (fs / 2);
wc_high_normalized = wc_high / (fs / 2);

% Design the filters
[b_low, a_low] = cheby1(order, rp, wc_low_normalized, 'low'); % Low-pass filter
[b_high, a_high] = cheby1(order, rp, wc_high_normalized, 'high'); % High-pass filter

% Frequency Response
f = linspace(1e9, 5e9, 1000); % Frequency range
[H_low, f] = freqz(b_low, a_low, f, fs); % Low-pass frequency response
[H_high, ~] = freqz(b_high, a_high, f, fs); % High-pass frequency response

```

```

% Magnitude Response (in dB)
magResponse_low = 20*log10(abs(H_low));
magResponse_high = 20*log10(abs(H_high));

% Plotting
figure;
subplot(2,1,1);
plot(f, magResponse_low);
grid on;
title('Magnitude Response of Low-Pass Filter');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

subplot(2,1,2);
plot(f, magResponse_high);
grid on;
title('Magnitude Response of High-Pass Filter');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

function interactiveSpuriousSignalPlot
% Initial values for fs and IF
fs_init = 3e9; % Starting value of fs in Hz
IF_init = 100e7; % Starting value of IF in Hz (1 GHz)

% Create a UI figure window
fig = uifigure('Name', 'Spurious Signal Visualizer', 'Position', [100 100 800 600]);

% Create sliders for fs and IF
fs_slider = uislider(fig, 'Position', [100, 50, 600, 3], ...
    'Limits', [3e9, 5e9], 'Value', fs_init, ...
    'MajorTicks', [3e9, 3.5e9, 4e9, 4.5e9, 5e9], ...
    'ValueChangedFcn', @(src, event) updatePlot());
fs_slider_label = uilabel(fig, 'Position', [100, 80, 150, 22], ...
    'Text', 'Tuned Frequency (fs): ');
fs_value_label = uilabel(fig, 'Position', [260, 80, 200, 22], ...
    'Text', sprintf('%.2f GHz', fs_init / 1e9));

IF_slider = uislider(fig, 'Position', [100, 120, 600, 3], ...
    'Limits', [50e6, 1.5e9], 'Value', IF_init, ...
    'MajorTicks', [50e6, 500e6, 1e9, 1.5e9], ...
    'ValueChangedFcn', @(src, event) updatePlot());
IF_slider_label = uilabel(fig, 'Position', [100, 150, 150, 22], ...
    'Text', 'Intermediate Frequency (IF): ');
IF_value_label = uilabel(fig, 'Position', [260, 150, 200, 22], ...
    'Text', sprintf('%.2f GHz', IF_init / 1e9));

% Create an axes to plot the spurious signals
ax = uiaxes(fig, 'Position', [50, 200, 700, 350]);
ax.XLabel.String = 'Frequency (Hz)';

```

```

ax.YLabel.String = 'Amplitude (Constant)';
ax.Title.String = 'Spurious Signals with Reference Frequency fs';
hold(ax, 'on');

% Initial plot
updatePlot();

% Nested function to update the plot and labels
function updatePlot()
    fs = fs_slider.Value; % Get current fs value from slider
    IF = IF_slider.Value; % Get current IF value from slider

    % Update label values for fs and IF
    fs_value_label.Text = sprintf('%.2f GHz', fs / 1e9);
    IF_value_label.Text = sprintf('%.2f GHz', IF / 1e9);

    % Define constant amplitude for all components in the stem plot
    amplitude = 1;

    % Highside components
    highside_freqs = [fs + IF, fs + 2 * IF, IF / 2, (fs + 2 * IF) / 2, ...
                     fs / 2, 2 * fs + 3 * IF, 2 * fs + IF, IF / 3];
    highside_labels = {'Highside LO', 'IMG(fs + 2IF)', 'IF/2', '(fs + 2IF)/2', ...
                      'fs/2', '2fs + 3IF', '2fs + IF', 'IF/3'};

    % Lowside components
    lowside_freqs = [fs - IF, fs - 2 * IF, IF / 2, (fs - 2 * IF) / 2, ...
                    fs / 2, 2 * fs - 3 * IF, 2 * fs - IF, IF / 3];
    lowside_labels = {'Lowside LO', 'IMG(fs - 2IF)', 'IF/2', '(fs - 2IF)/2', ...
                     'fs/2', '2fs - 3IF', '2fs - IF', 'IF/3'};

    % Clear and update plot
    cla(ax); % Clear previous plot
    hold(ax, 'on');

    % Plot fs as a reference point
    stem(ax, fs, amplitude, 'k', 'DisplayName', 'fs', 'LineWidth', 1.5);

    % Plot highside components
    for i = 1:length(highside_freqs)
        stem(ax, highside_freqs(i), amplitude, 'b', 'DisplayName', highside_labels{i});
    end

    % Plot lowside components
    for i = 1:length(lowside_freqs)
        stem(ax, lowside_freqs(i), amplitude, 'r', 'DisplayName', lowside_labels{i});
    end

    % Adjust plot properties for better visibility
    ax.YLim = [0, amplitude * 1.2]; % Set Y-axis limit slightly above amplitude

```



```

    ax.XLabel.String = 'Frequency (Hz)';
    ax.YLabel.String = 'Amplitude (Constant)';
    ax.Title.String = 'Spurious Signals with Reference Frequency fs';
    legend(ax, 'show');
    hold(ax, 'off');
end
end

interactiveSpuriousSignalPlot;

% Selectable preselection filters for visualization in fvtool
Fs = 20e9;           % Sampling frequency (20 GHz)
Rp = 0.5;           % Passband ripple in dB
Rs = 50;            % Stopband attenuation in dB
n = 6;              % Filter order

% Define each filter with specified frequency ranges

% Filter 1: 3-3.5 GHz
Fpass1_1 = 3e9;
Fpass2_1 = 3.5e9;
bpFilt1 = designfilt('bandpassiir', 'FilterOrder', n, ...
    'PassbandFrequency1', Fpass1_1, 'PassbandFrequency2', Fpass2_1, ...
    'PassbandRipple', Rp, 'StopbandAttenuation1', Rs, 'SampleRate', Fs);

% Filter 2: 3.5-4 GHz
Fpass1_2 = 3.5e9;
Fpass2_2 = 4e9;
bpFilt2 = designfilt('bandpassiir', 'FilterOrder', n, ...
    'PassbandFrequency1', Fpass1_2, 'PassbandFrequency2', Fpass2_2, ...
    'PassbandRipple', Rp, 'StopbandAttenuation1', Rs, 'SampleRate', Fs);

% Filter 3: 4-4.5 GHz
Fpass1_3 = 4e9;
Fpass2_3 = 4.5e9;
bpFilt3 = designfilt('bandpassiir', 'FilterOrder', n, ...
    'PassbandFrequency1', Fpass1_3, 'PassbandFrequency2', Fpass2_3, ...
    'PassbandRipple', Rp, 'StopbandAttenuation1', Rs, 'SampleRate', Fs);

% Filter 4: 4.5-5 GHz
Fpass1_4 = 4.5e9;
Fpass2_4 = 5e9;
bpFilt4 = designfilt('bandpassiir', 'FilterOrder', n, ...
    'PassbandFrequency1', Fpass1_4, 'PassbandFrequency2', Fpass2_4, ...
    'PassbandRipple', Rp, 'StopbandAttenuation1', Rs, 'SampleRate', Fs);

% Use fvtool to overlay all filter responses on a single plot
h = fvtool(bpFilt1, bpFilt2, bpFilt3, bpFilt4, 'Analysis', 'magnitude');
legend(h, 'Filter 1: 3-3.5 GHz', 'Filter 2: 3.5-4 GHz', 'Filter 3: 4-4.5 GHz', 'Filter 4:
    4.5-5 GHz');

```

```
title('Magnitude Response of Parallel Bandpass Filters');
```

Listing 1: Appendix All Code